Simultaneous Lattice Surgery for fast two-level factories with biased-noise qubits

Hugo Jacinto[1] [2], **Élie Gouzien[1]**

[1]Alice & Bob
[2]Institut de Physique Théorique, CEA Saclay

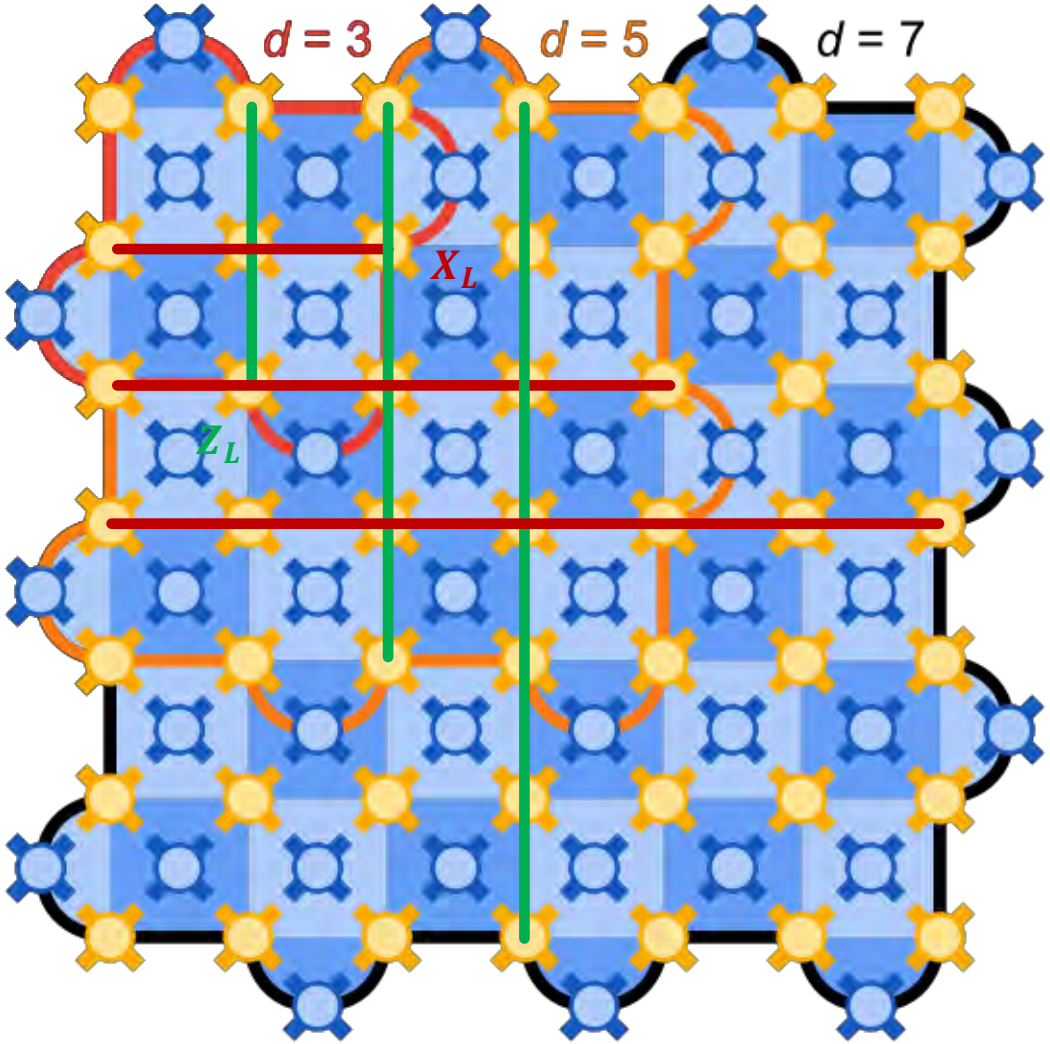
ALICE & BOB

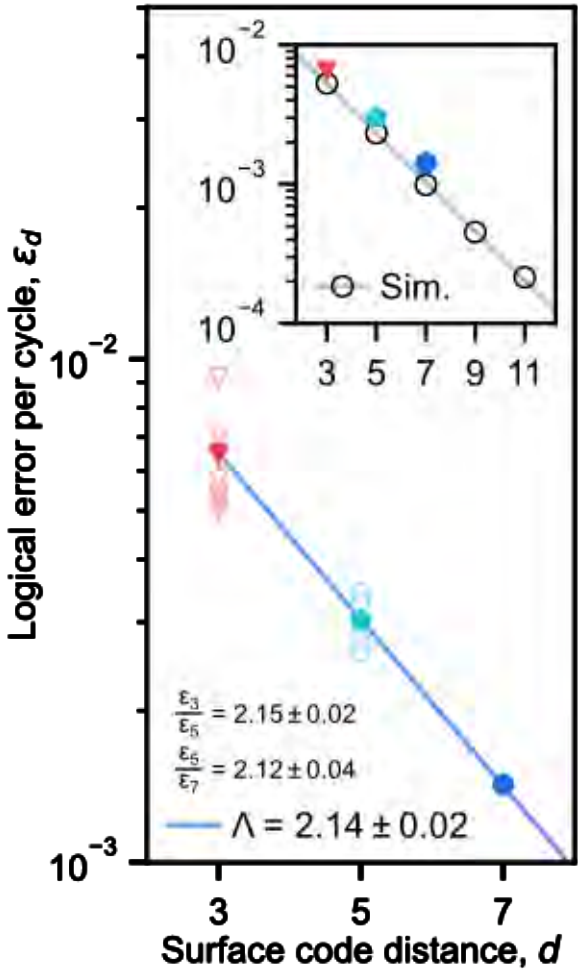
cea

IPhT SACLAY

Cat qubits architecture

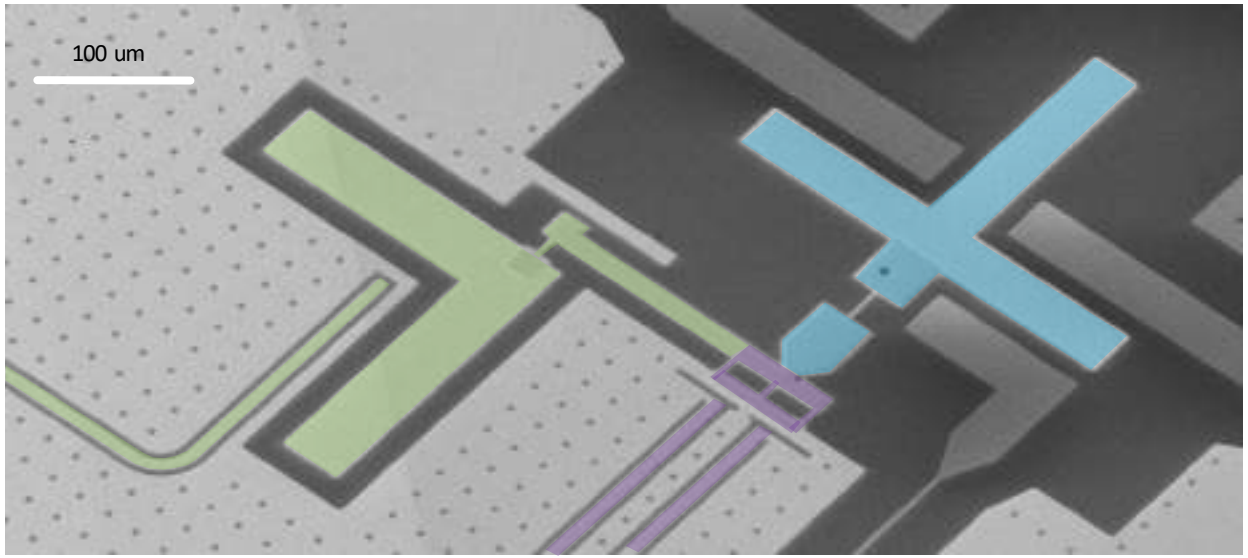
Quantum error correction with surface code

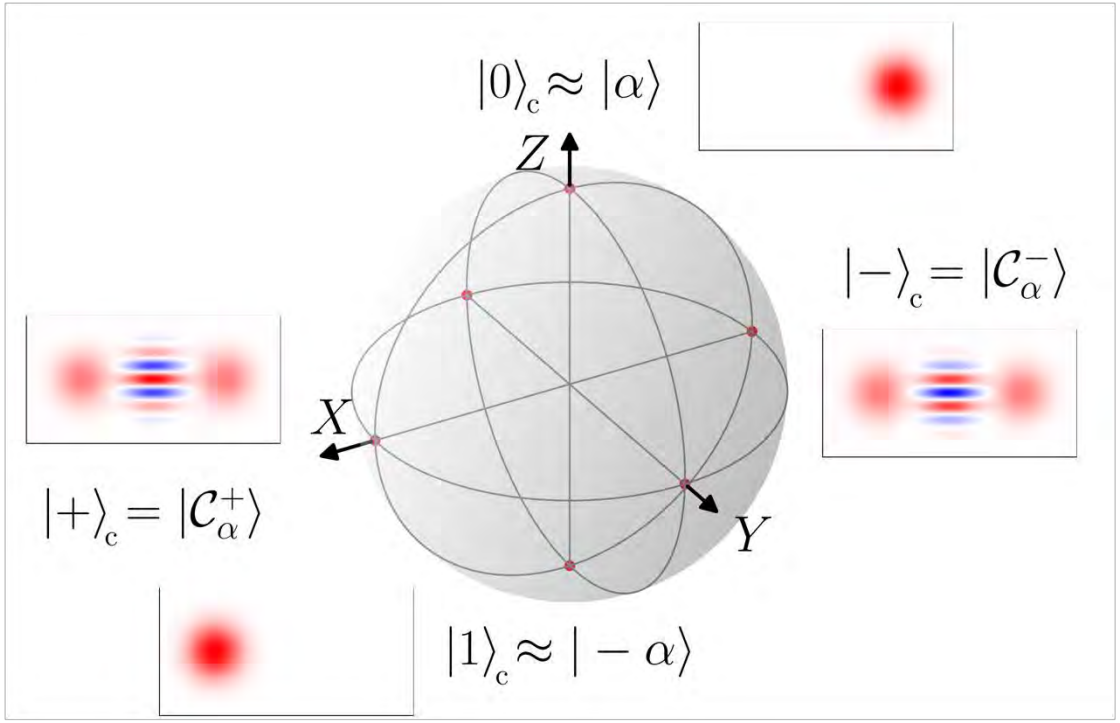




Google Quantum AI and Collaborators, Nature **638**, 920 (2025)

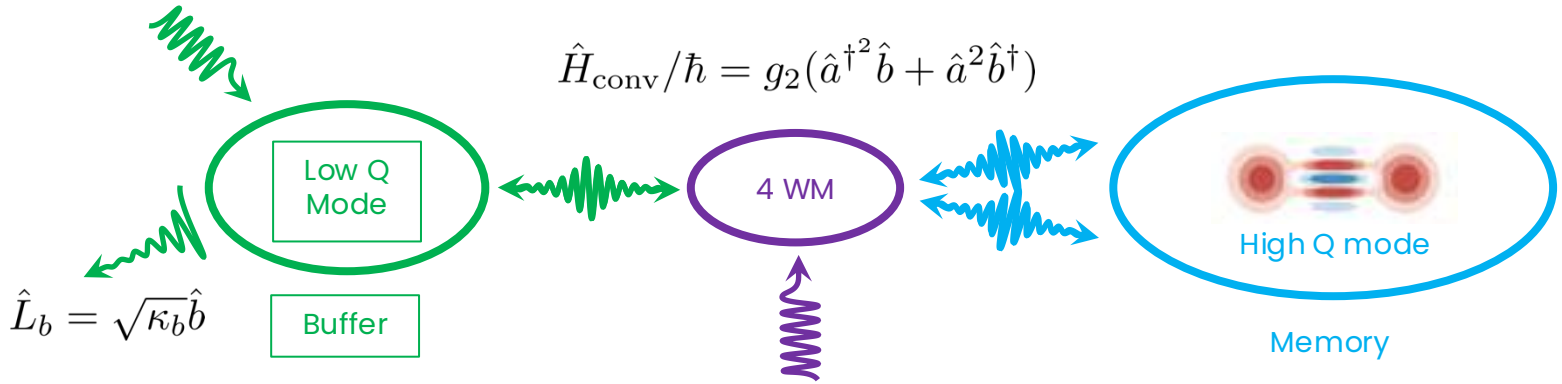
Dissipative cat qubits





$$\hat{H}_{\text{drive}}/\hbar = \epsilon_d(\hat{b}^\dagger + \hat{b})$$

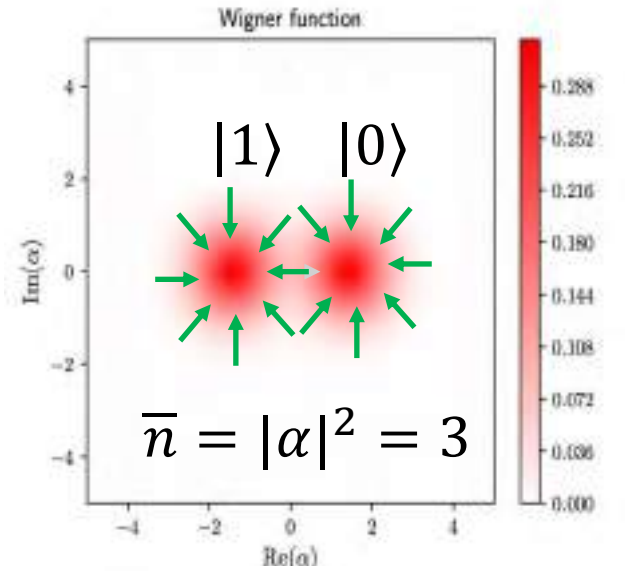
$$\hat{H}_{\text{conv}}/\hbar = g_2(\hat{a}^{\dagger 2}\hat{b} + \hat{a}^2\hat{b}^\dagger)$$

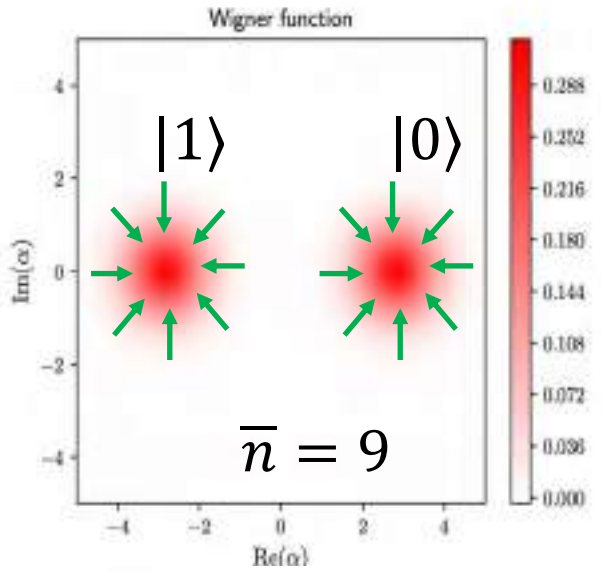


A biased noise, with « tunable bias »

Exponential reduction of bit-flips

Typical noise acts locally in phase-space

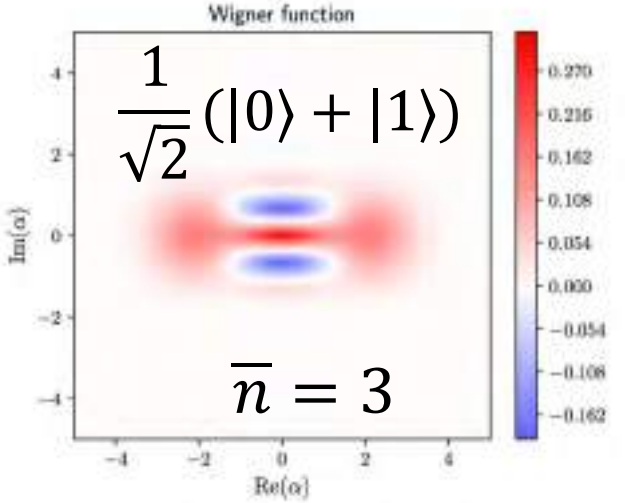


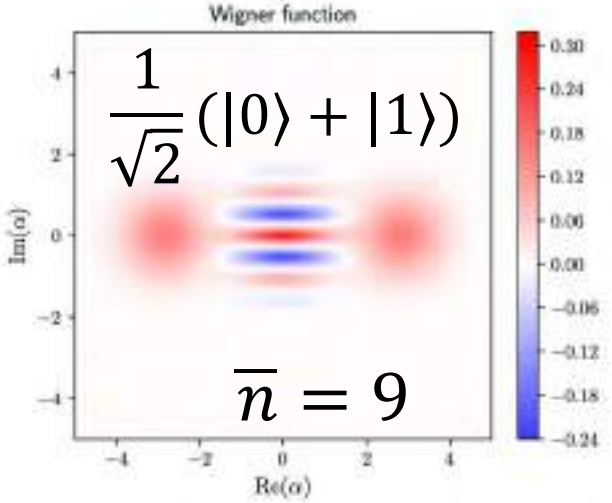


$+Z$

Linear increase of phase-flips

Interference patterns get smaller with cat size





$+X$

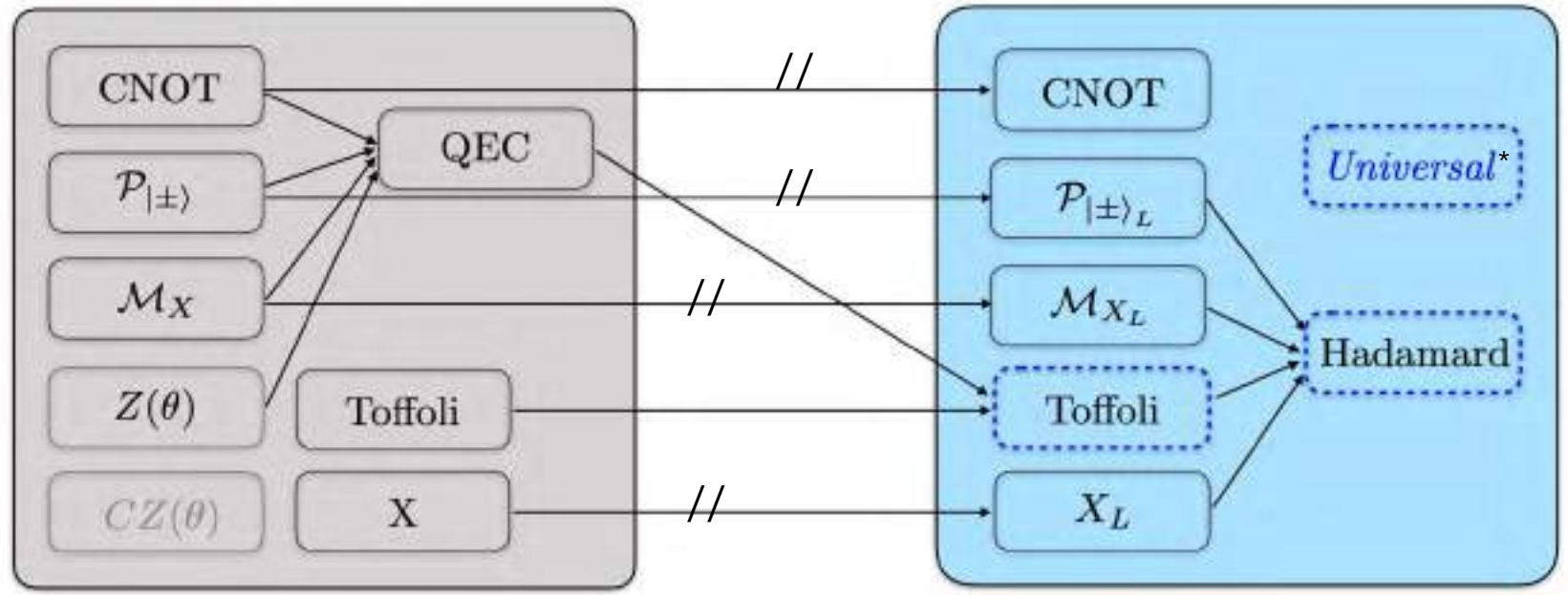
see J. Cohen, PhD thesis, 2017

Repetition-cat architecture

Accessible Quantum Gates on Classical Stabilizer Codes, Barizien, Jacinto and Sangouard, arXiv: 2507.05408

« Bias-preserving » operations

Fault-tolerant logical operations

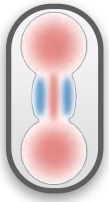


// Transversal

Single cat-qubit level

$|+\rangle = |\mathcal{C}_\alpha^+\rangle$

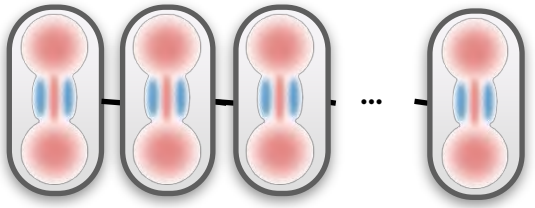
$|-\rangle = |\mathcal{C}_\alpha^-\rangle$



Repetition cat-qubit level

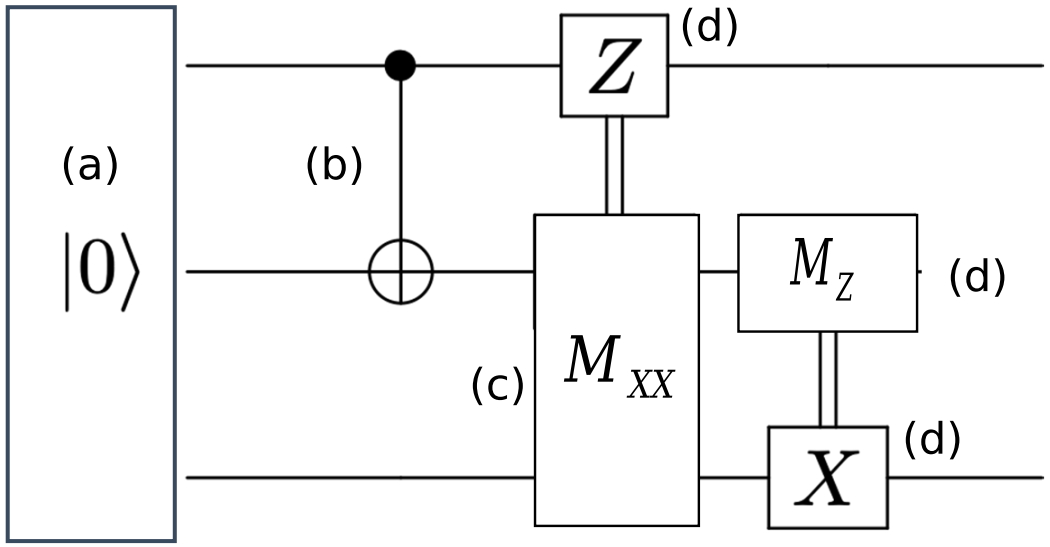
$|+\rangle_L = |\mathcal{C}_\alpha^+\rangle^{\otimes n}$

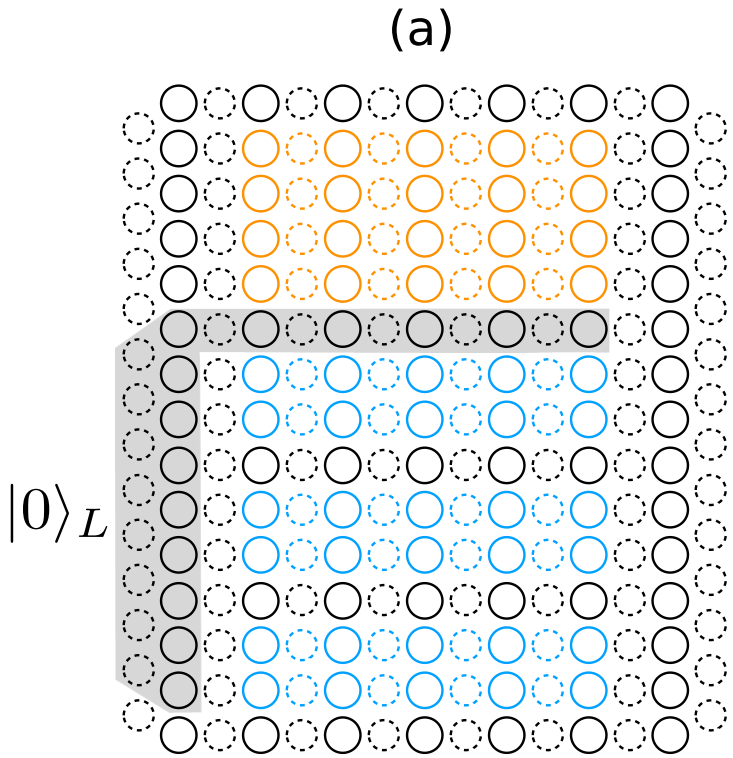
$|-\rangle_L = |\mathcal{C}_\alpha^-\rangle^{\otimes n}$



JG and MM, Phys. Rev. X 9, 041053

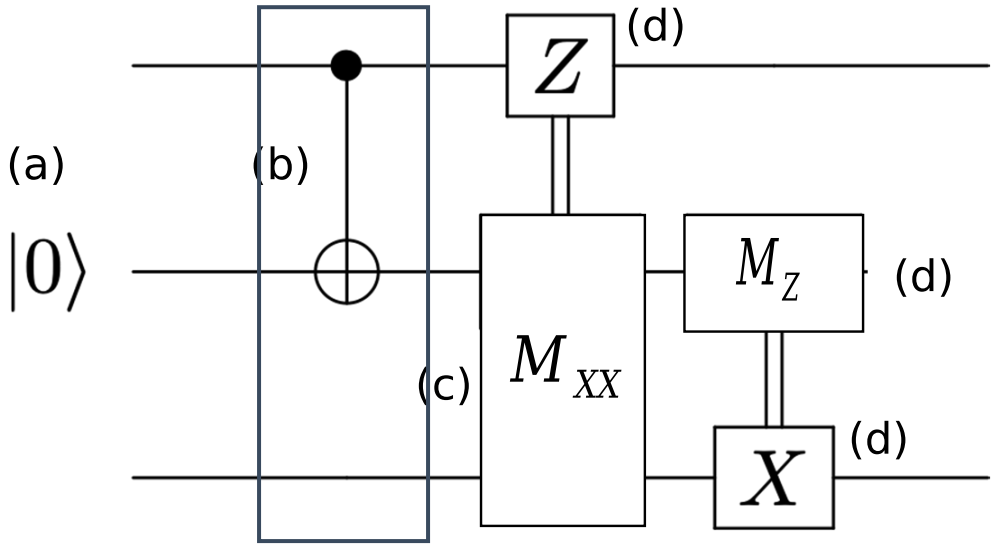
LONG RANGE CNOT



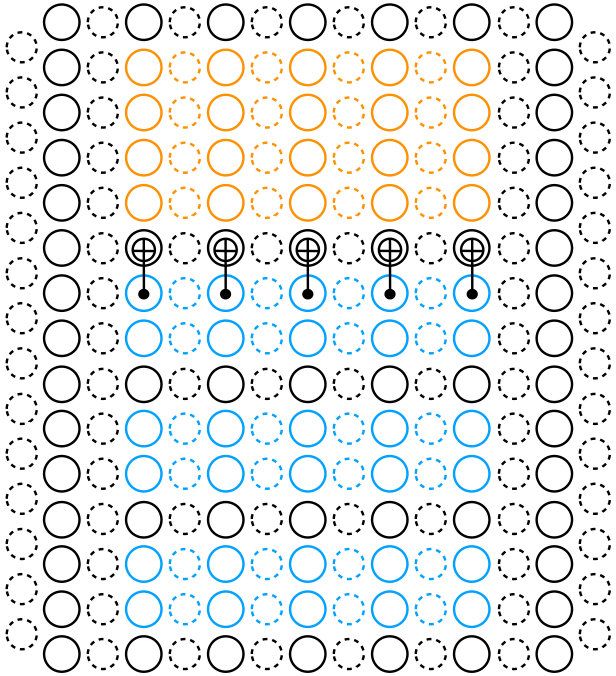


Gouzien et al., PRL 131 040602 (2023)

LONG RANGE CNOT

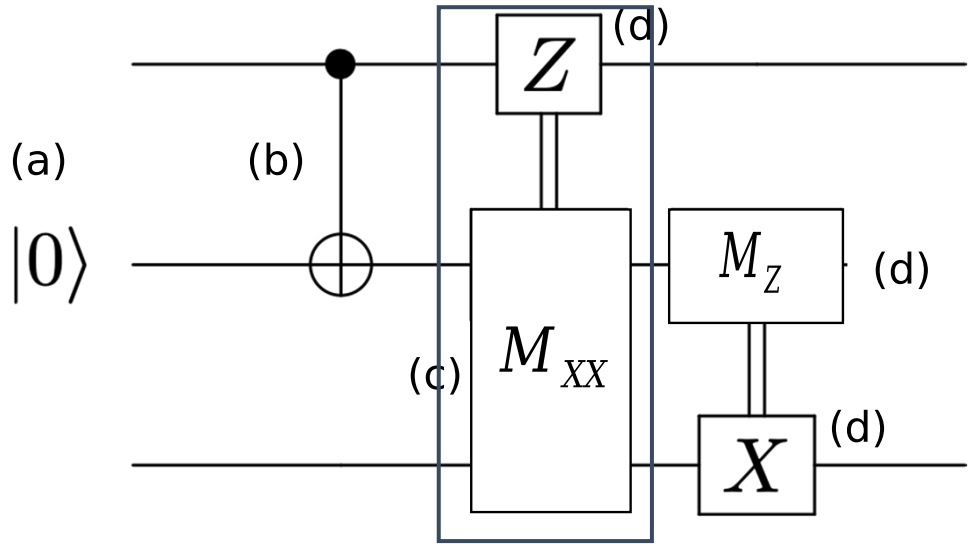


(b)

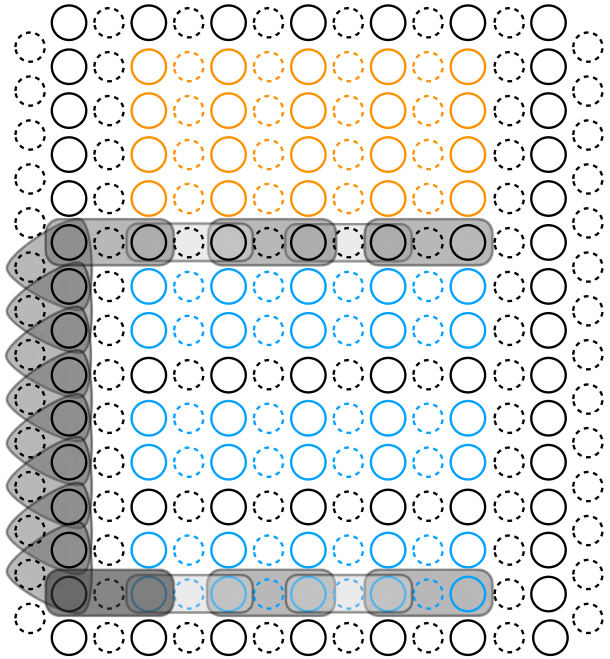


Gouzien et al., PRL 131 040602 (2023)

LONG RANGE CNOT

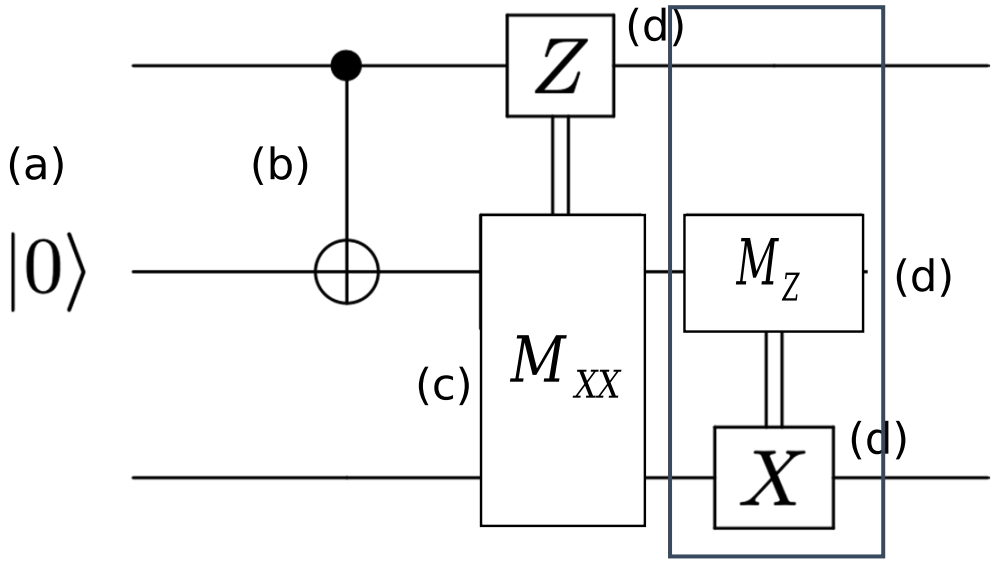


(c)

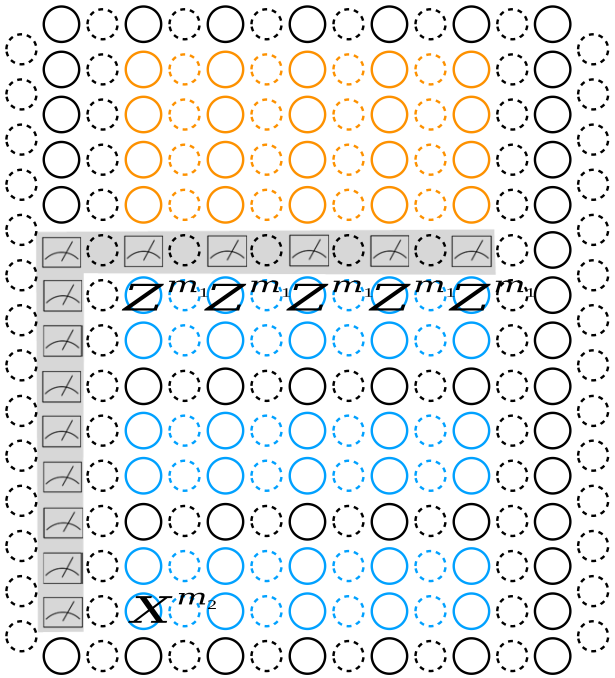


Gouzien et al., PRL 131 040602 (2023)

LONG RANGE CNOT



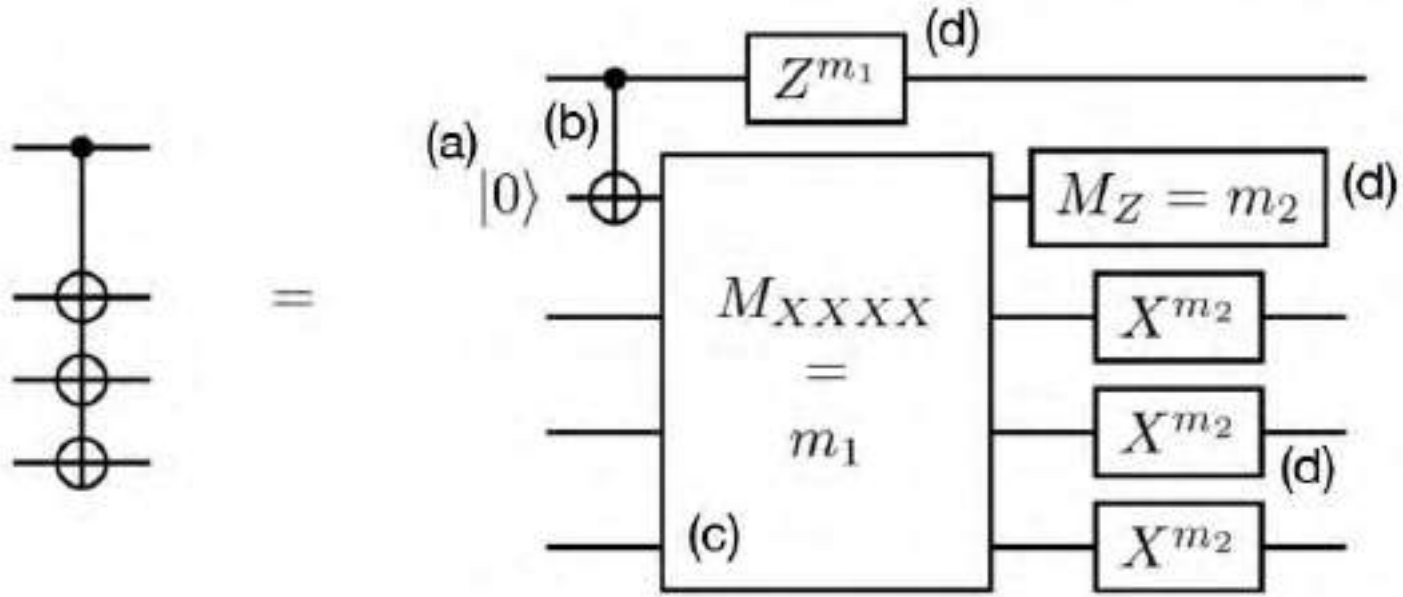
(d)

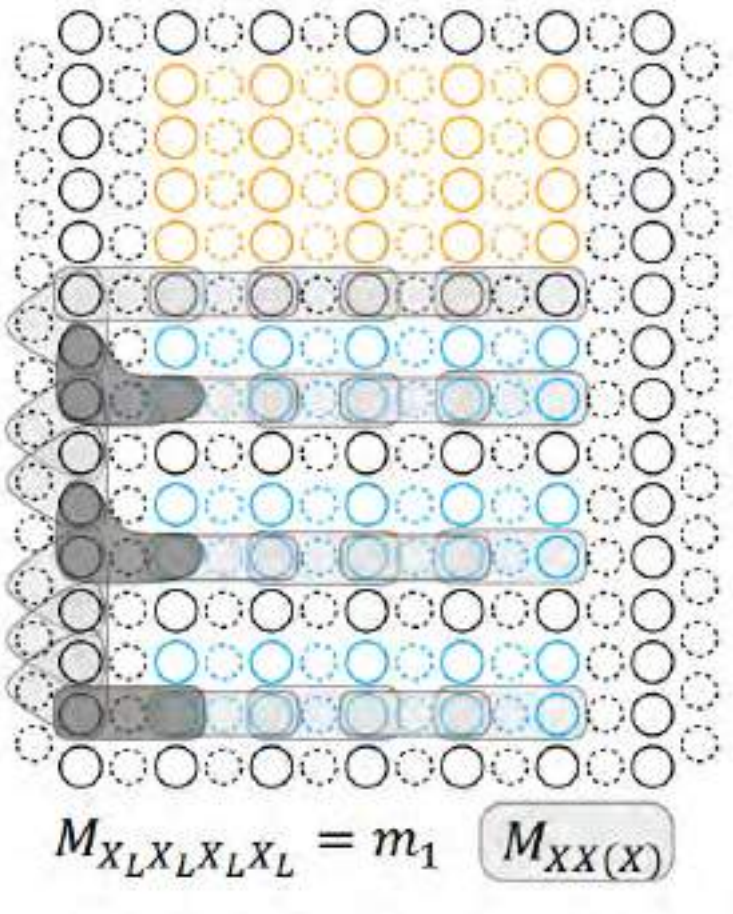


$M_{Z_L} = m_2$

Gouzien et al., PRL 131 040602 (2023)

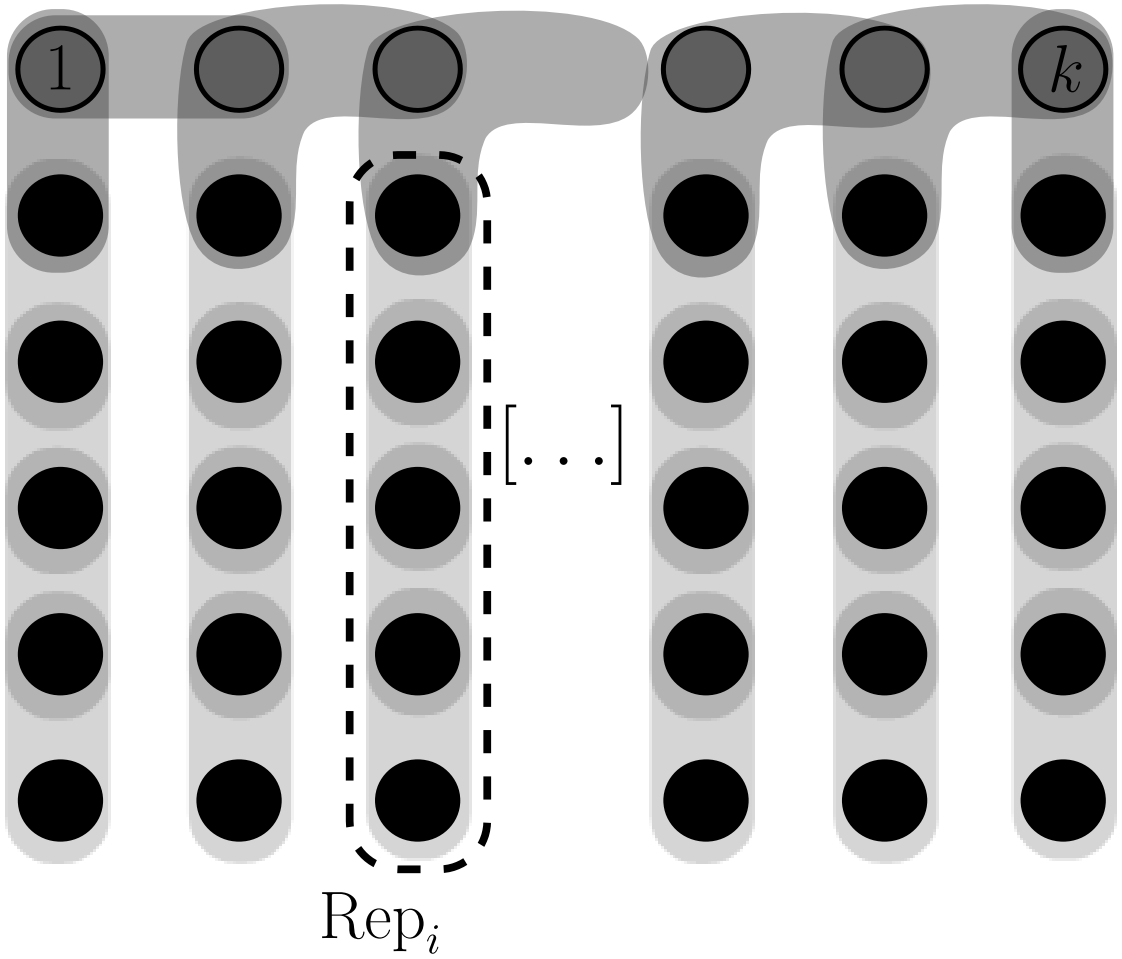
Multi-target CNOT





Gouzien et al., PRL 131 040602 (2023)

Zoom on the fork code

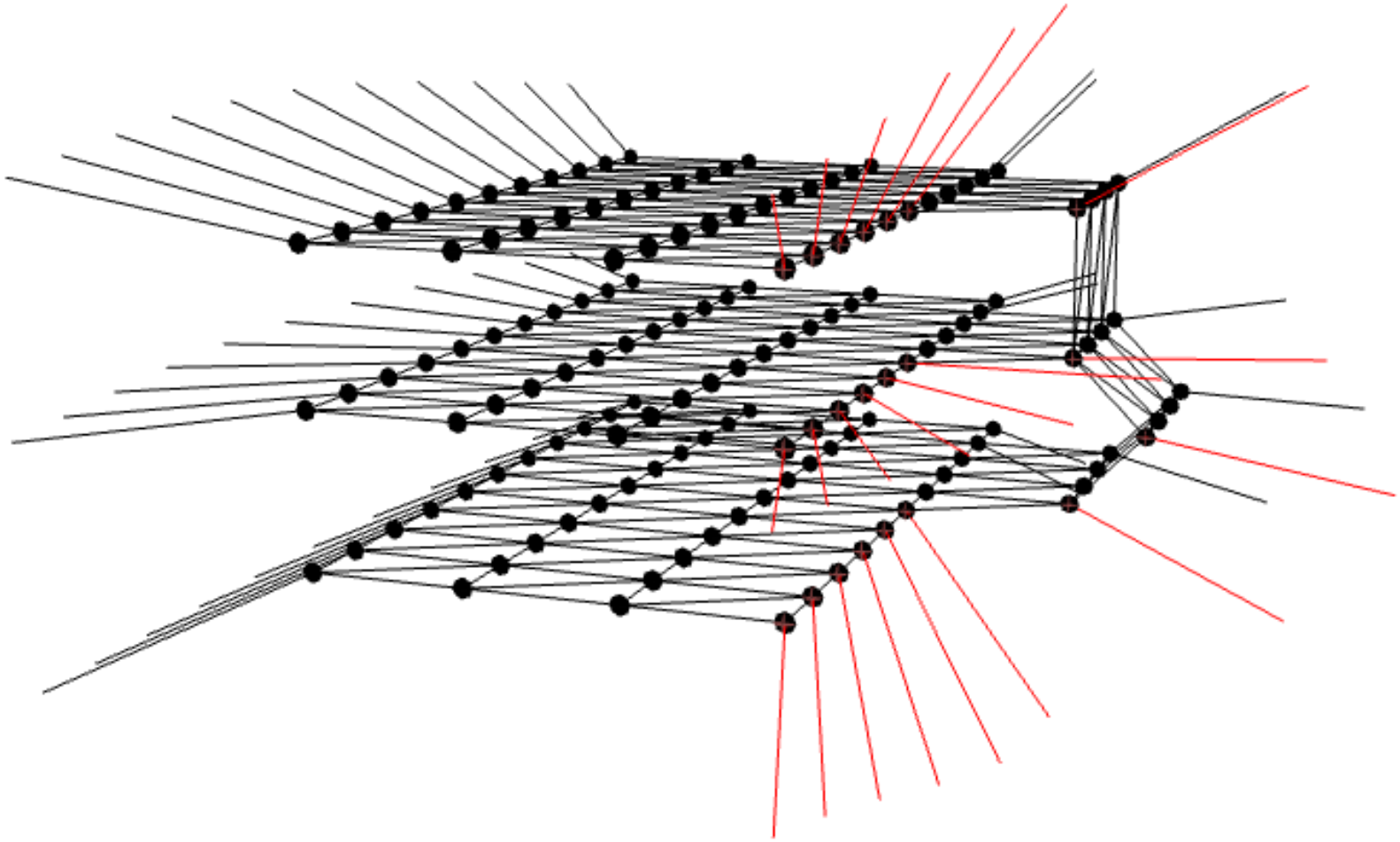


$$
\begin{aligned}
S_0 &= X_{L_1} X_1 \\
S_1 &= X_1 X_2 \\
S_2 &= X_{L_2} X_2 X_3 \\
S_3 &= X_{L_3} X_3 X_4 \\
&\vdots \\
S_i &= X_{L_i} X_i X_{i+1} \\
&\vdots \\
S_{k-1} &= X_{L_{k-1}} X_{k-1} X_k \\
S_k &= X_{L_k} X_k
\end{aligned}
$$

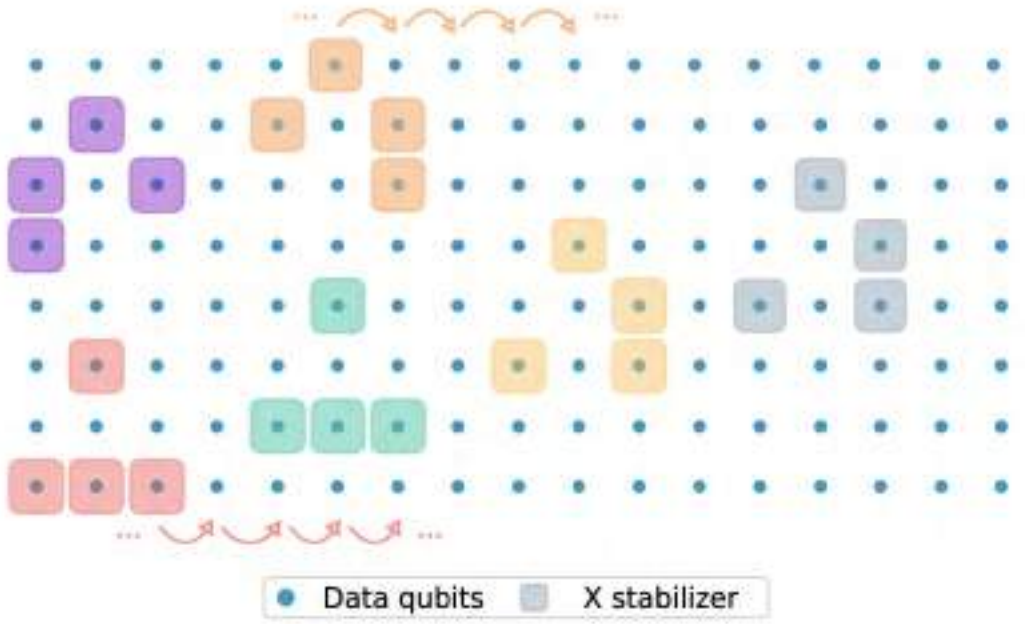
$$\prod_i S_i = \prod_i X_{L_i}$$

Distance: $d = 2d_0 + 1$

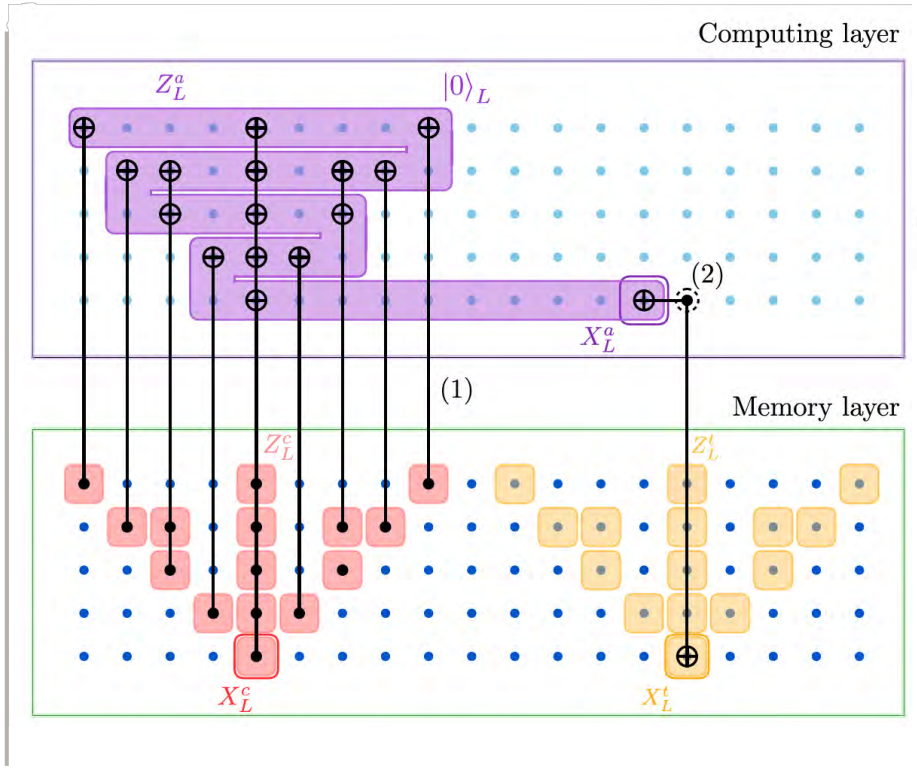
Fork code fault-tolerance

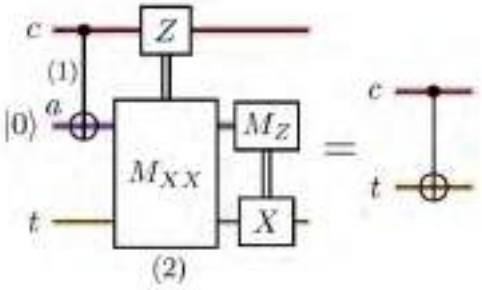


LDPC-cat code



Cellular automaton code



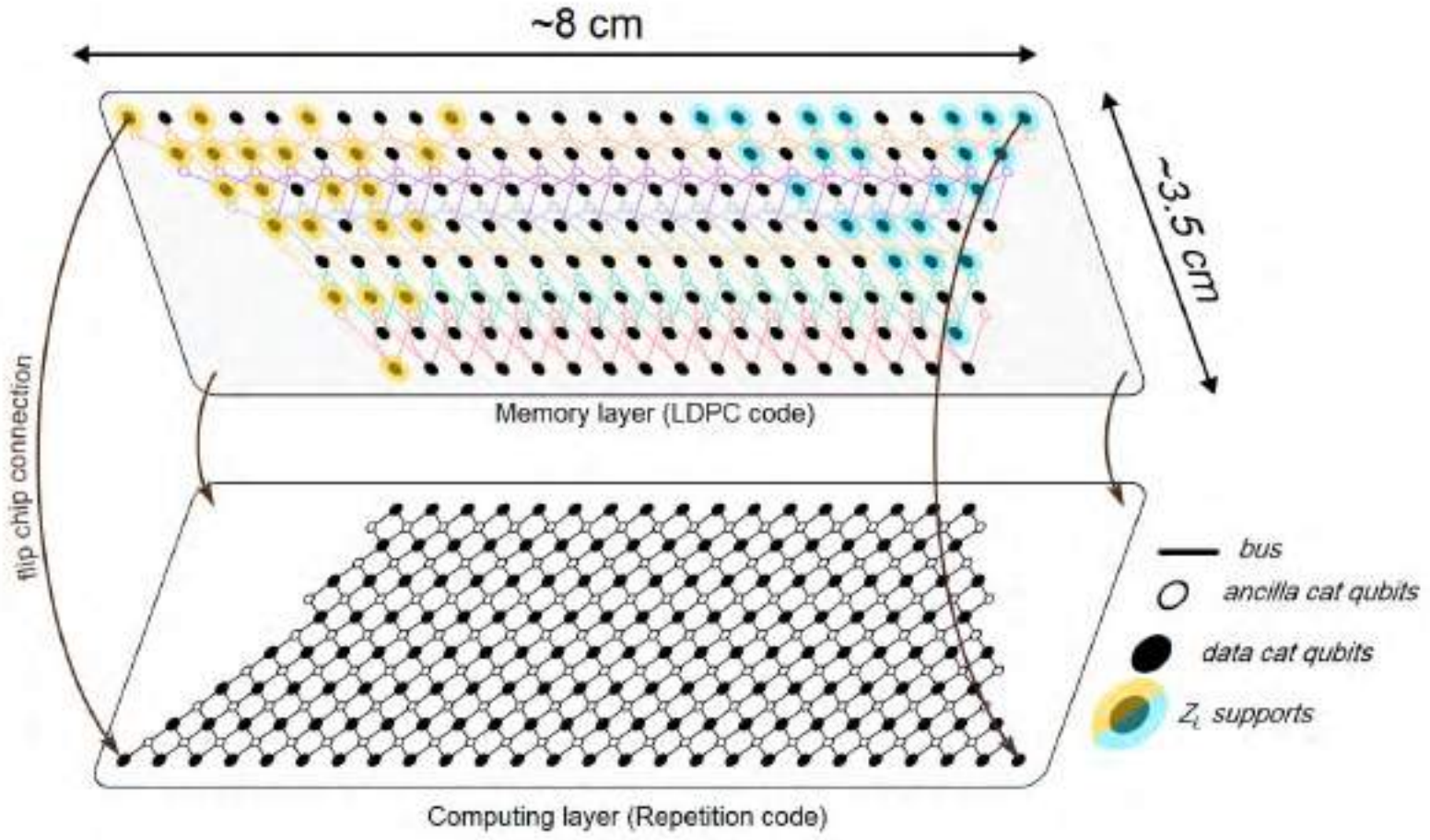


- Parallelizable for almost all logical qubits
- Flip-chip architecture

	Surface code + sc qubits [,]	High-rate qLDPC codes + sc qubits [ ]	Repetition code + cat qubits [ ]	**High-rate LDPC code + cat qubits**
Short-range interactions	yes (2D)	no (2D)	yes (1D)	yes (2D)
Tanner graph degree	3-4	6	2	4
$N_L = 100$ footprints, $\epsilon = 10^{-3}$, $\kappa_1/\kappa_2 = 10^{-4}$ $\rightarrow \epsilon_L \leq 10^{-8}$	N = 33,700	2,400 (N/14)	-	-
	-	-	2,100 (N/16)	758 (N/44)

Towards a 100 logical qubits processor, And beyond

Flip-chip architecture



100 logical qubits
[429, 100, 22] code
10^{-8} logical error rate

Theoretical targets
$\kappa_1/\kappa_2 = 10^{-4}$
$T_{\text{bit-flip}} = 13$ minutes
Bias-preserving CNOT

Current state of the art
$\kappa_1/\kappa_2 = 6.5\times10^{-3}$
$T_{\text{bit-flip}} = 15$ seconds
CNOT gate WIP

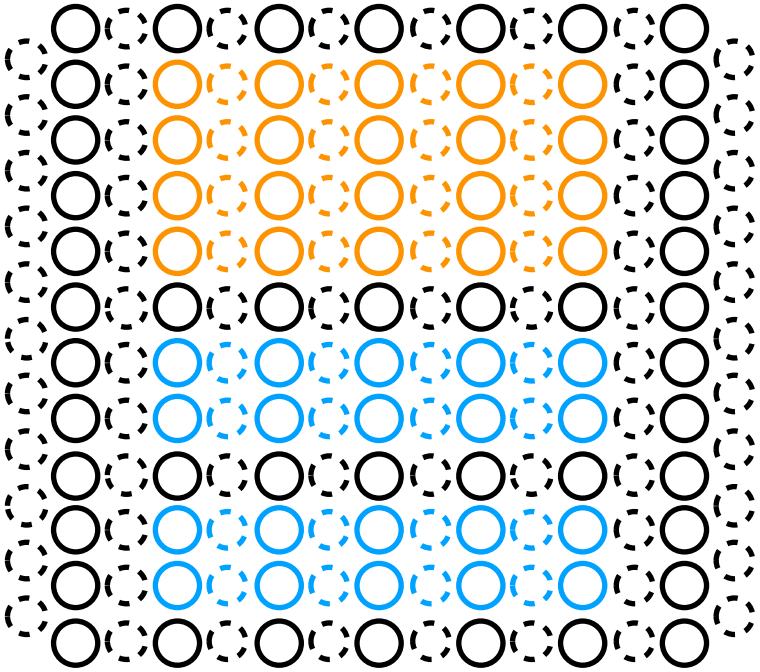
2048-bit factorization
Less than 100 000 cats
[24885, 6214, 22] code

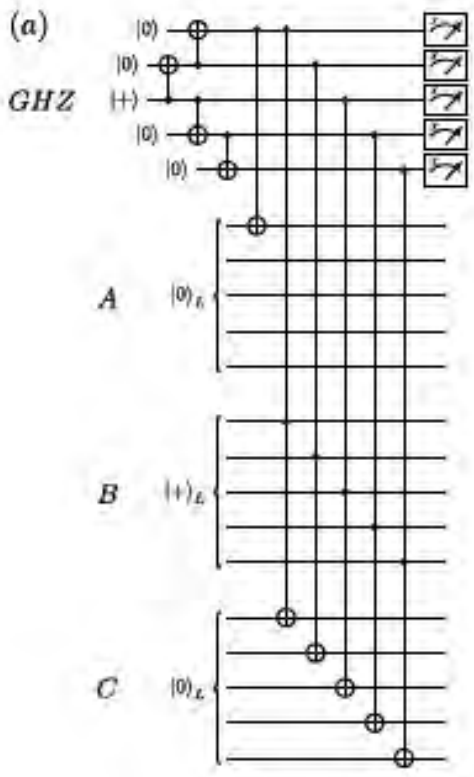


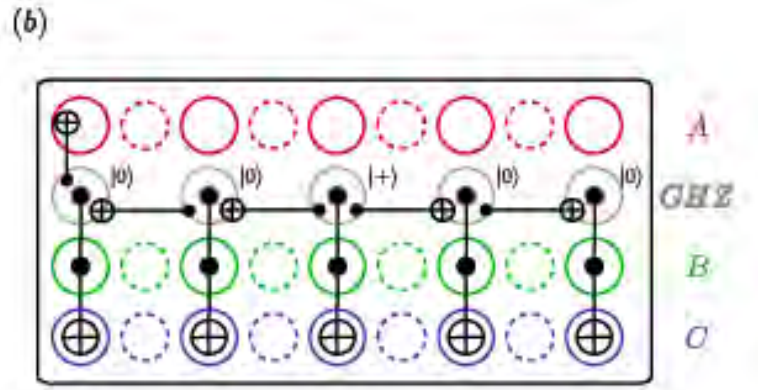
Magic state factories for cats previous work

Toffoli magic state factories

Toffoli magic state factories

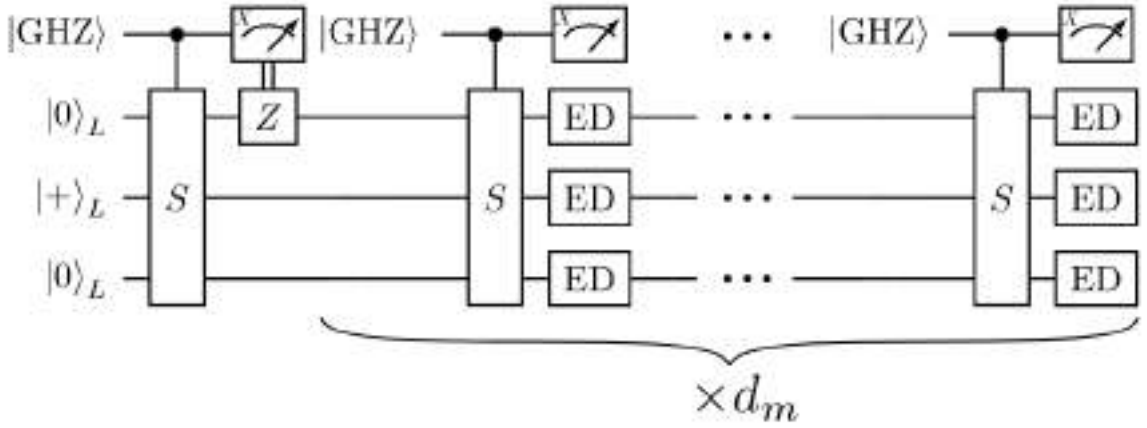




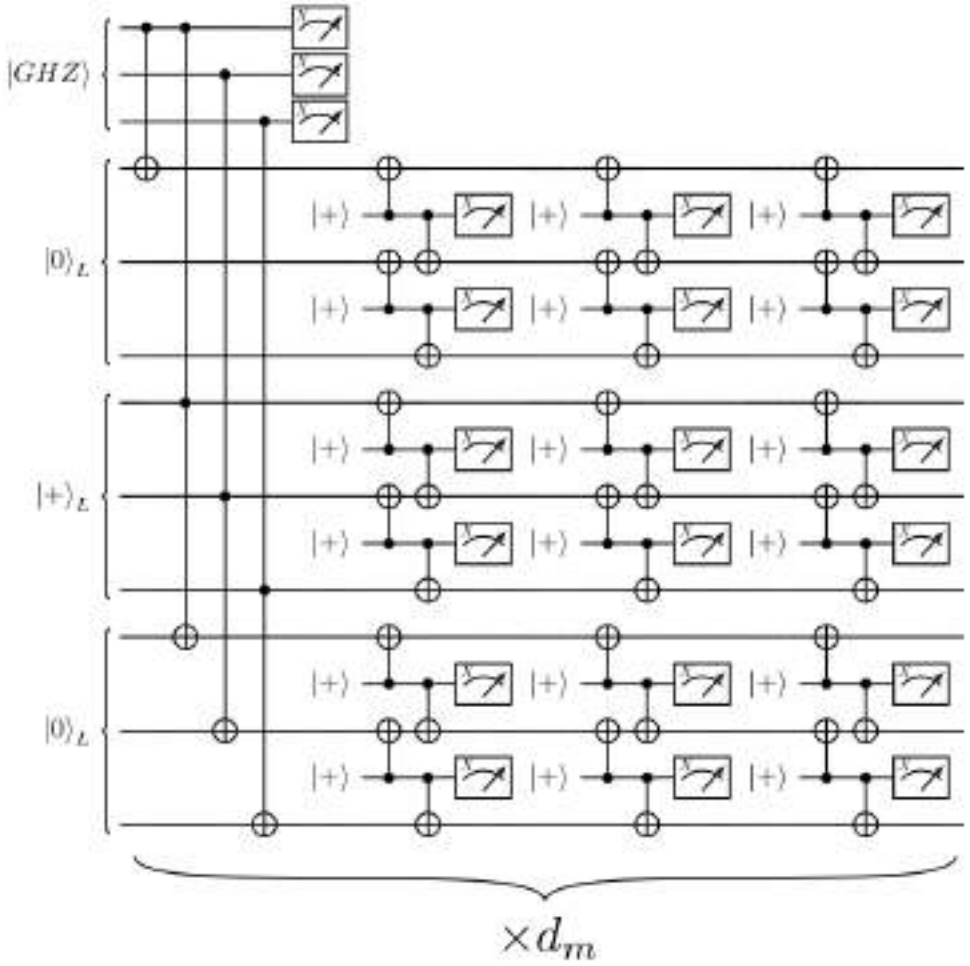


Toffoli magic state factories

Error detecting scheme



Error correcting scheme



Toffoli magic state factories

arXiv:2302.00639

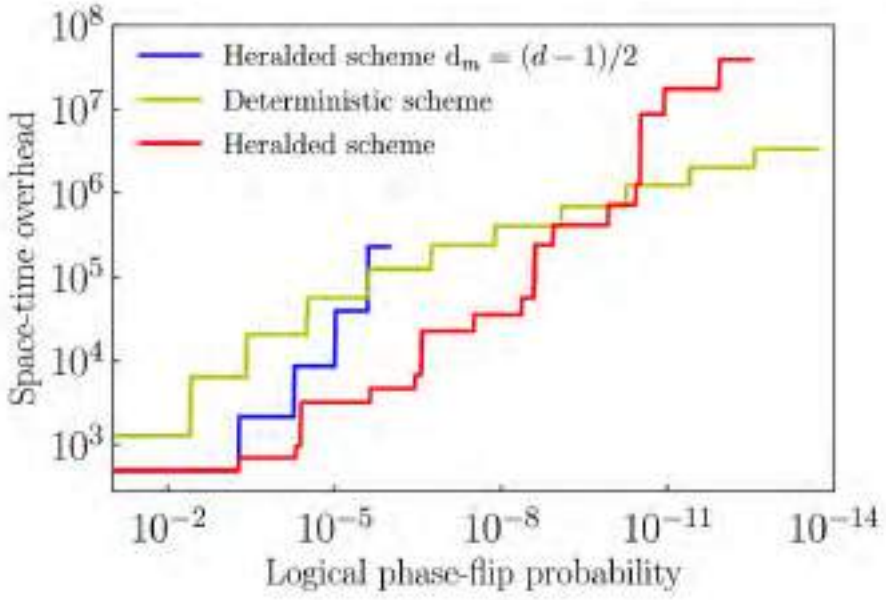
doi:10.1103/PhysRevLett.131.040602

Performance Analysis of a Repetition Cat Code Architecture: Computing 256-bit Elliptic Curve Logarithm in 9 Hours with 126 133 Cat Qubits

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(Dated: December 12, 2023)



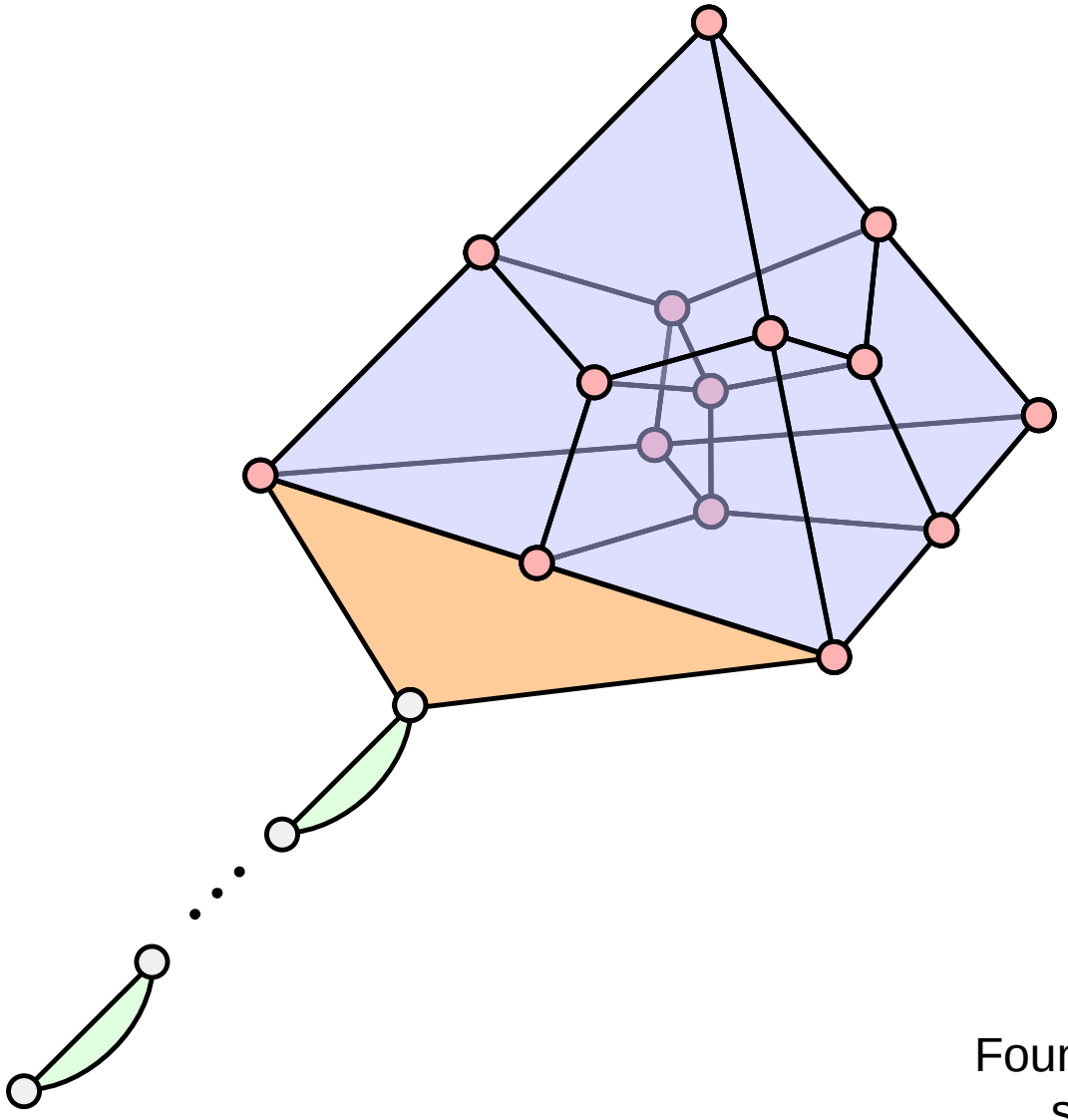
i	d	α^2	error prob.	steps	time	accept. prob. (%)
0	3	3.75	1.05×10^{-3}	23	54.7 µs	84
1	3	3.93	1.02×10^{-4}	29	65.8 µs	74.5
2	3	5.32	8.14×10^{-5}	35	58.7 µs	66
3	5	7.15	4.62×10^{-6}	46	57.4 µs	45.6
4	5	8.18	7.00×10^{-7}	53	57.8 µs	36.2
5	5	8.38	5.36×10^{-7}	60	63.9 µs	28.8
6	7	9.71	6.14×10^{-8}	73	67.1 µs	14.8
7	7	10.76	8.40×10^{-9}	81	67.2 µs	10.5
8	7	11.06	5.16×10^{-9}	89	71.8 µs	7.27
9	9	11.64	2.28×10^{-9}	104	79.7 µs	2.62
10	9	12.83	2.30×10^{-10}	113	78.6 µs	1.54
11	9	13.44	7.36×10^{-11}	122	81 µs	0.975
12	19	17.35	7.90×10^{-12}	9576	4.92 ms	100
13	21	18.94	5.40×10^{-13}	14112	6.65 ms	100
14	23	20.53	3.74×10^{-14}	21344	9.27 ms	100

Unfolded distillation

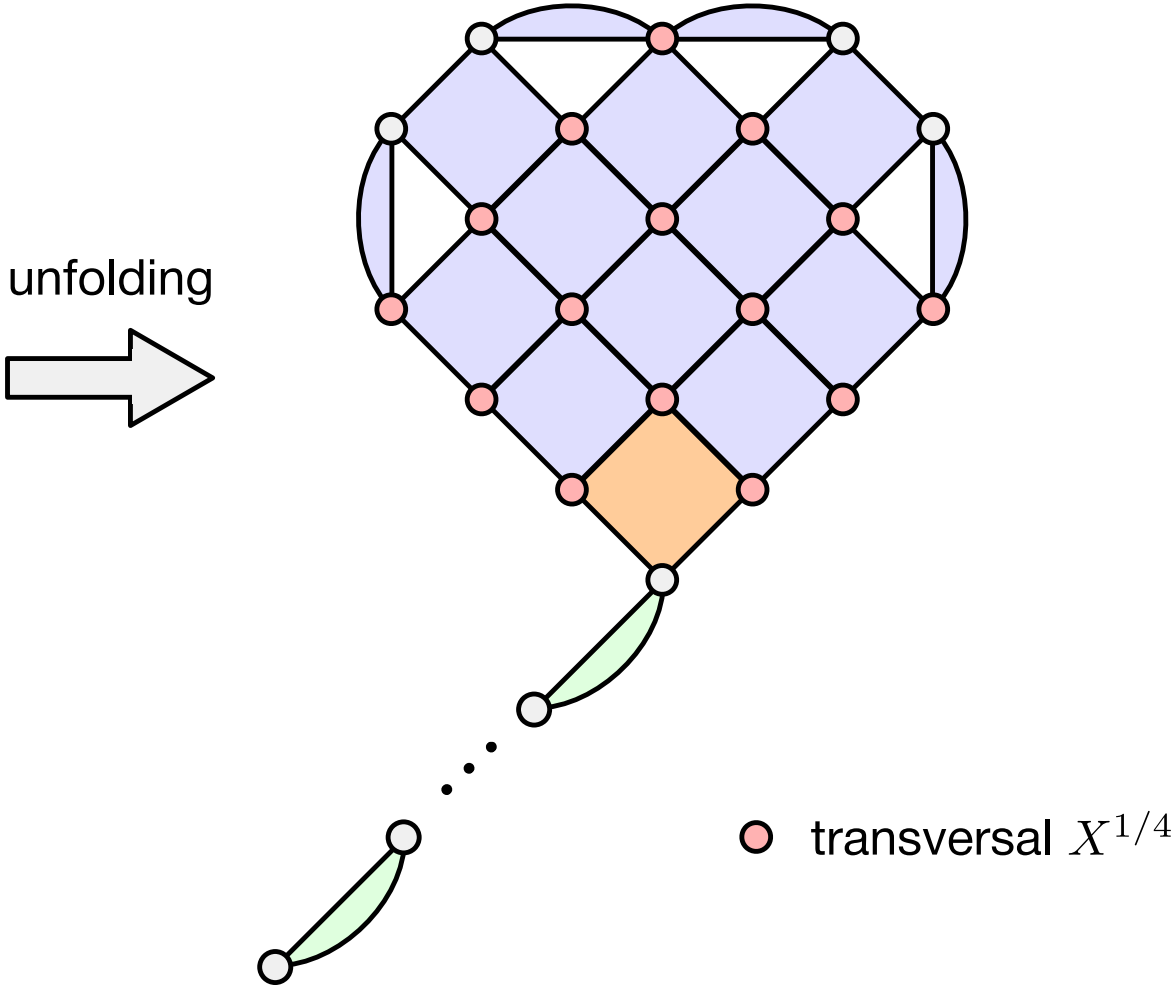
Ruiz *et al*, arXiv:2507.12511, 2025

Biased-noise qubit reduce the spatial dimension

 distillation at the physical level



unfolding

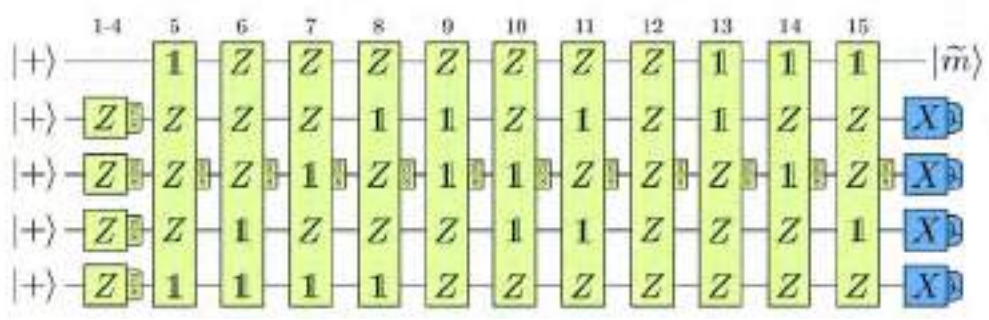


transversal $X^{1/4}$

Found with the SAT solver *kissat*

Unfolded distillation

Magic State Distillation: Not as Costly as You Think

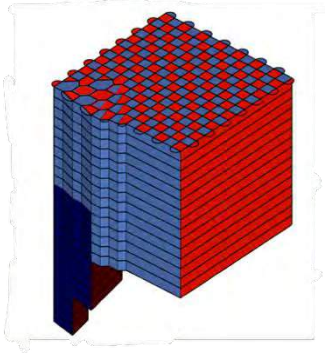


Litinski, Quantum 3, 205, 2019

4,620 qubits
42.6 rounds
197 000 Qubitcycles

$\Rightarrow \quad 10^{-3} \to 4.5 \times 10^{-8}$

Magic state cultivation

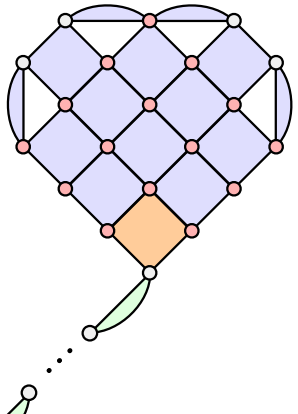


Gidney *et al*, arXiv:2409, 2024

500 qubits
50 rounds
25 000 Qubitcycles $(/10)$

$\Rightarrow \quad 10^{-3} \to 10^{-6}$

Unfolded distillation Ruiz *et al*, arXiv:2507.12511, 2025



53 qubits (incl rep code)
5.5 rounds
300 Qubitcycles $(/650)$

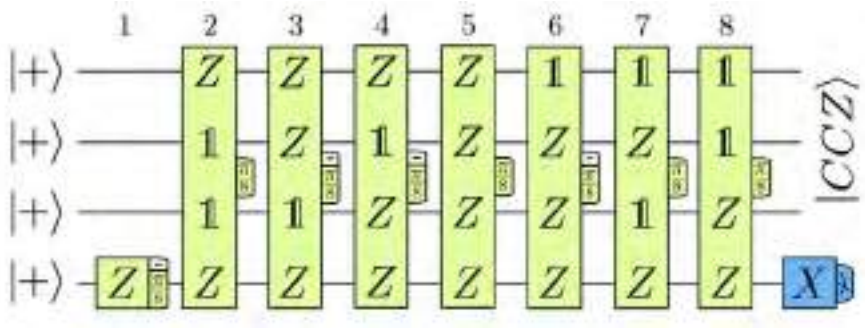
$\Rightarrow \quad \begin{cases} p_{Z,\text{in}} = 10^{-3} \\ \eta = 5 \times 10^{6} \end{cases} \longrightarrow p_{\text{out}} = 3 \times 10^{-7}$

Not enough for Shor, chemistry…

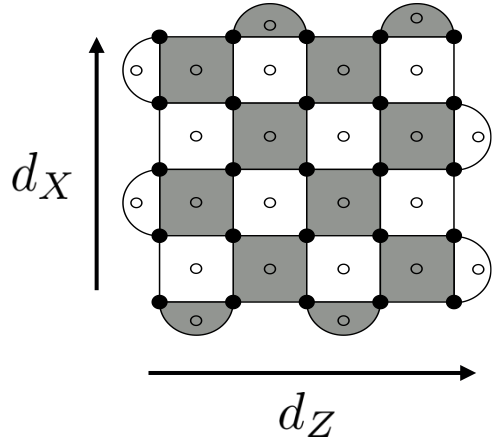


Second distillation stage

Second-level of distillation as a compilation problem

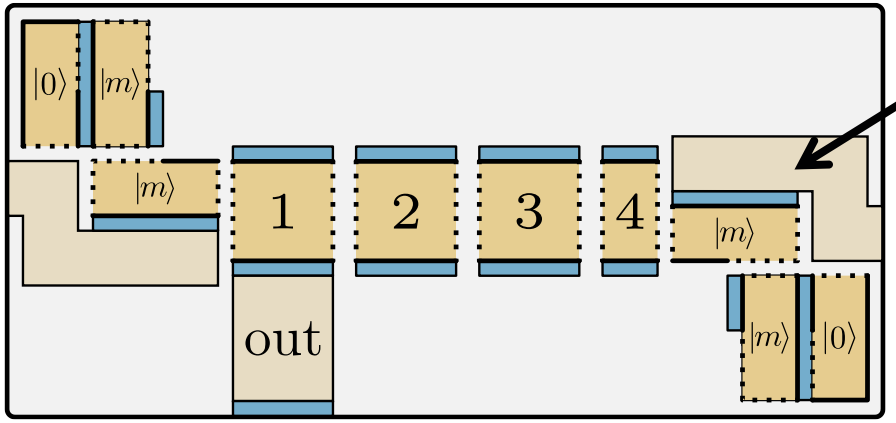


$$p_{\text{out},1} \longrightarrow p_{\text{out},2} \geq 28(p_{\text{out},1})^2$$

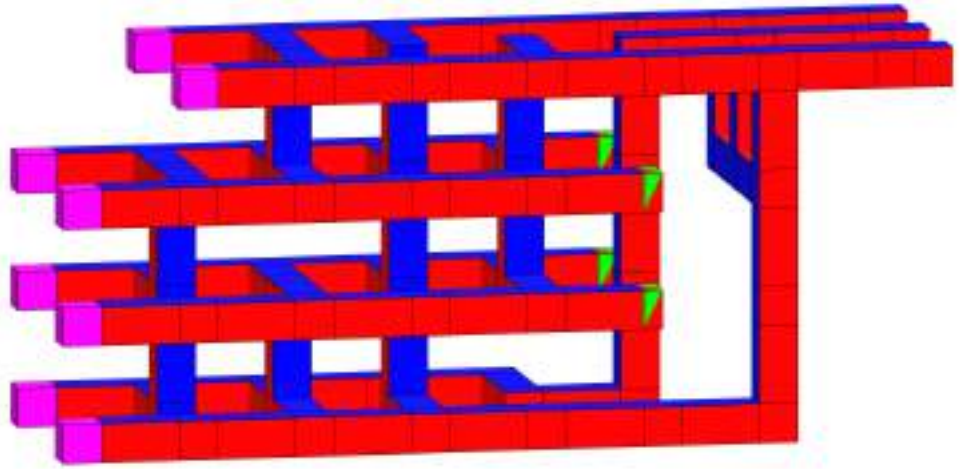


Surface code architectures :

First level factories outputs

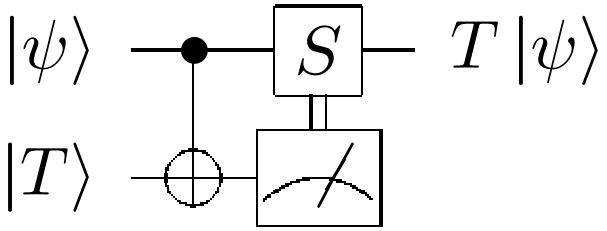


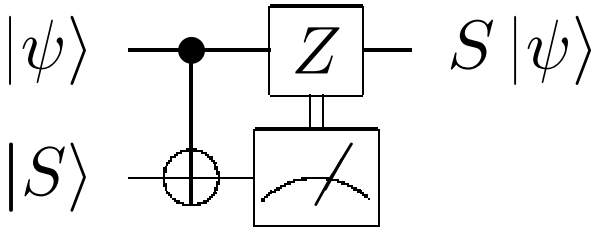
Litinski, Quantum 3, 205, 2019

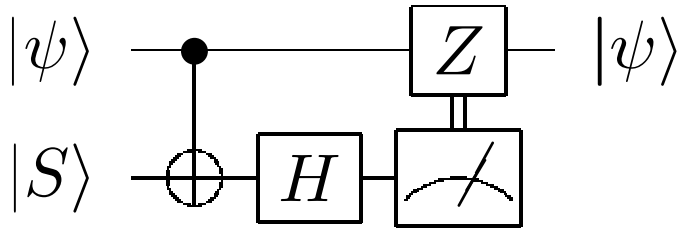


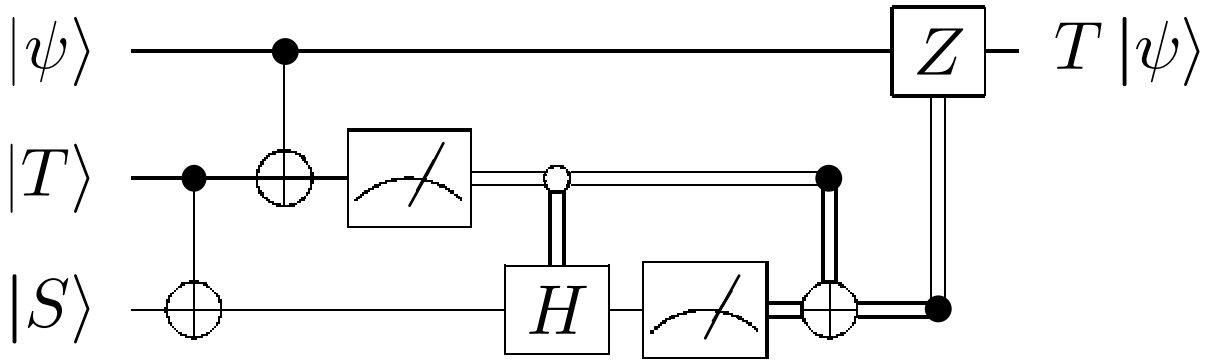
Gidney, arxiv:2409.17595

Autocorrected T injection





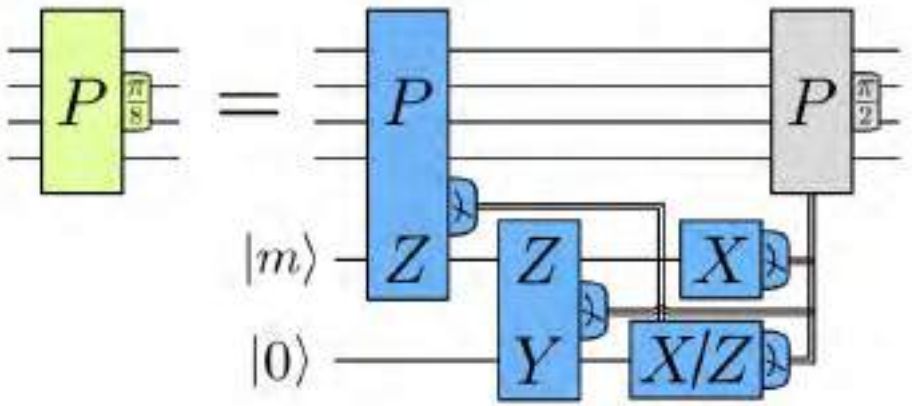




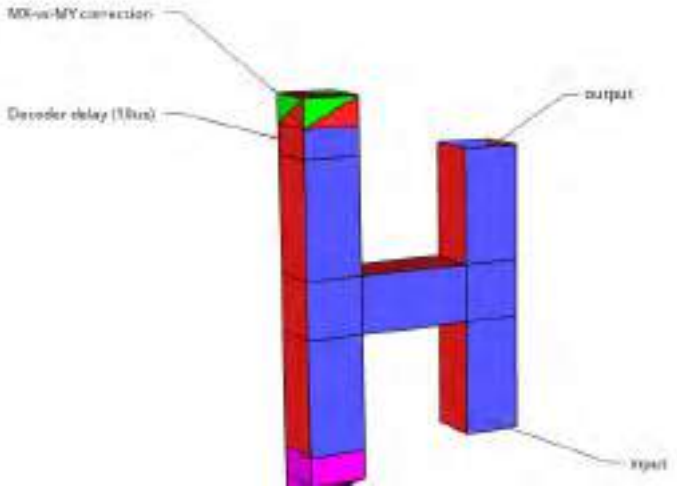
Replaced “apply a gate” by “chose measurement basis”

Lattice-surgery autocorrected $\pi/8$ injection

Surface code

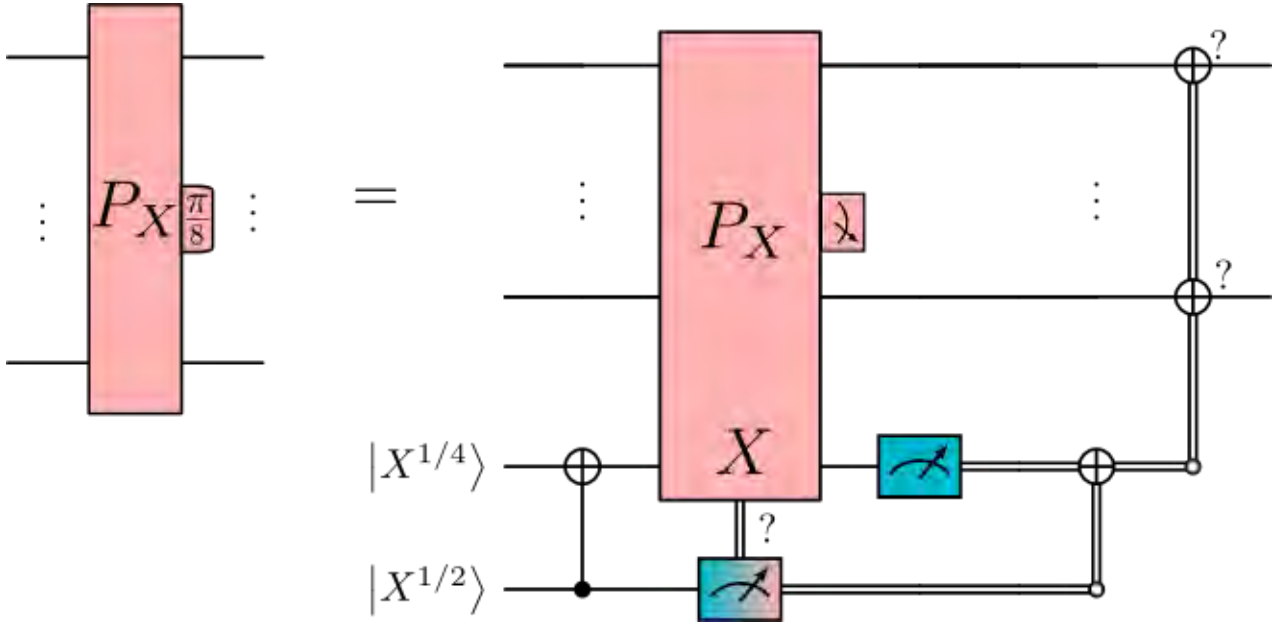


Litinski, Quantum 3, 205, 2019

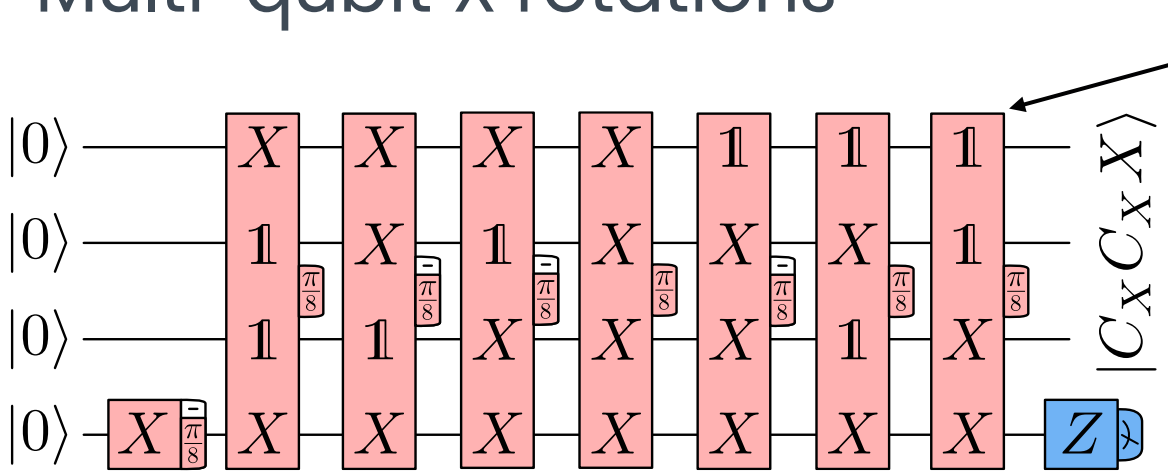


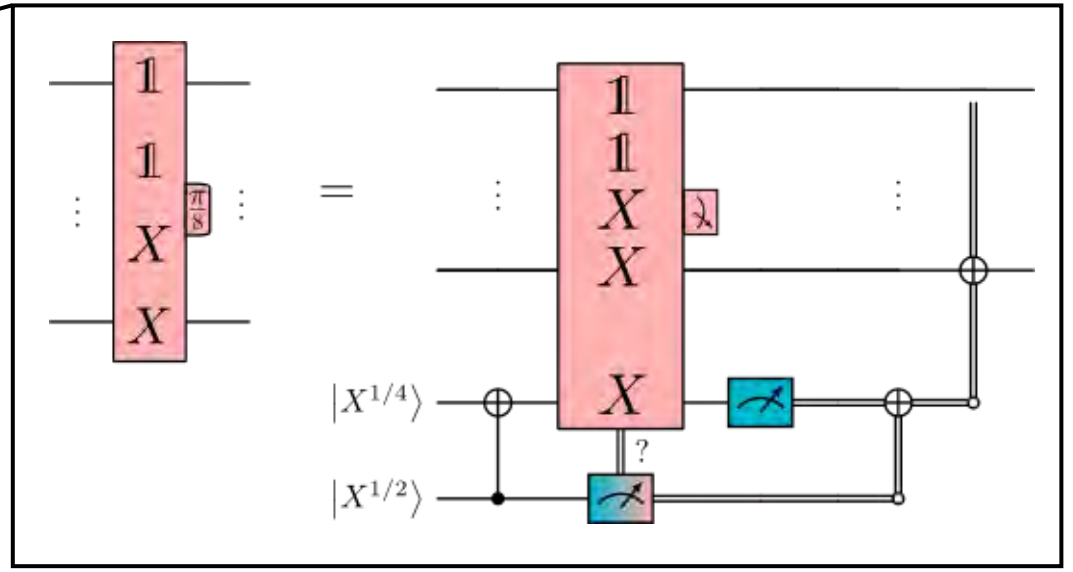
Fowler, TQEC talks, 2025

Cat-repetition code



Multi-qubit X rotations





Lattice surgery for phase-flip repetition code:

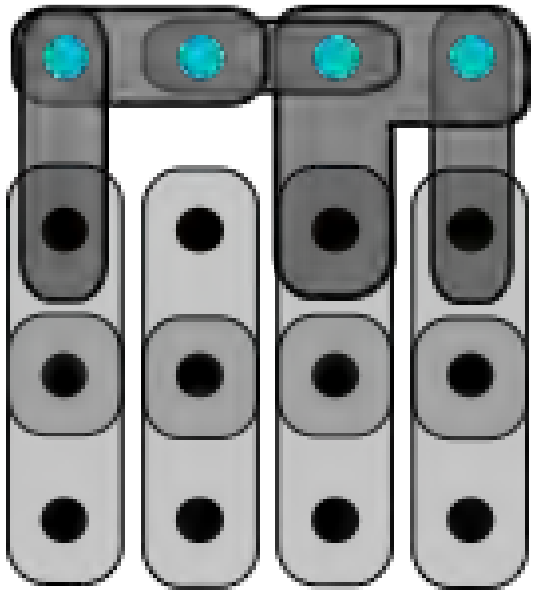
data qubits

Rep. Code
X stabilizers

routing qubits
Initialized in $|0\rangle$

Merging 2 qubits
X stabilizers

Merging 3 qubits
X stabilizer



$$\bigoplus_{m \in \text{dark grey stabs}} M_m$$

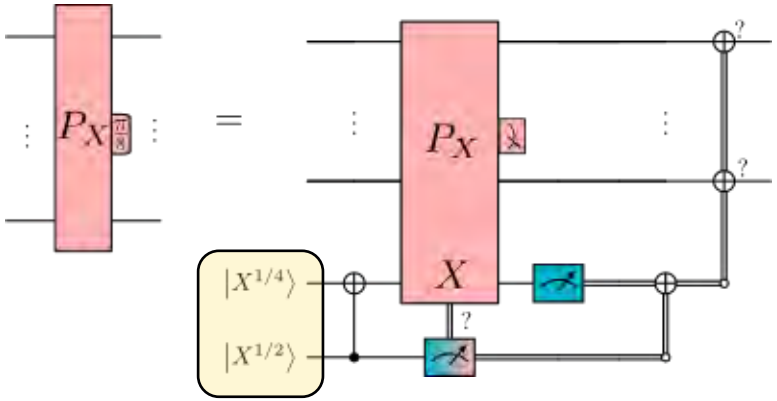
$$\Downarrow$$

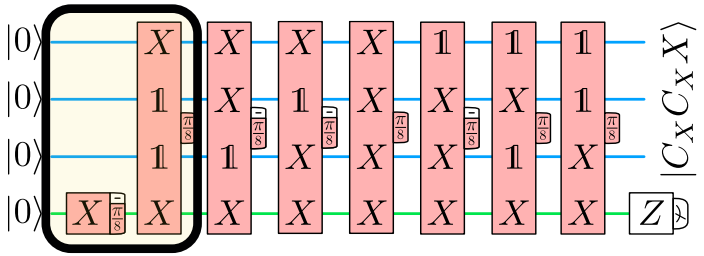
$$M_{X_1 X_3 X_4}$$

Timelike distance : d_m

Gouzien *et al*, Physical Review Letters 131, 2023

Multi-qubit X rotations



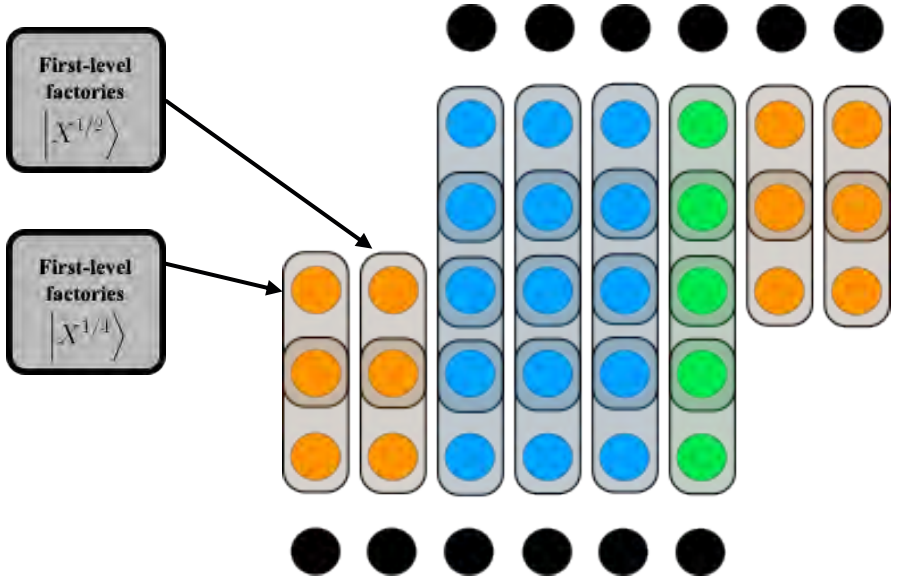


Level 1 magic states

Level 2 Check qubit

Level 2 magic states

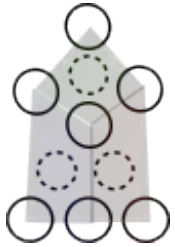
Routing qubits



Unfolded distillation :



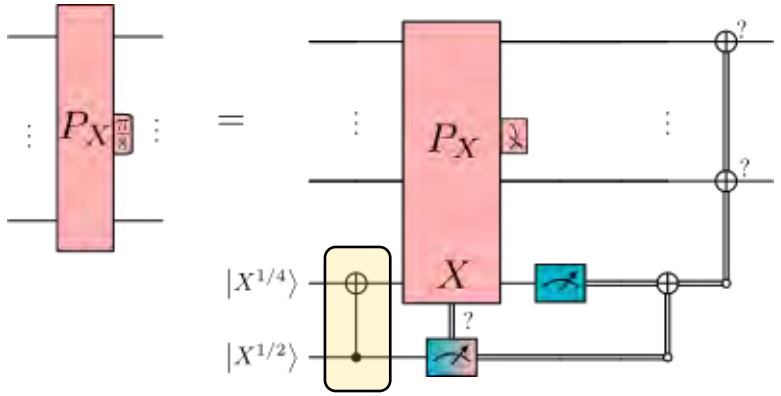
$$\left.\begin{array}{l} 5.5 \text{ cycles} \\ 35 \text{ cat qubits} \\ p_{\text{in},Z} = 1 \times 10^{-3} \\ p^{(1)}_{\text{out}} = 3 \times 10^{-7} \end{array}\right\|$$

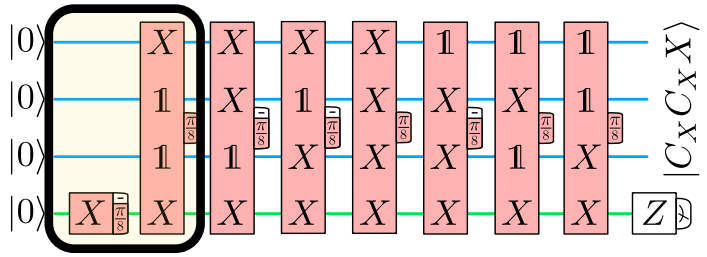


$$\left.\begin{array}{l} 5 \text{ cycles} \\ 10 \text{ cat qubits} \\ p_{\text{in},Z} = 1 \times 10^{-3} \\ p^{(1)}_{\text{out}} = 6 \times 10^{-8} \end{array}\right\|$$

Ruiz *et al*, arXiv:2507.12511, 2025

Multi-qubit X rotations







Level 1
magic states



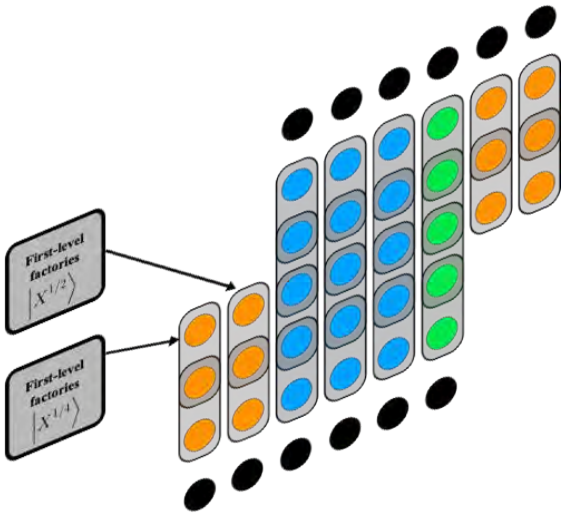
Level 2
Check qubit

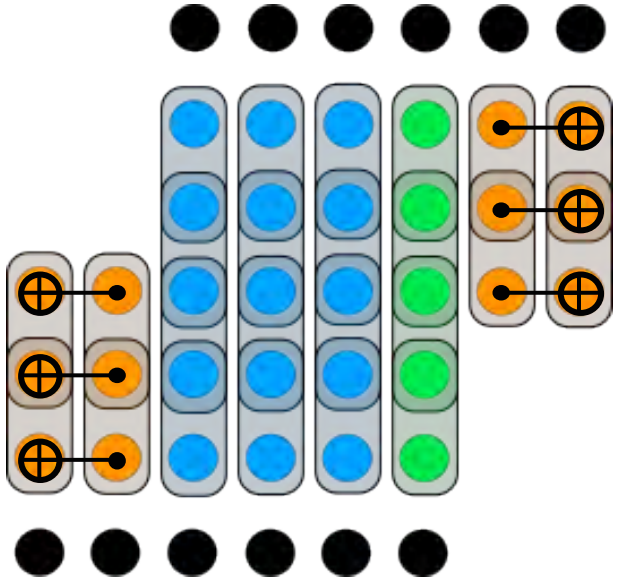


Level 2
magic states

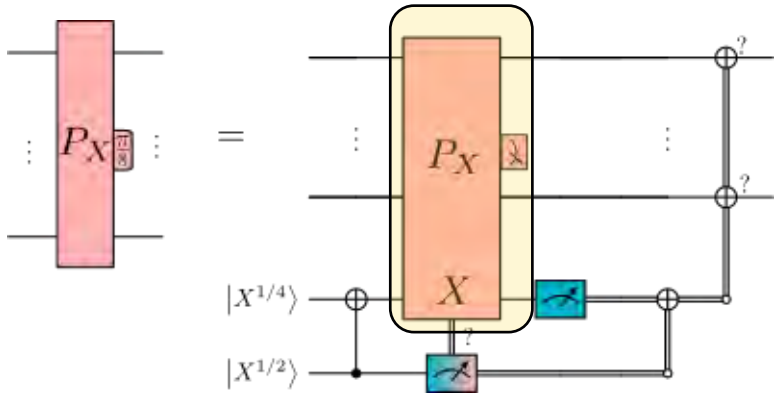


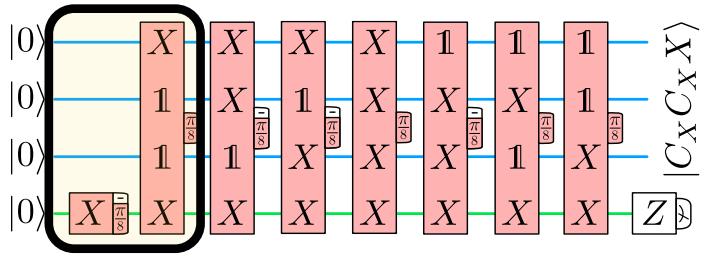
Routing qubits





Multi-qubit X rotations







Level 1
magic states



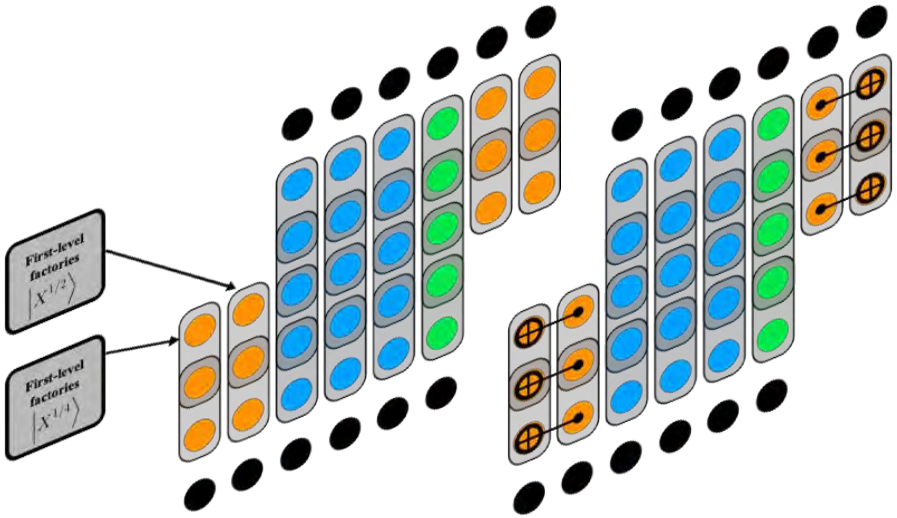
Level 2
Check qubit

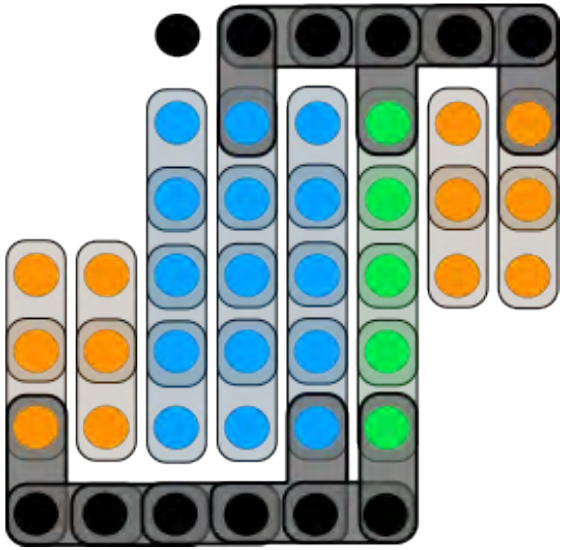


Level 2
magic states

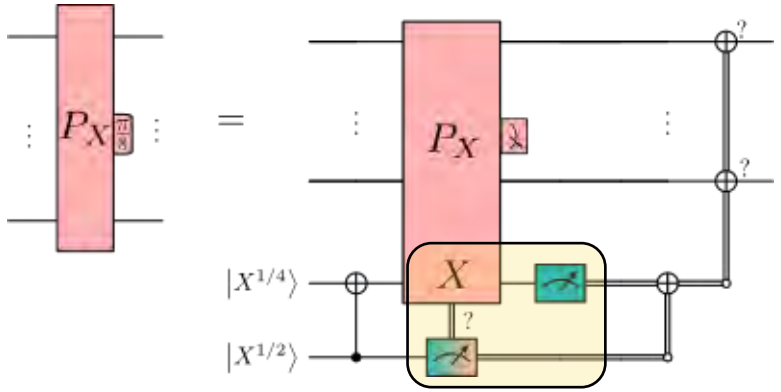


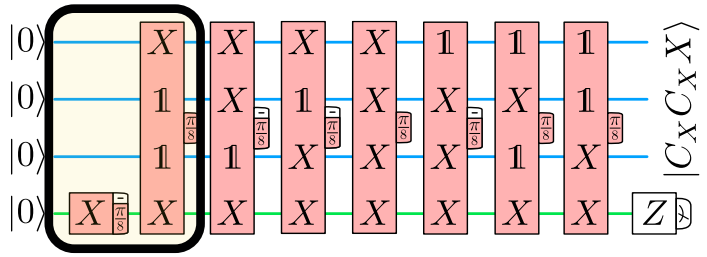
Routing qubits





Multi-qubit X rotations







Level 1
magic states



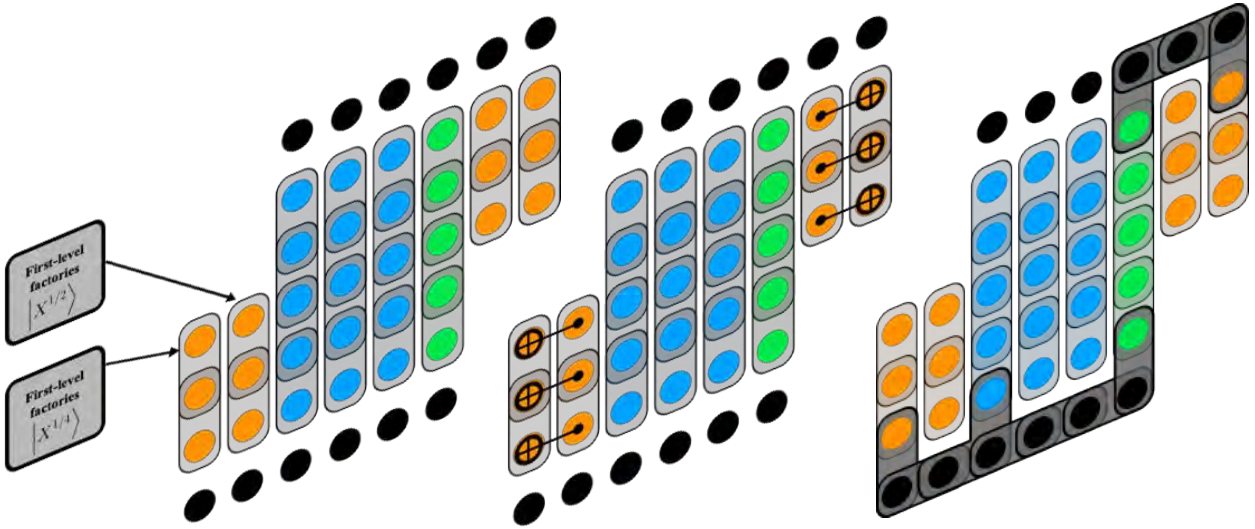
Level 2
Check qubit

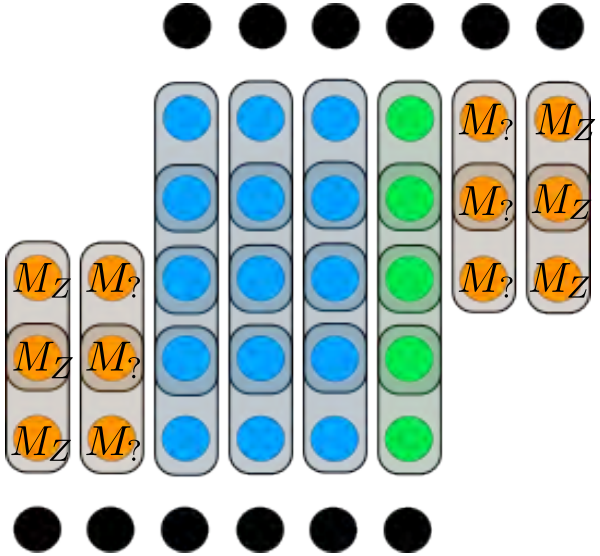


Level 2
magic states

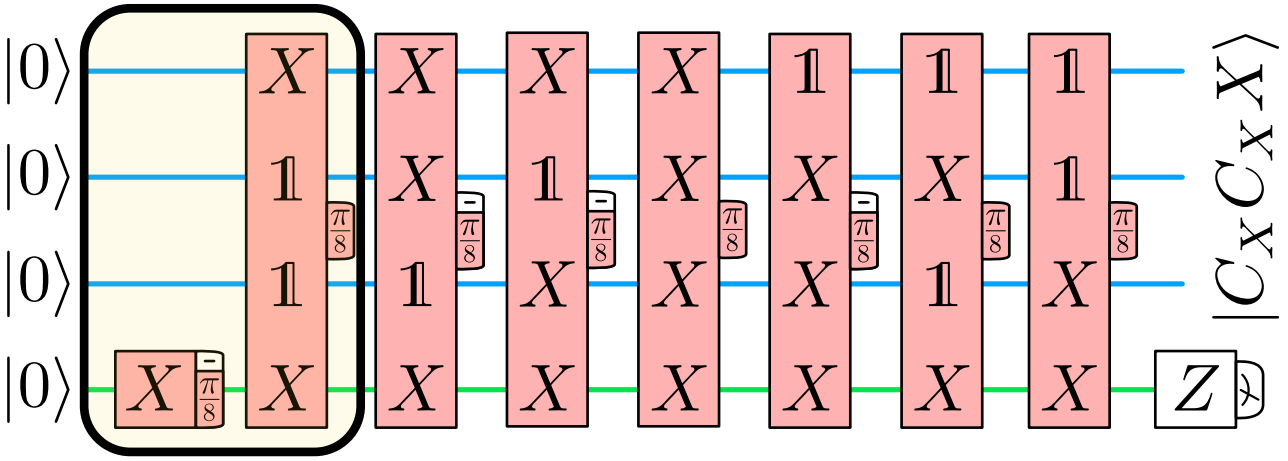


Routing qubits





Multi-qubit X rotations





Level 1
magic states



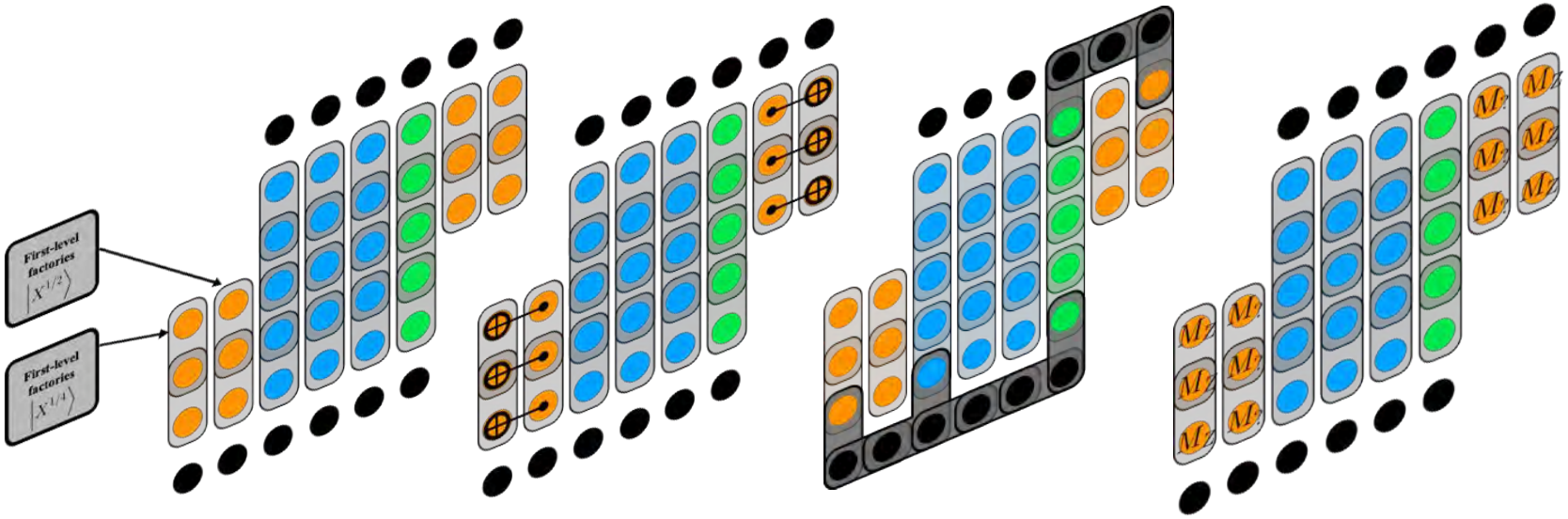
Level 2
Check qubit



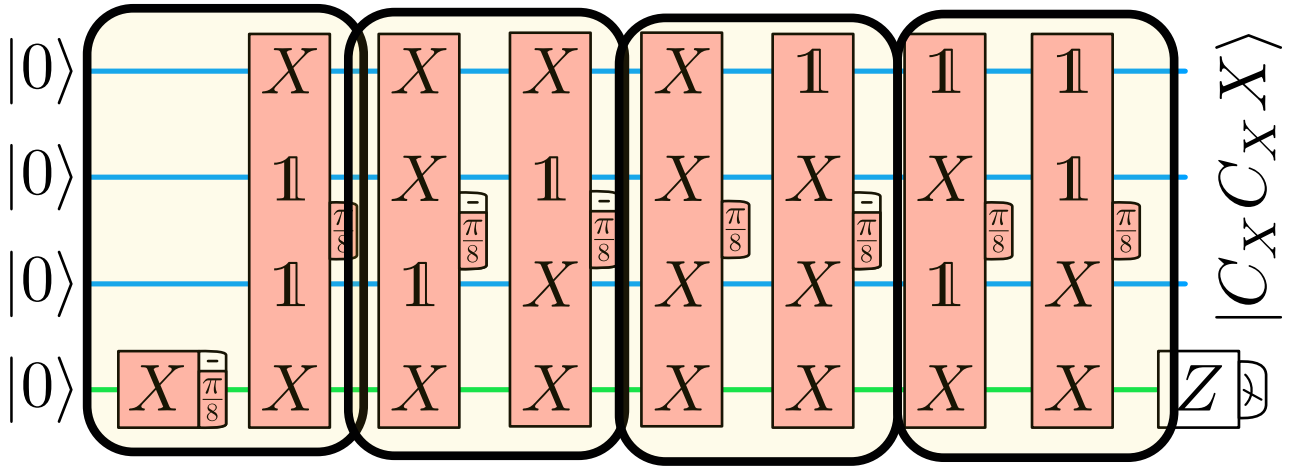
Level 2
magic states

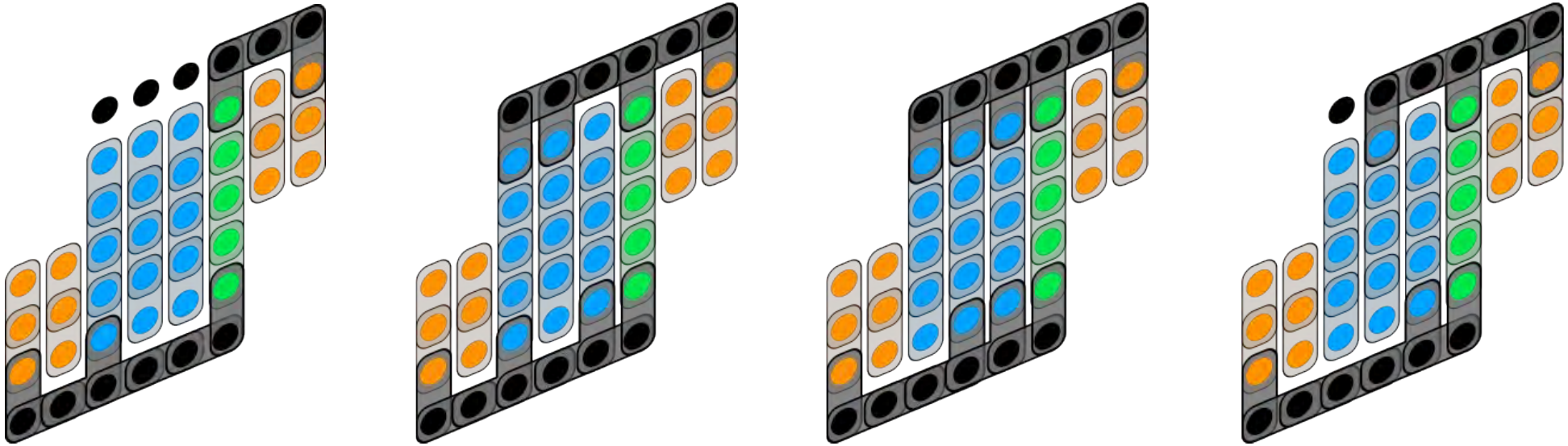


Routing qubits



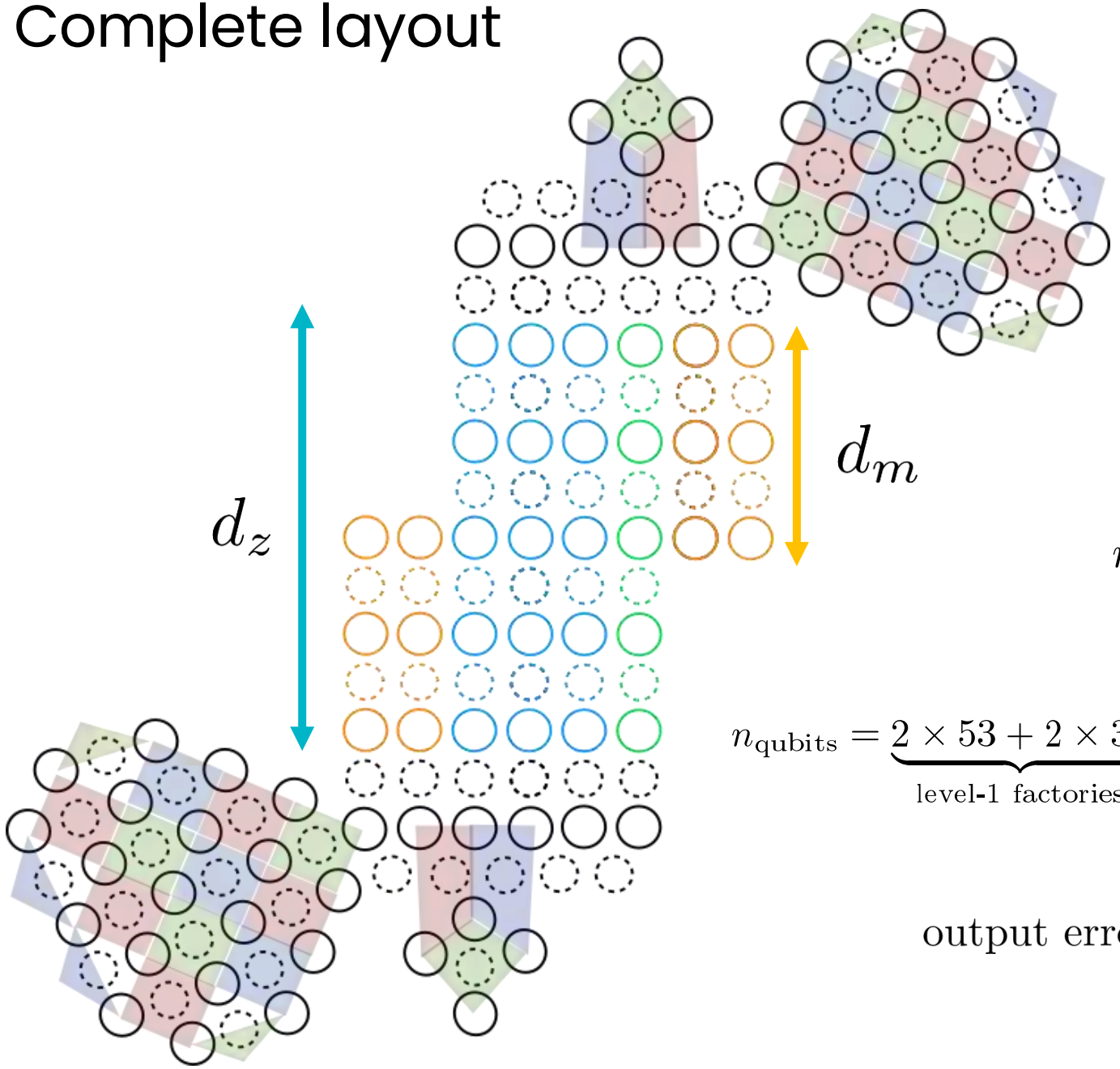
Multi-qubit X rotations





4 lattice surgery operations duration $\sim 4d_m$

Complete layout



$$n_{\text{cycle}} = 4 \times (5.5 + d_m + t_r/t_c) + 1$$

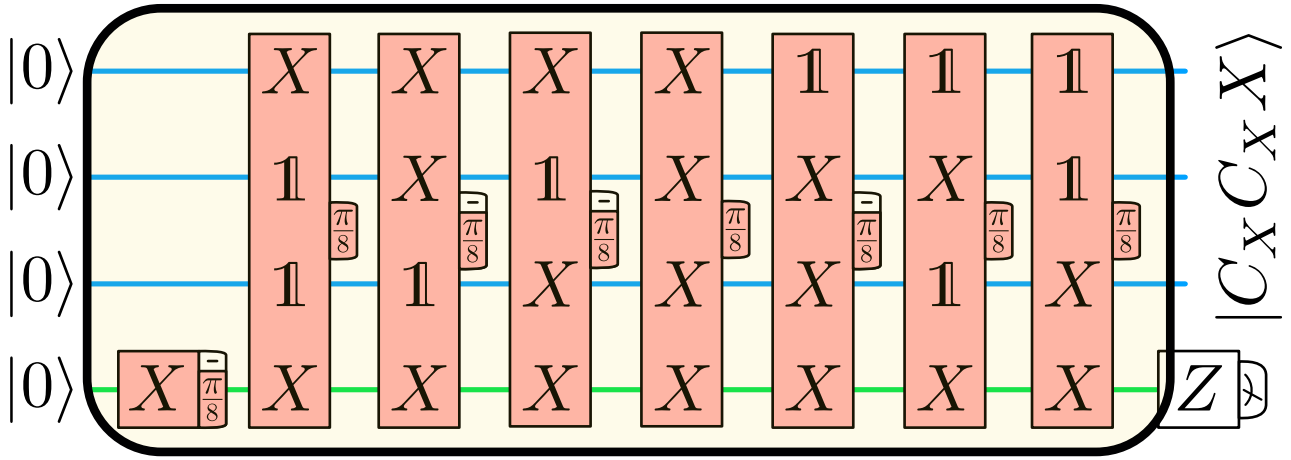
$$n_{\text{qubits}} = \underbrace{2 \times 53 + 2 \times 33}_{\text{level-1 factories}} + 2 \times \left(\underbrace{4 \times d_Z}_{\text{log. qubits}} + \underbrace{2 \times 6}_{\text{routing areas}} \right) \approx 420 \quad (d_z \approx 27)$$

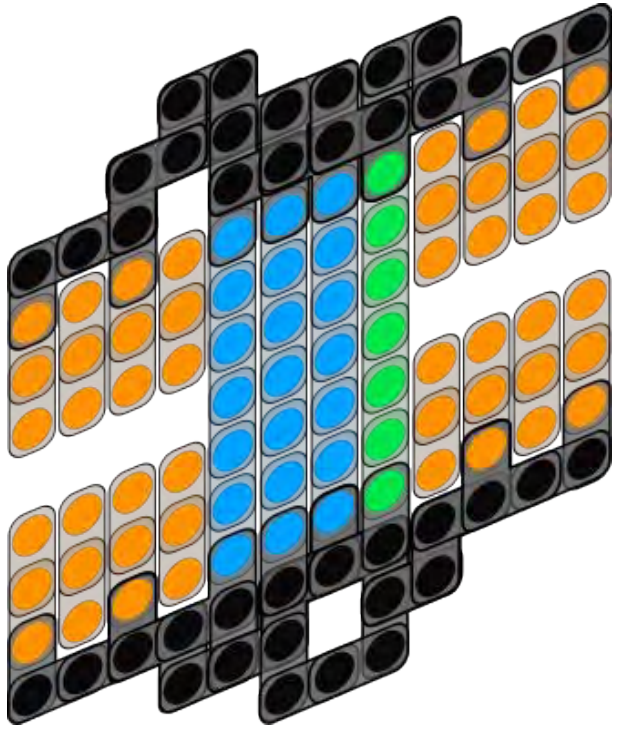
output error saturates at $p_{out} = 28p^2 \approx 2.3 \times 10^{-12}$



Second distillation level with simultaneous lattice surgery

Simultaneous lattice surgery





1 simultaneous surgery operation,
duration: $\sim d'_m$

SMT formulation

Notations:

$\mathcal{S}_N$ all possible weight 2 and 3 stabilizers

Objective: measure $X_L(\alpha)$ and $X_L(\beta)$, $\alpha, \beta \in \mathbb{F}_2^N$
the binary mask vectors defining the targets

$\mathcal{S}_1 \subset \mathcal{S}_N$ set of stabilizers used to measure $X_L(\alpha)$

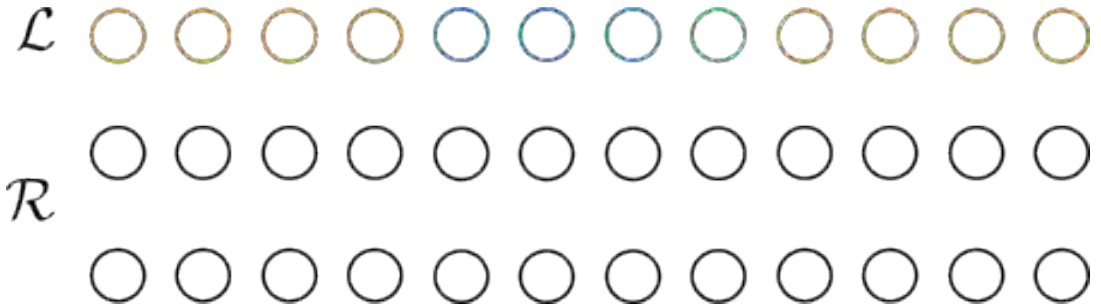
$\mathcal{S}_2 \subset \mathcal{S}_N$ set of stabilizers used to measure $X_L(\beta)$

$U = \bigcup_i S_i$ set of added stabilizers

s_1 and s_2 binary mask vectors defining the selected stabilizers in $\mathcal{S}_1$ and $\mathcal{S}_2$

$u = s_1 \vee s_2$

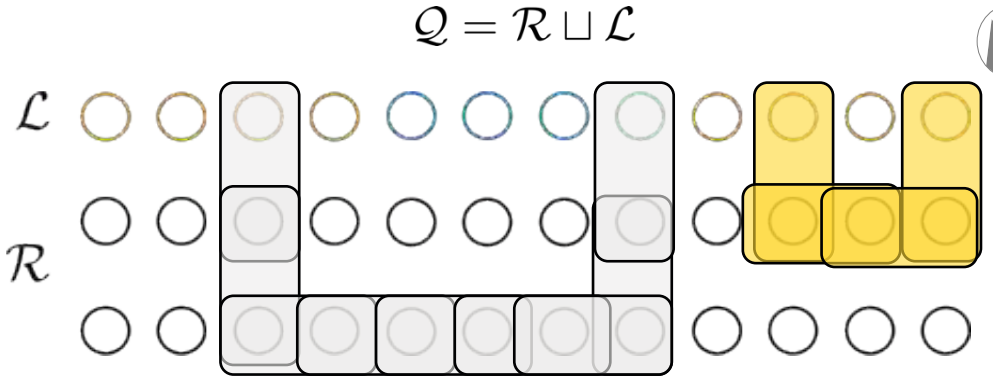
$\mathcal{Q} = \mathcal{R} \sqcup \mathcal{L}$



SMT formulation

- Measure the targets

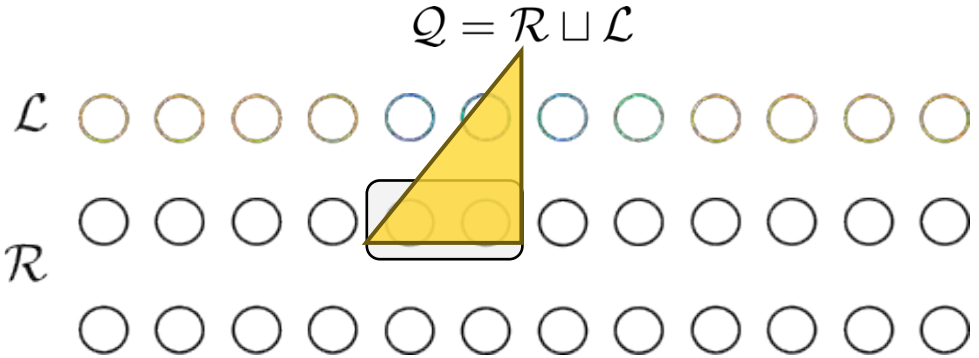
$$\forall i, \prod_{X \in S_i} X = X_L(\alpha_i)$$



$$\forall i, \forall q \in Q,\ [q \notin \mathrm{Supp}(X_L(\alpha_i))] \implies \neg \bigoplus_k [s_i[k] \wedge (q \in \mathcal{S}_N[k])]$$

$$[q \in \mathrm{Supp}(X_L(\alpha_i))] \implies \bigoplus_k [s_i[k] \wedge (q \in \mathcal{S}_N[k])]$$

- Don't measure any other logical operator



$$\forall I \subset U, \left[\mathrm{Supp}\left(\prod_{s \in I} s\right) \subset \mathcal{L}\right] \implies$$
$$\left(\prod_{s \in I} s = X_L(\alpha)\right) \vee \left(\prod_{s \in I} s = X_L(\beta)\right) \vee \left(\prod_{s \in I} s = X_L(\alpha) X_L(\beta)\right)$$

- Other simple conditions to reduce number of qubits and make circuit short.

SMT formulation, 4 simultaneous operators

- Measure the targets

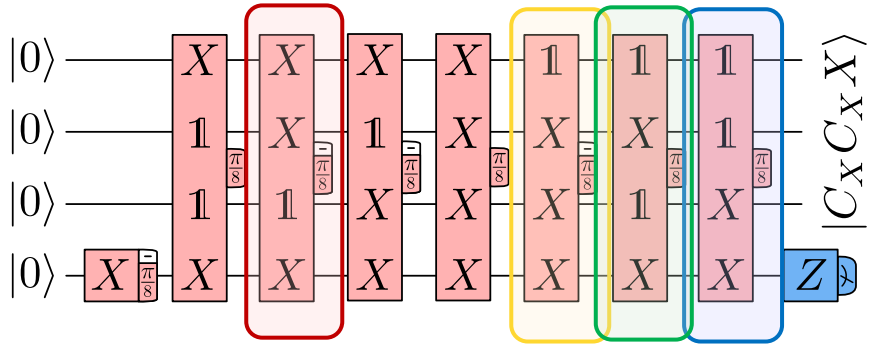
$$\forall i, \prod_{X \in S_i} X = X_L(\alpha_i)$$

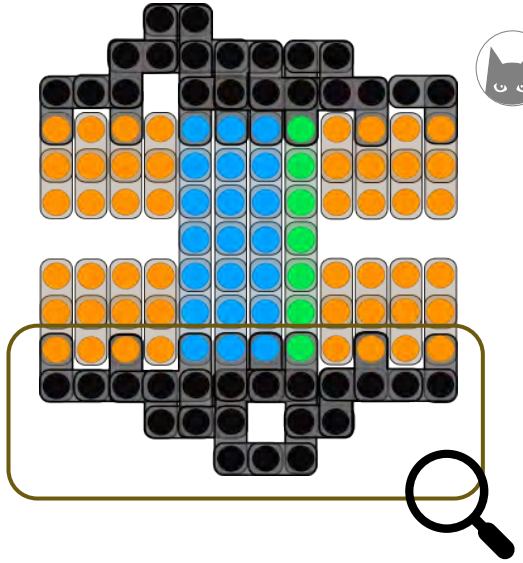
- Don't measure any other logical operator

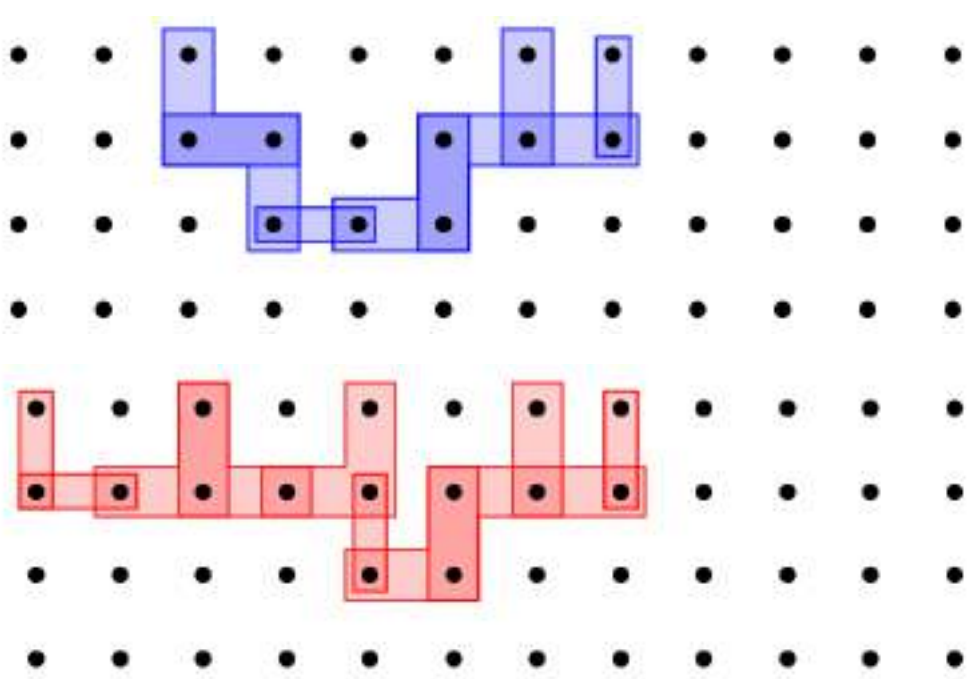
$$\forall I \subset U, \left[\text{Supp} \left(\prod_{s \in I} s \right) \subset \mathcal{L} \right] \implies$$
$$\bigvee_i \left(\prod_{s \in I} s = X_L(\alpha_i) \right) \vee \bigvee_{i,j,i<j} \left(\prod_{s \in I} s = X_L(\alpha_i) X_L(\alpha_j) \right)$$
$$\vee \bigvee_{i,j,k,i<j<k} \left(\prod_{s \in I} s = X_L(\alpha_i) X_L(\alpha_j) X_L(\alpha_k) \right)$$
$$\vee \left(\prod_{s \in I} s = X_L(\alpha_1) X_L(\alpha_2) X_L(\alpha_3) X_L(\alpha_4) \right)$$

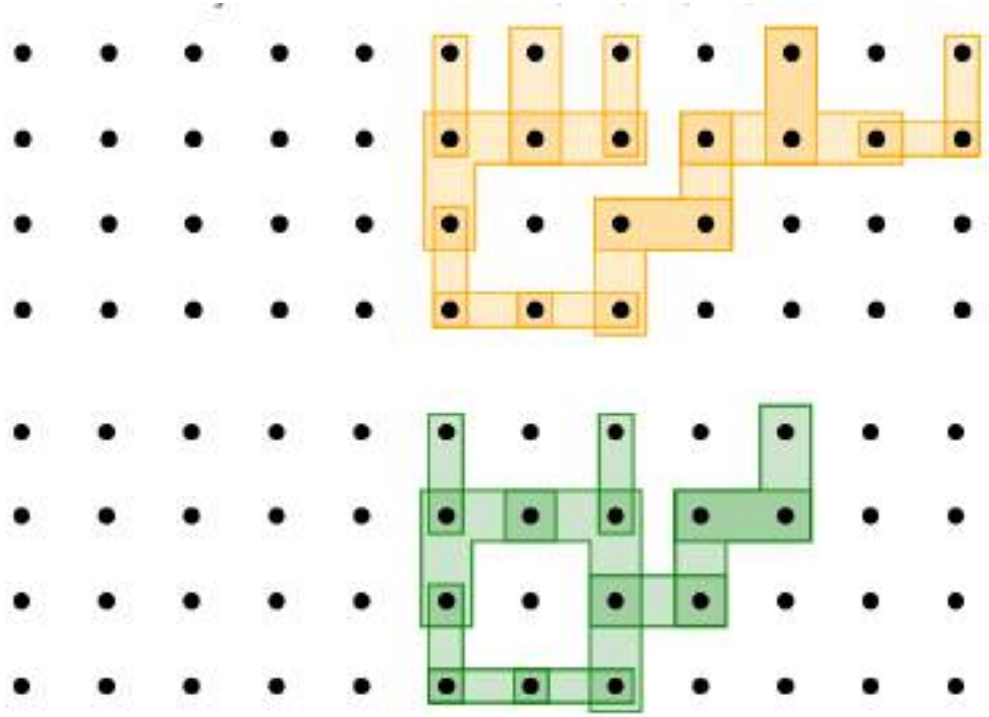
Simultaneous lattice surgery

Surgery code found with **SAT solver** z3



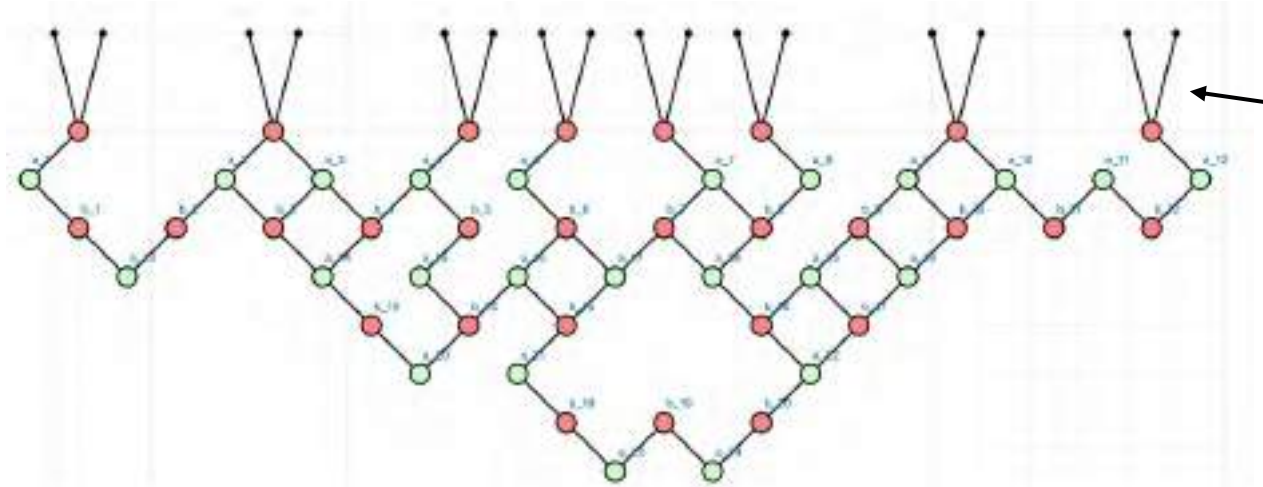


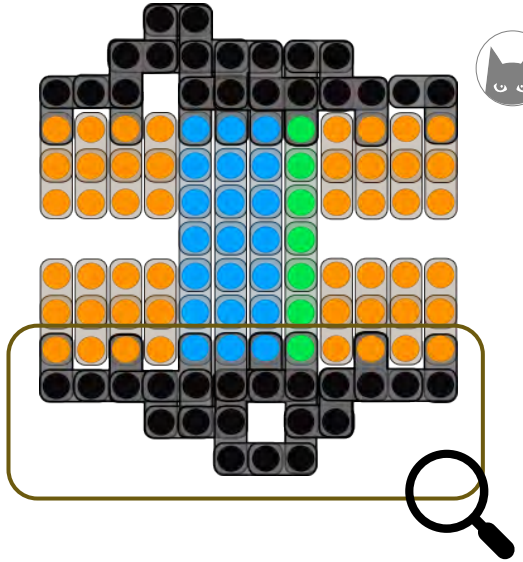




Simultaneous lattice surgery

ZX-calculus for verification



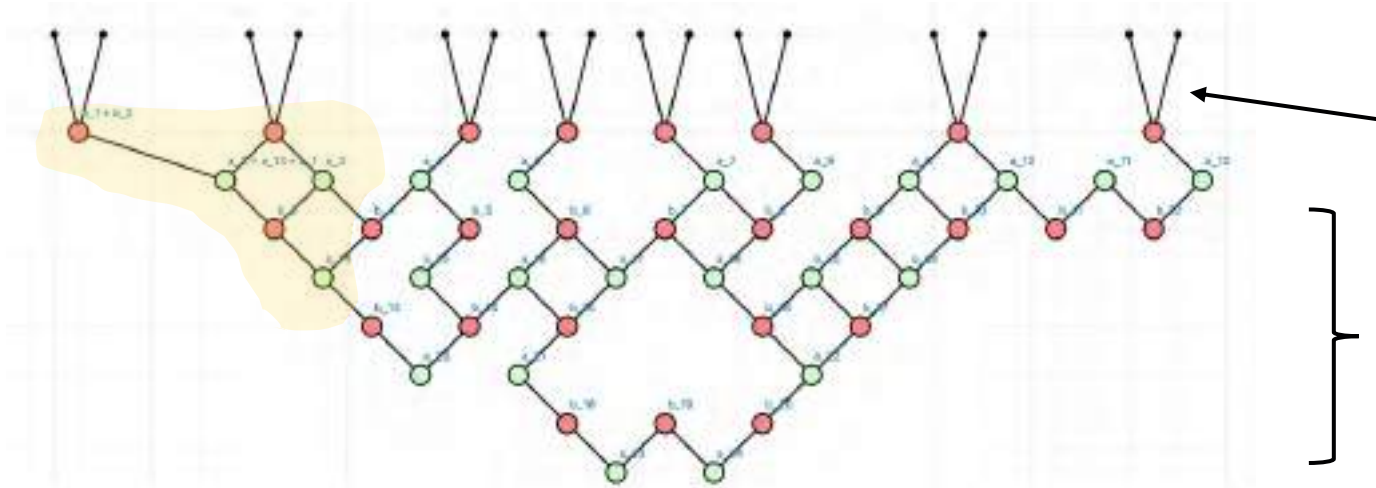


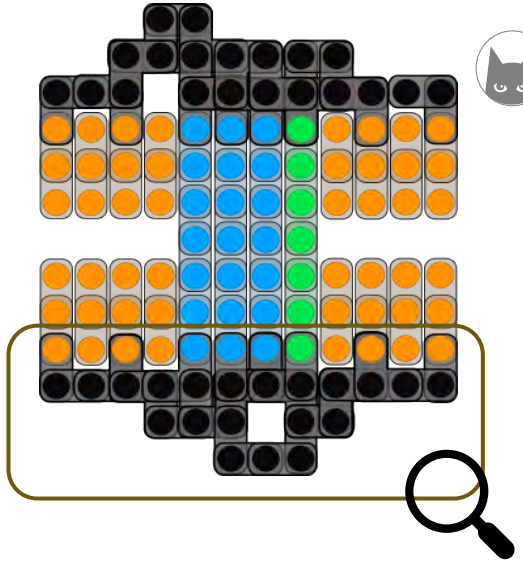
Repetition code qubits

Routing qubits

Simultaneous lattice surgery

ZX-calculus for verification

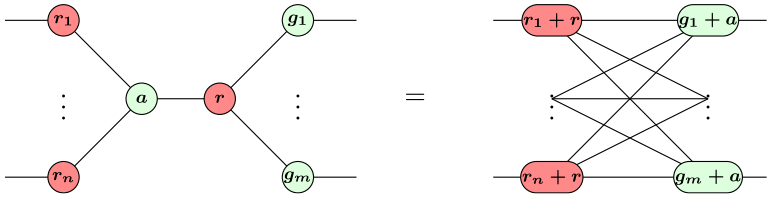




Repetition code qubits

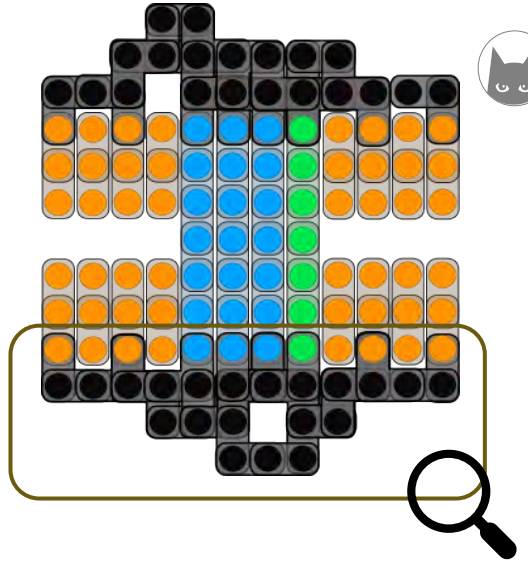
Routing qubits

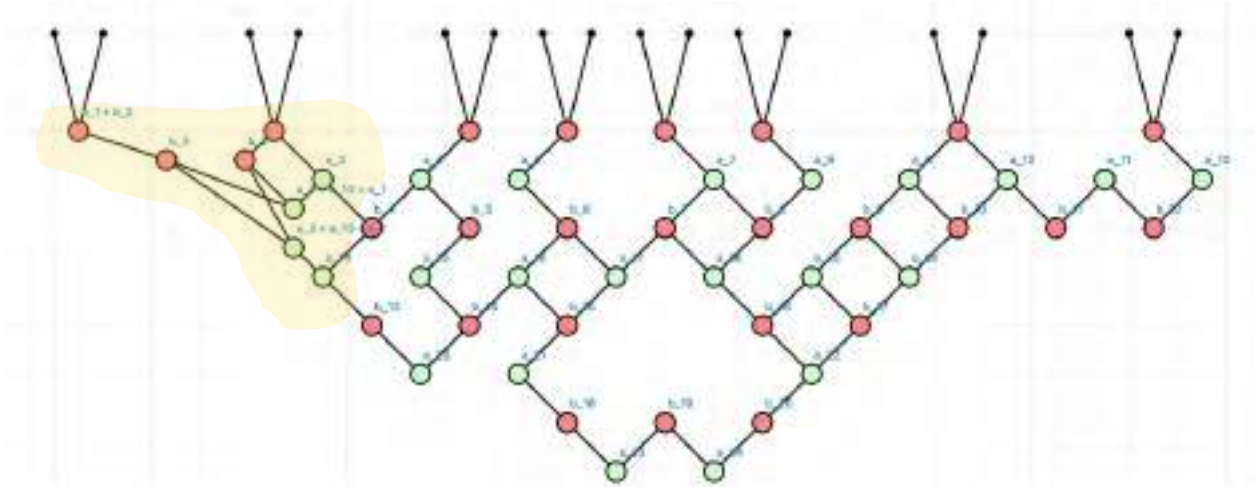
Bialgebra rule



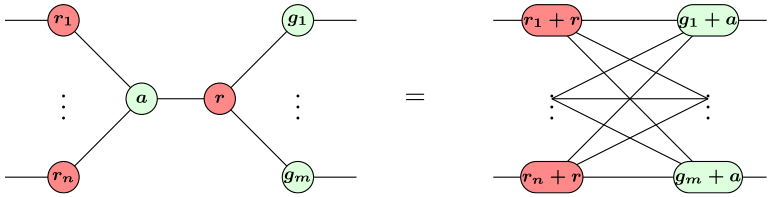
Simultaneous lattice surgery

ZX-calculus for verification



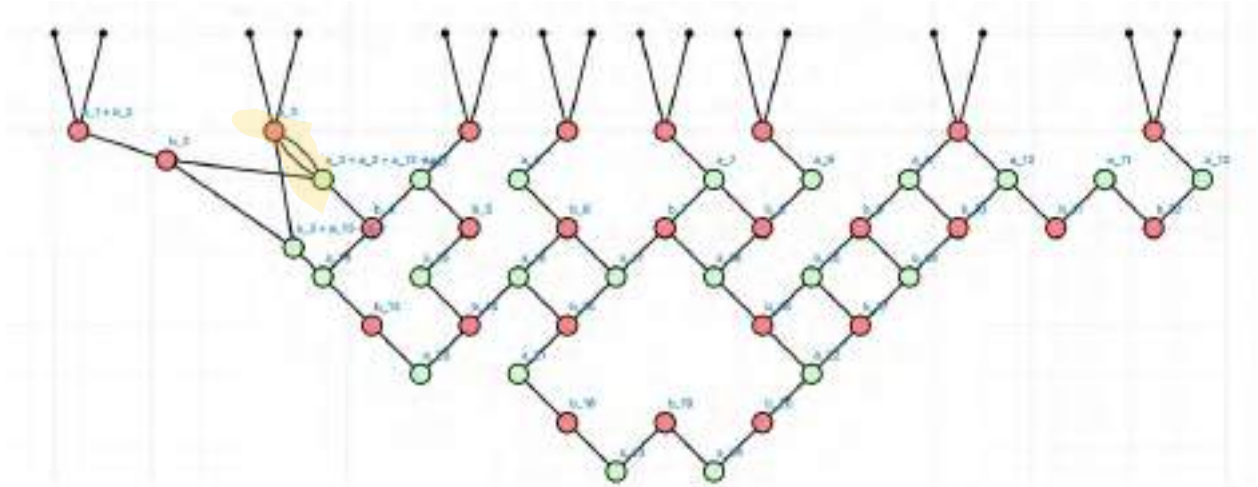


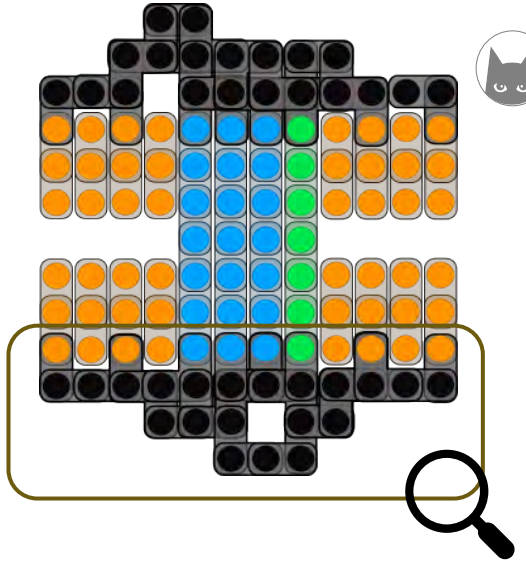
Bialgebra rule



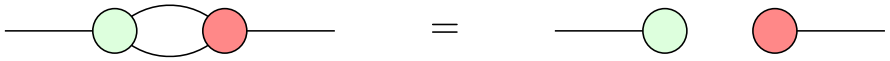
Simultaneous lattice surgery

ZX-calculus for verification



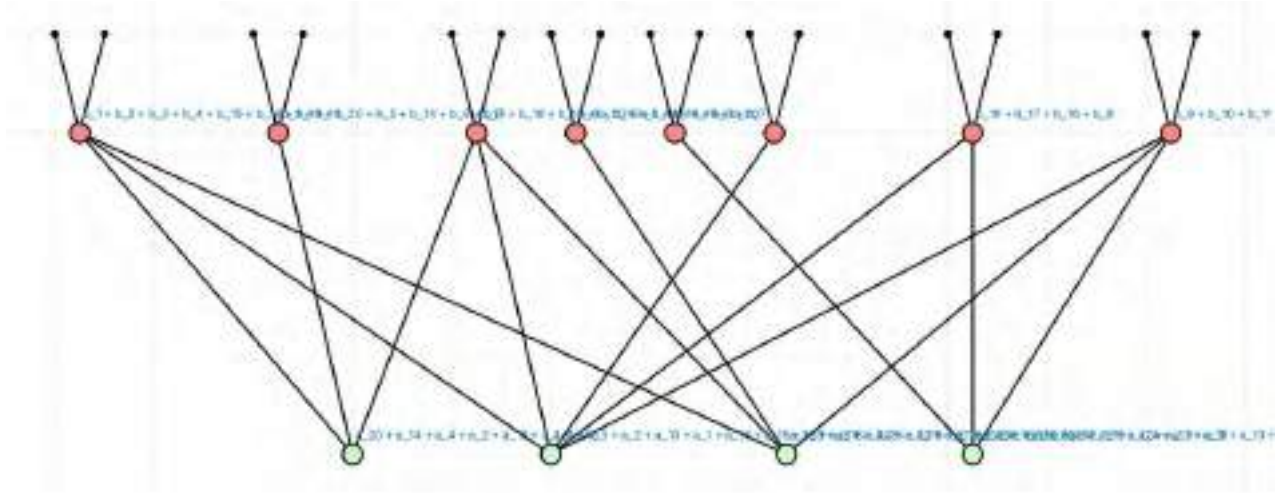


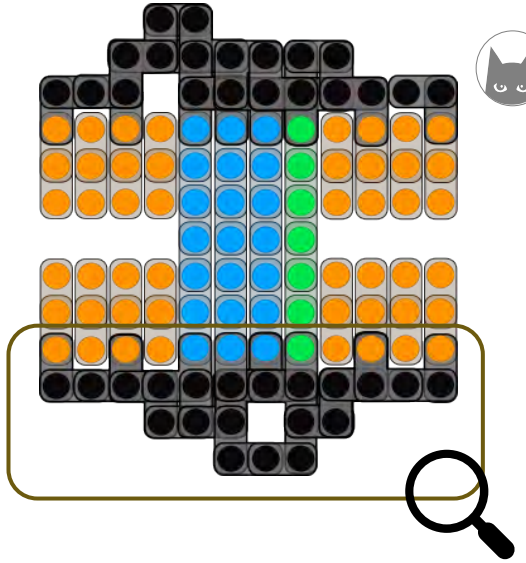
Hopf rule



Simultaneous lattice surgery

ZX-calculus for verification

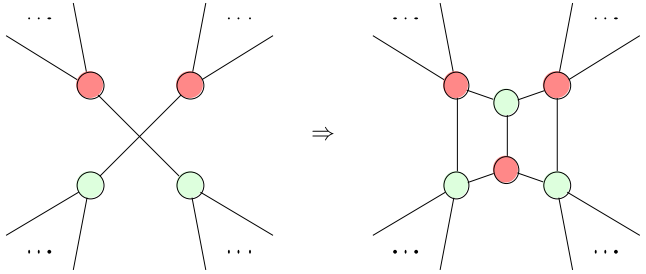


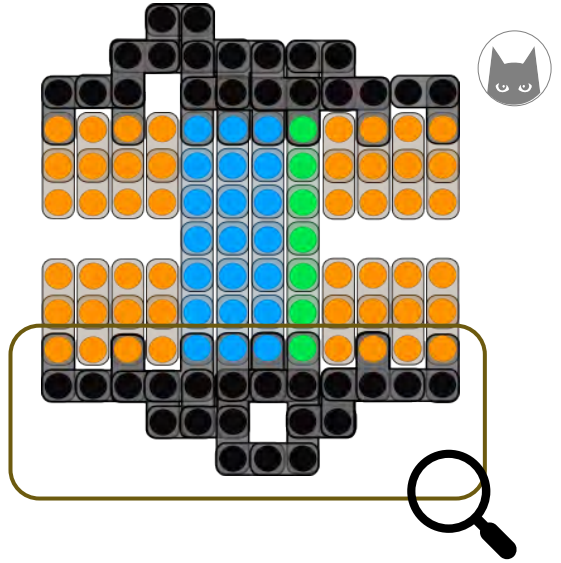


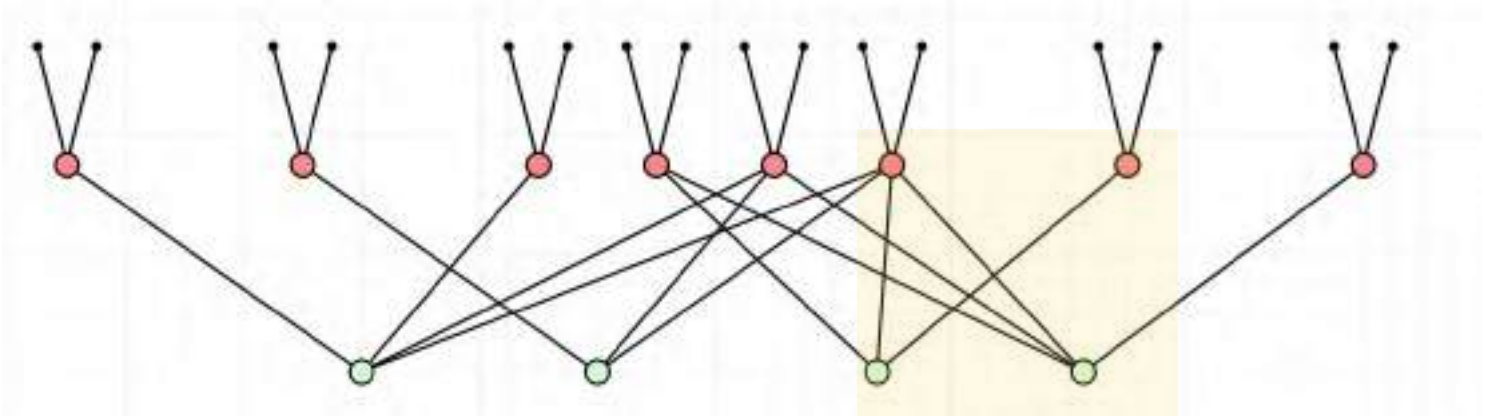
Simultaneous lattice surgery

Constructive approach

Bialgebra (+Hopf) rule again



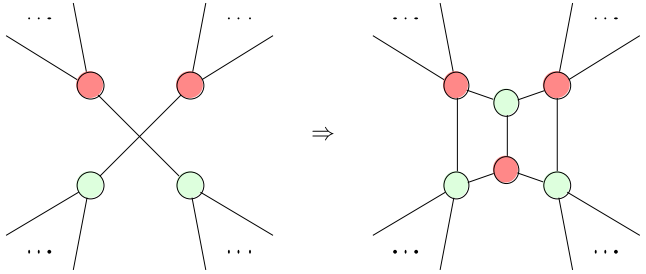


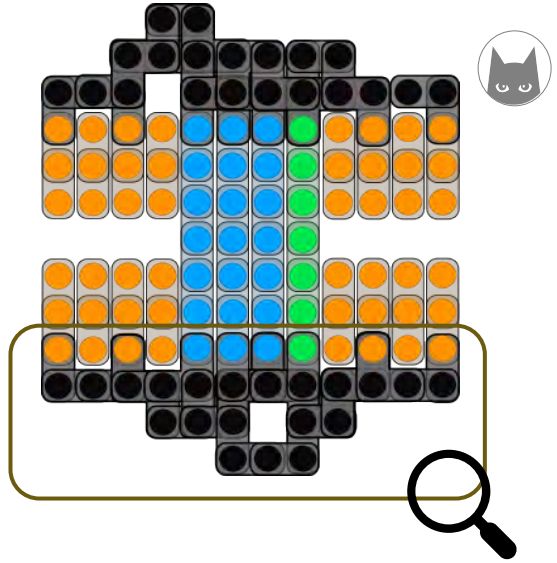


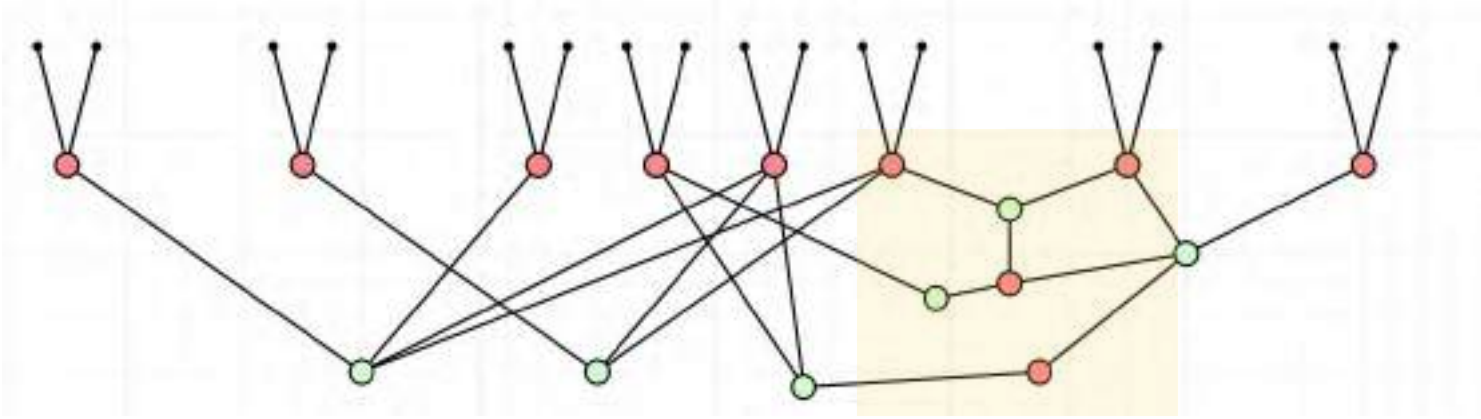
Simultaneous lattice surgery

Constructive approach

Bialgebra (+Hopf) rule again



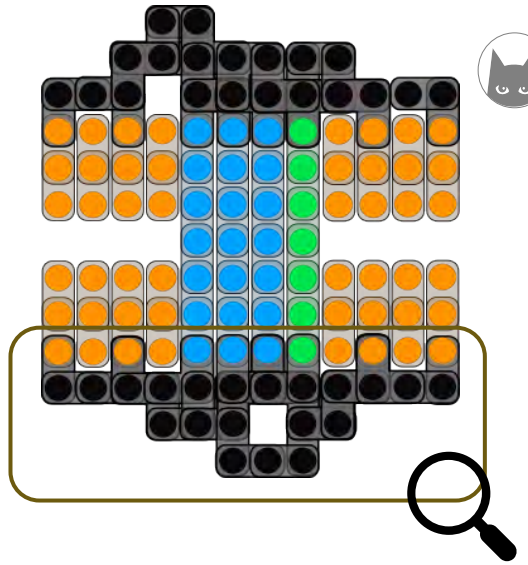


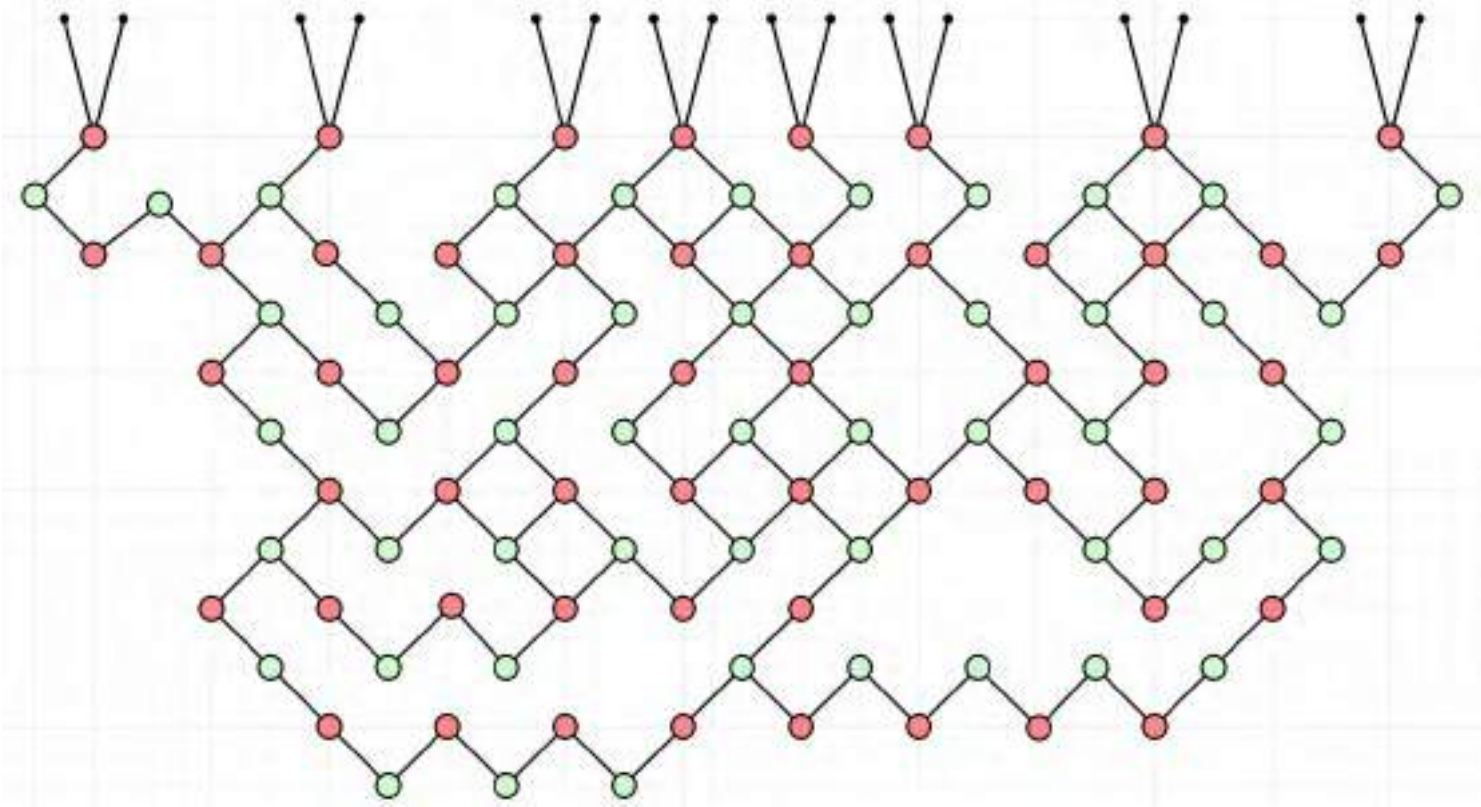


Simultaneous lattice surgery

Constructive approach

A lot **more routing** qubits (compared to SAT result)

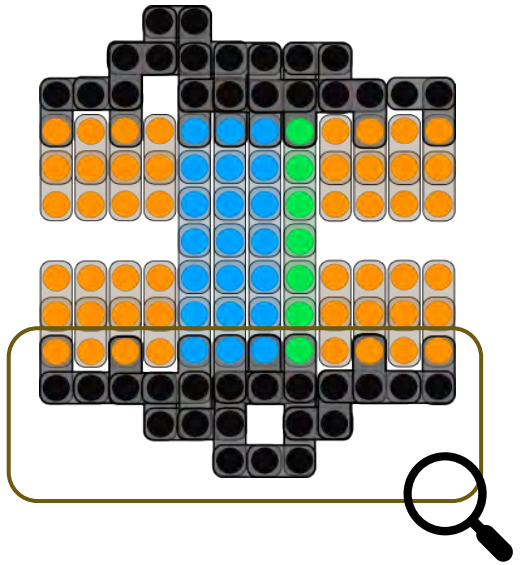




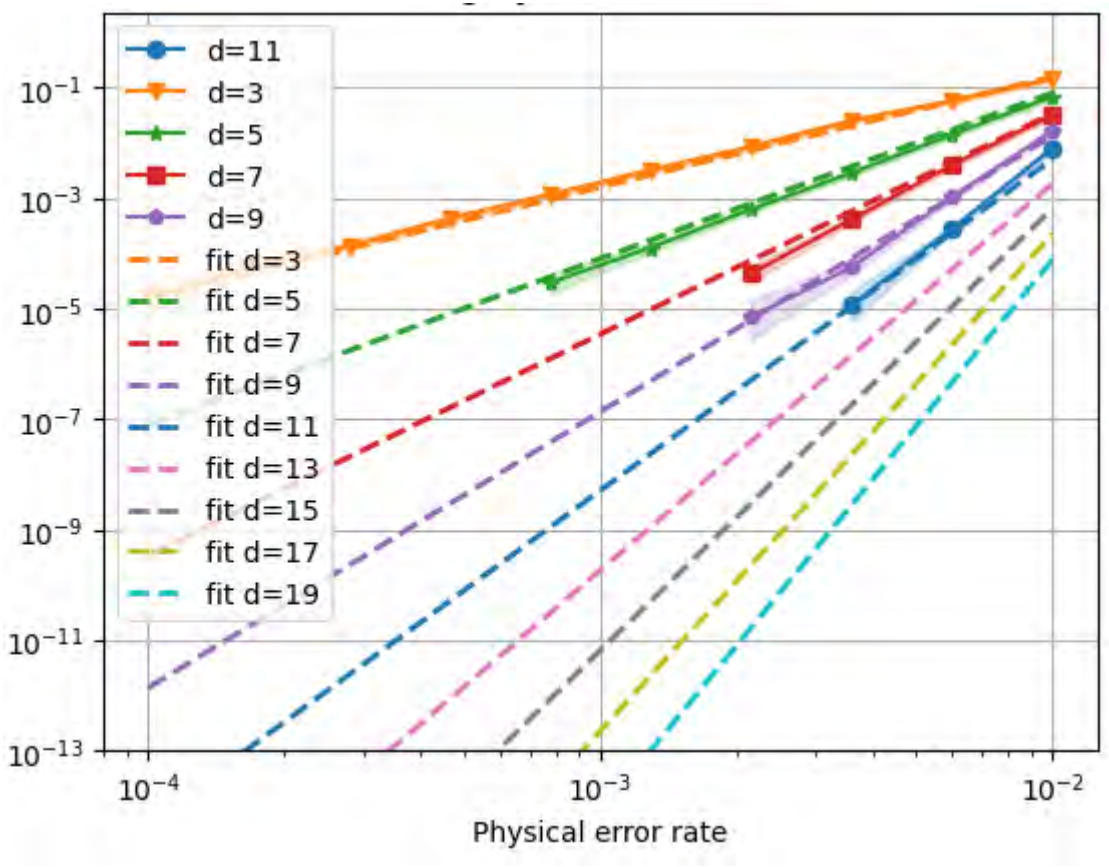
Simulation of simultaneous lattice surgery

Decoding is harder => not matchable anymore

Yet good performances with Minimum Weight Parity Factor (mwpf)



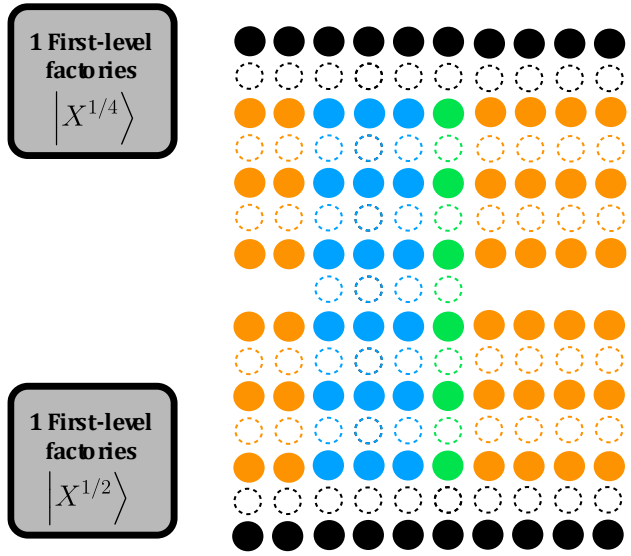
$$\mathbb{P}(M_{X\ldots X} \neq \bigoplus M_X)$$



Choice of $d_m' = d_2$=17 to avoid correlated logical timelike errors

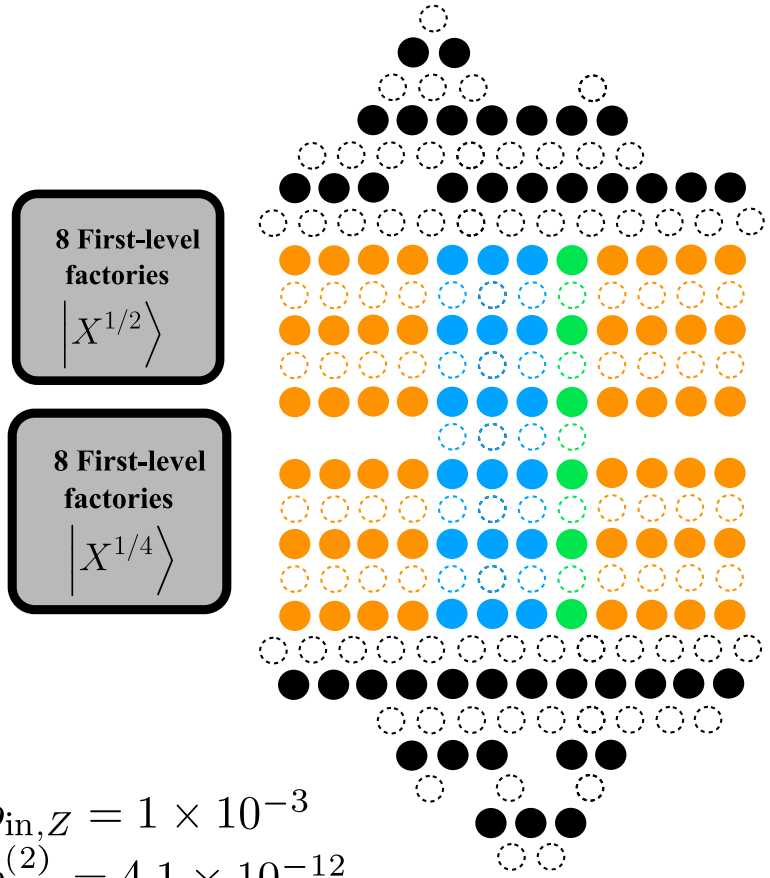
$$\mathbb{P}_L \approx 0.56 d_2 \left(\frac{p}{3.2 \times 10^{-2}} \right)^{\frac{d_m'+1}{2}}$$

Regular surgery



$p_{\text{in},Z} = 1 \times 10^{-3}$
$p_{\text{out}}^{(2)} = 4.7 \times 10^{-12}$
4×11 cycles
437 cat qubits
volume $= 19228$ cat qubits $\times$ cycles

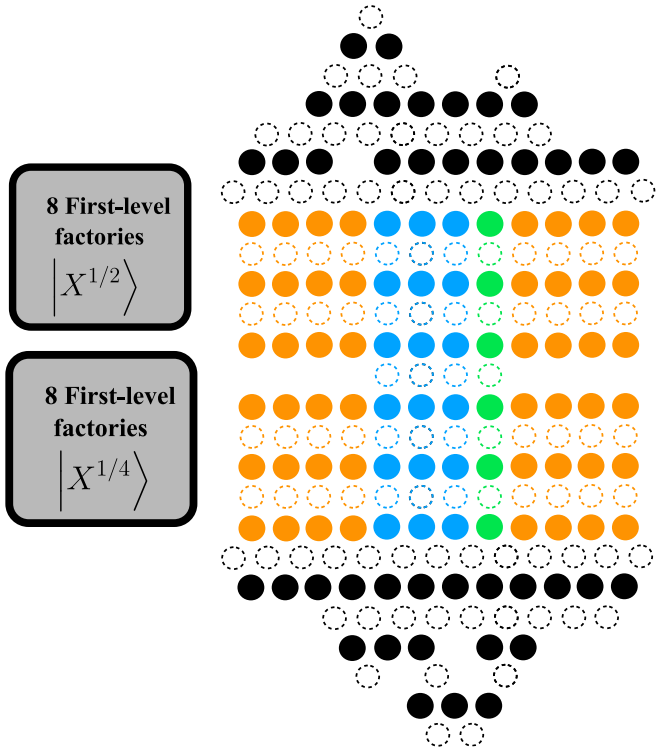
Simultaneous surgery



$p_{\text{in},Z} = 1 \times 10^{-3}$
$p_{\text{out}}^{(2)} = 4.1 \times 10^{-12}$
1×17 cycles
864 cat qubits
volume $= 14688$ cat qubits $\times$ cycles

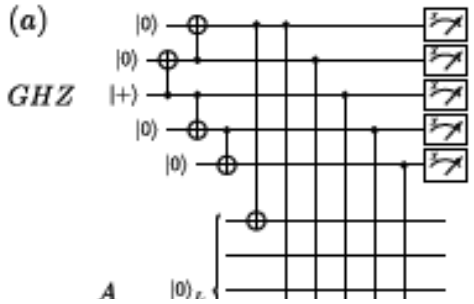
Comparison to previous works

Simultaneous surgery

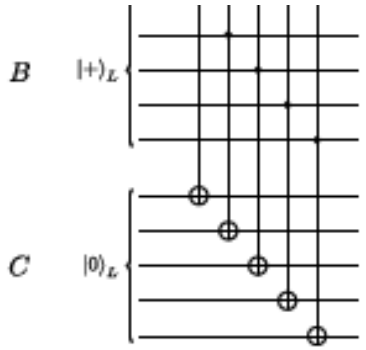


$p_{\text{in},Z} = 1 \times 10^{-3}$
$p_{\text{out}}^{(2)} = 4.1 \times 10^{-12}$
1×17 cycles
864 cat qubits
volume $= 14\,688$ cat qubits $\times$ cycles

Toffoli cultivation



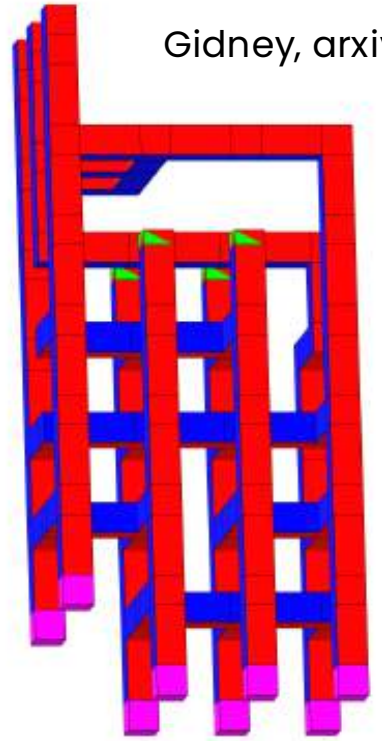
Physical Toffoli



$p_{\text{out}} = 7.90 \times 10^{-12}$
2394 cycles
213 cat qubits
volume $= 509\,922$ cat qubits $\times$ cycles

Surface code

Gidney, arxiv:2505.15917



$p_{\text{out}} = 2.8 \times 10^{-13}$
114.7 cycles
16224 qubits
volume $= 1\,860\,893$ qubits $\times$ cycles

Summary & Outlook



Summary & Outlook



- Reach the $<10^{-11}$ logical error rate regime
- Single lattice surgery operation per distillation protocol
- New compilation tool for strongly biased noise qubit
- WIP: 15 to 1 second stage and optimized layout
- WIP: constructive way to build surgery codes

Summary & Outlook

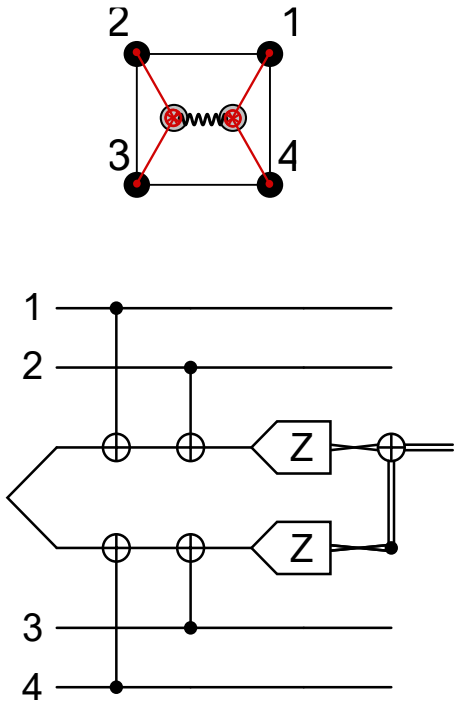


- Live decoding?
- Codes designed for logical operators' pattern/operation volume, not encoding rate
- Directly in LDPC?
- Simultaneous distillation of a lot of magic states?
- …

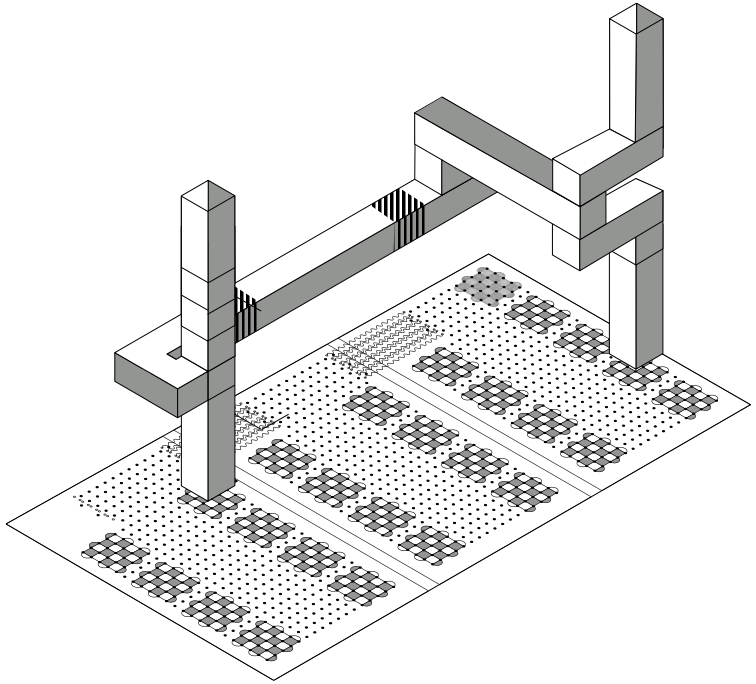
Distributed architecture

[H. Jacinto *et al.*, arXiv:2504.08891 (2025)]

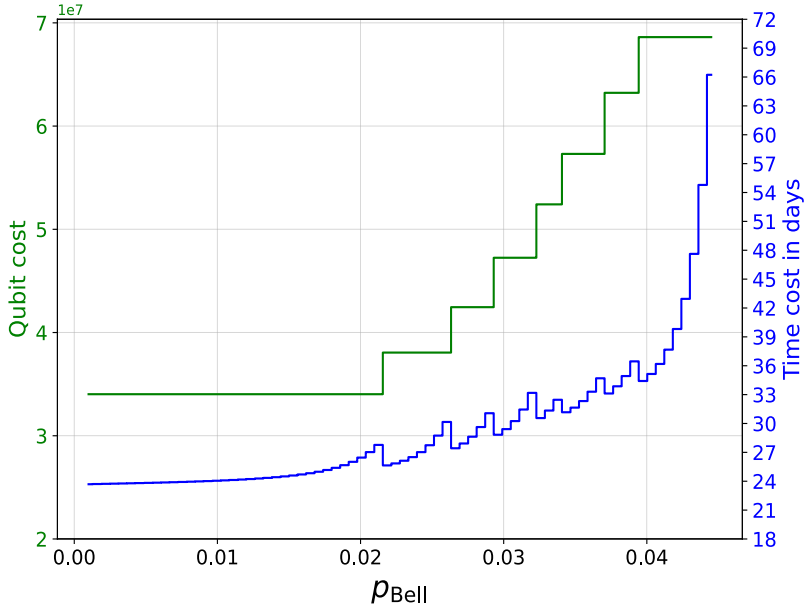
Measurement teleportation:



Architecture proposal:



Resource estimation:

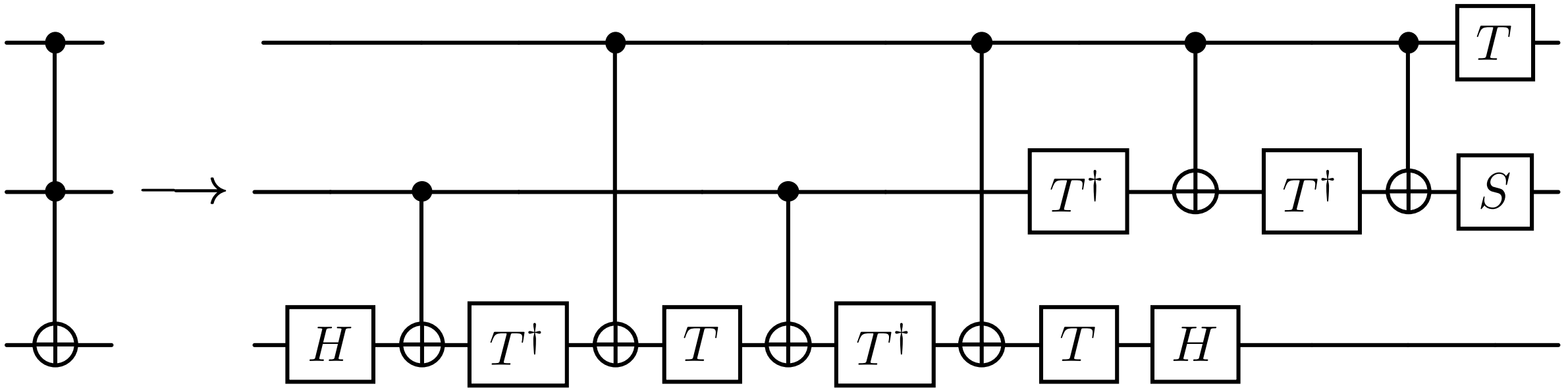


Provably optimal exact gate synthesis from a discrete gate set

$$\mathrm{CNOT} = \frac{1}{2}\begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 2 & 0 \end{pmatrix} \qquad H = \frac{1}{2}\begin{pmatrix} \sqrt{2} & \sqrt{2} \\ \sqrt{2} & -\sqrt{2} \end{pmatrix} \qquad T = \frac{1}{2}\begin{pmatrix} 2 & 0 \\ 0 & \sqrt{2}+i\sqrt{2} \end{pmatrix}$$

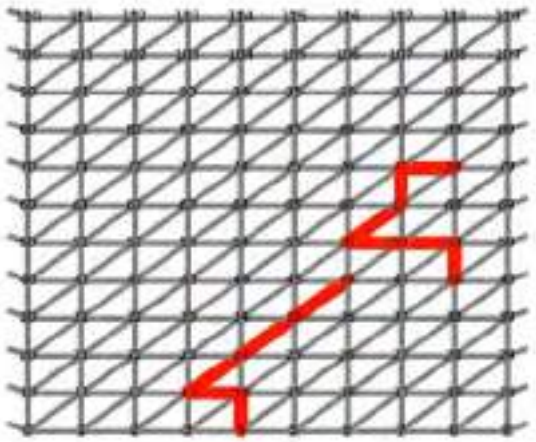
$$e^{i\frac{\pi}{4}} = \frac{\sqrt{2}+i\sqrt{2}}{2}$$

$$\left[\bigvee_j x_{i,j}\right] \wedge \left[\bigwedge_{j_1,j_2} \neg x_{i,j_1} \vee \neg x_{i,j_2}\right]$$

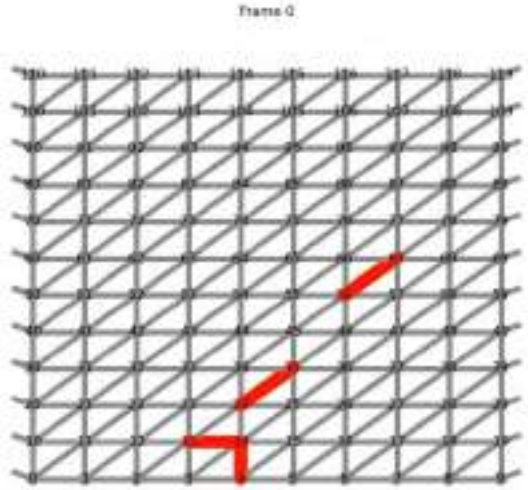


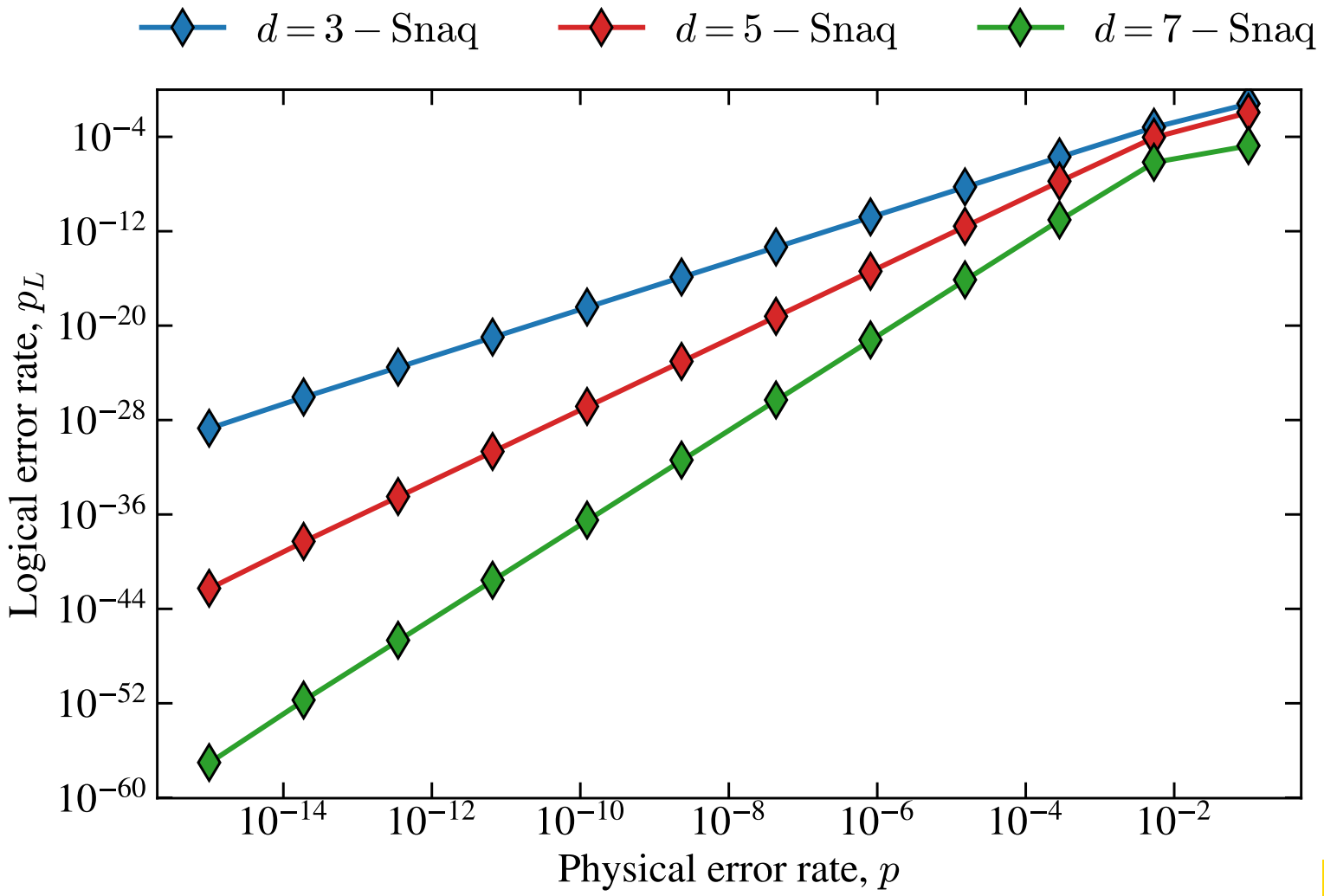
Importance sampling for (ridiculously) low logical error estimation

Full snaq



Skin





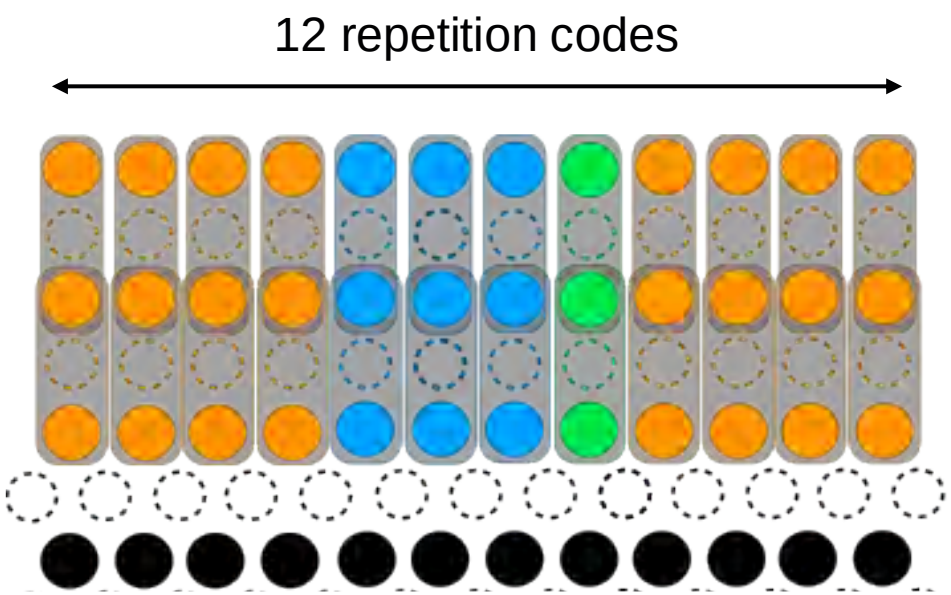
Summary & Outlook



THANK YOU.

Additional Slides/Figures

How many line of routing qubits on the layout ?

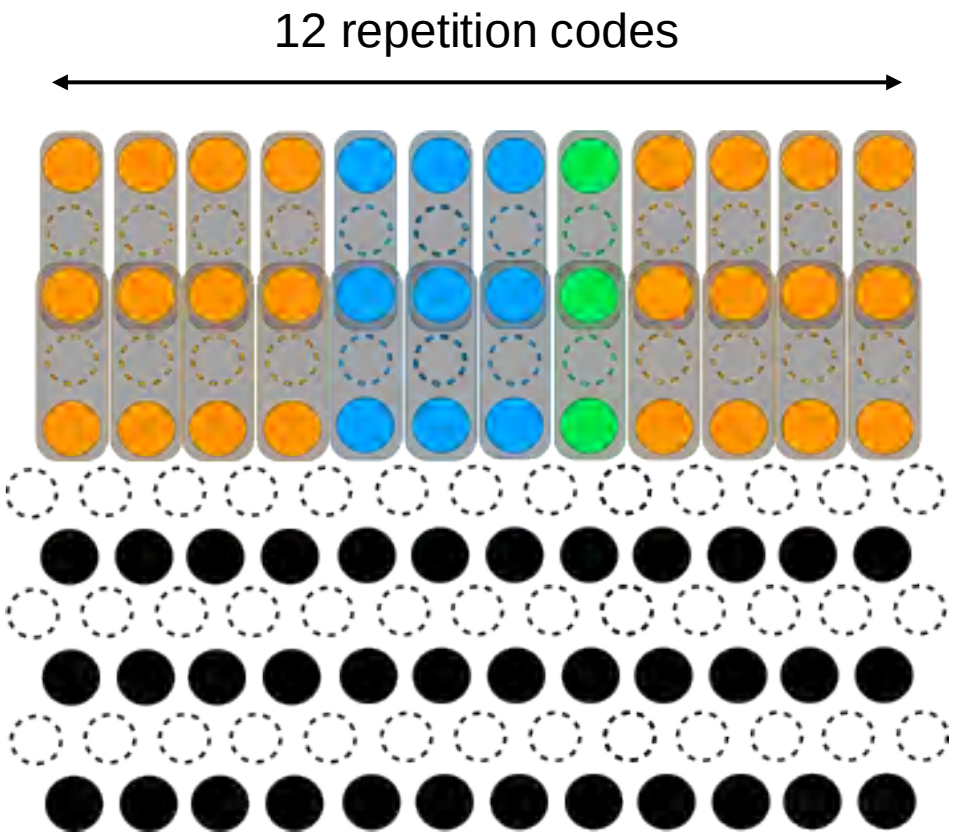


Bit-sequences of the 4 X-Pauli strings to measure

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Additional Slides/Figures

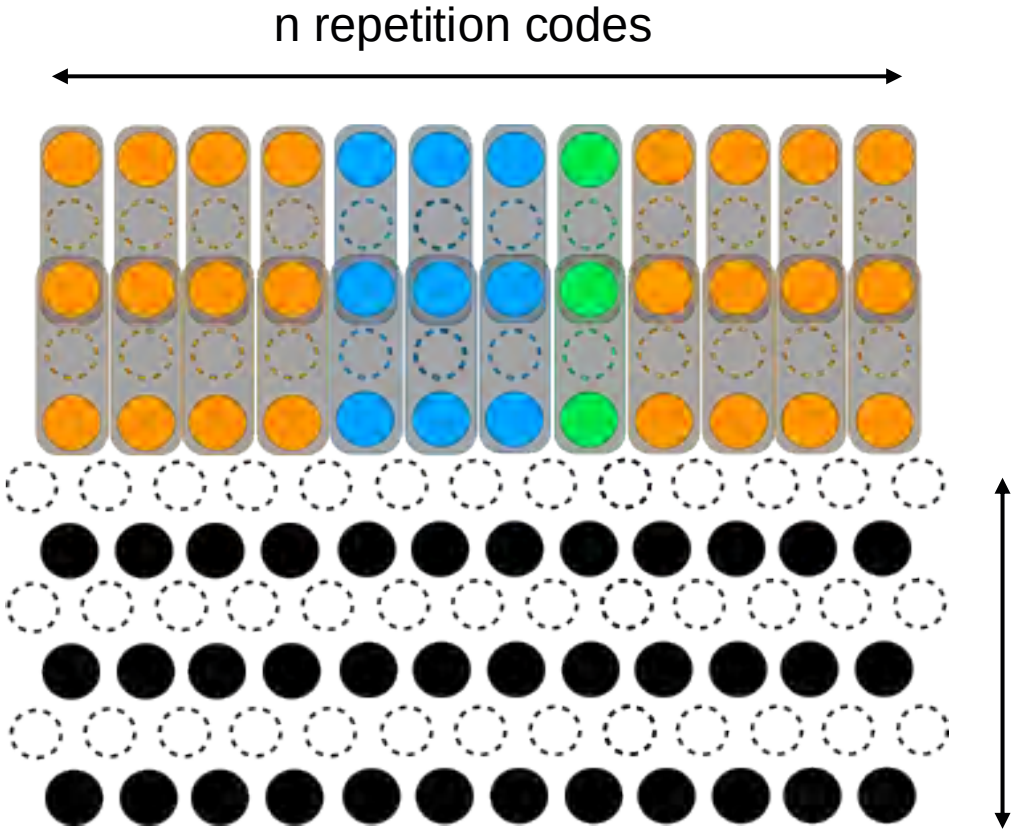
How many line of routing qubits on the layout ?



Bit-sequences of the 4 X-Pauli strings to measure

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Additional Slides/Figures



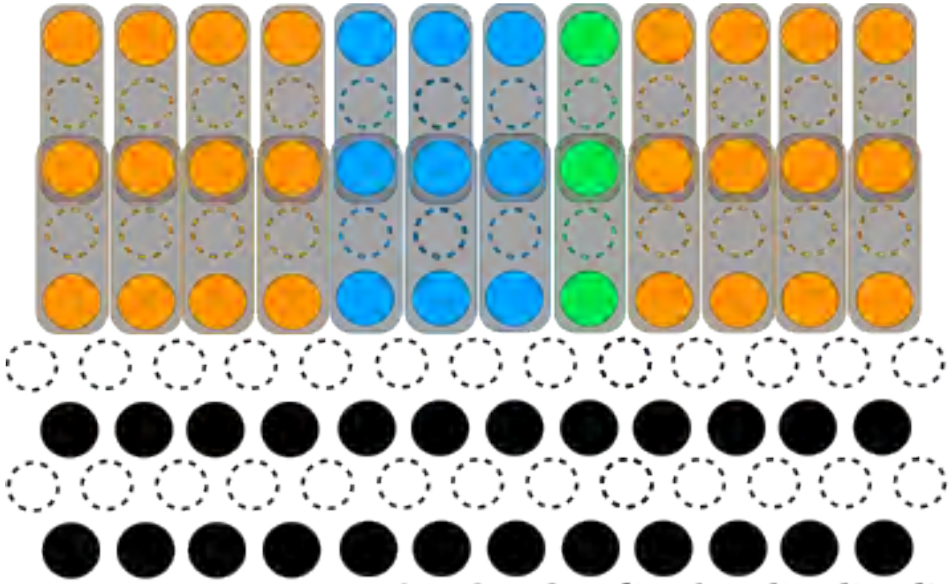
Bit-sequences of the 4 X-Pauli strings to measure

$$A = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \end{pmatrix}$$

Forget about unused logical qubit (columns of 0)

Additional Slides/Figures

How many line of routing qubits on the layout ?

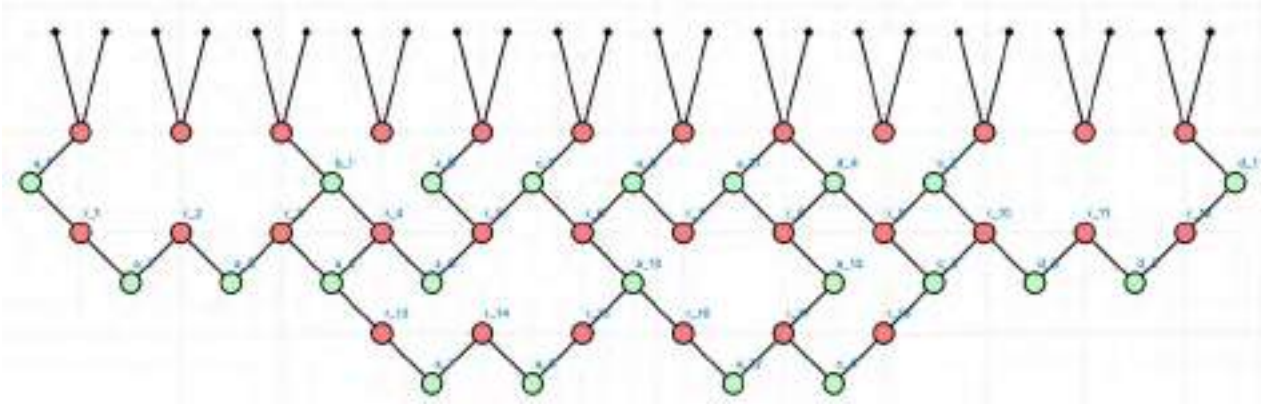


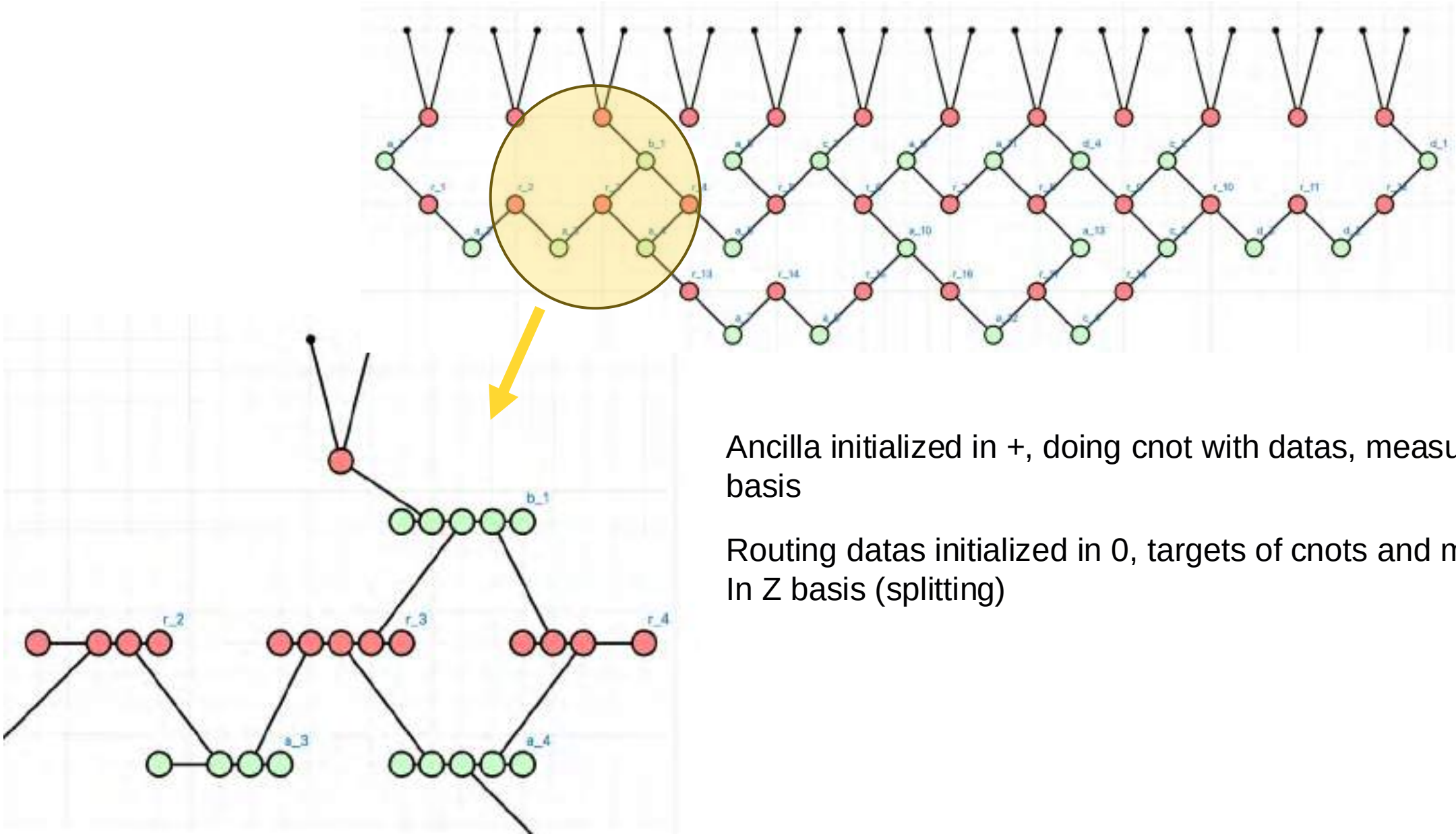
h = 2 lines

Bit-sequences of the 4 X-Pauli strings to measure

$$A = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \end{pmatrix}$$

Ask the SAT solver :





Ancilla initialized in +, doing cnot with datas, measured in X basis

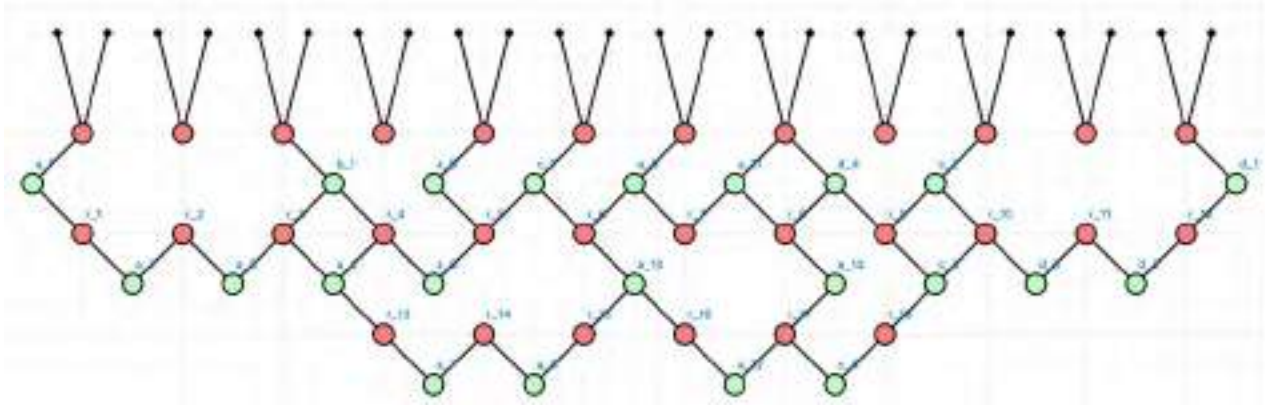
Routing datas initialized in 0, targets of cnots and measured In Z basis (splitting)

Additional Slides/Figures

How many line of routing qubits on the layout ?

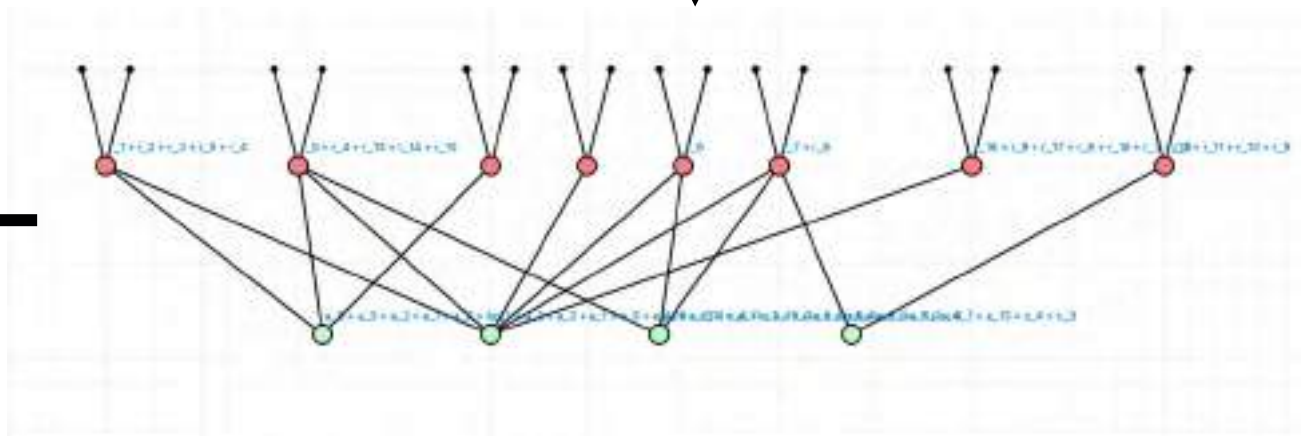
Ask the SAT solver :

$$A = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \end{pmatrix}$$



Same RowSpan

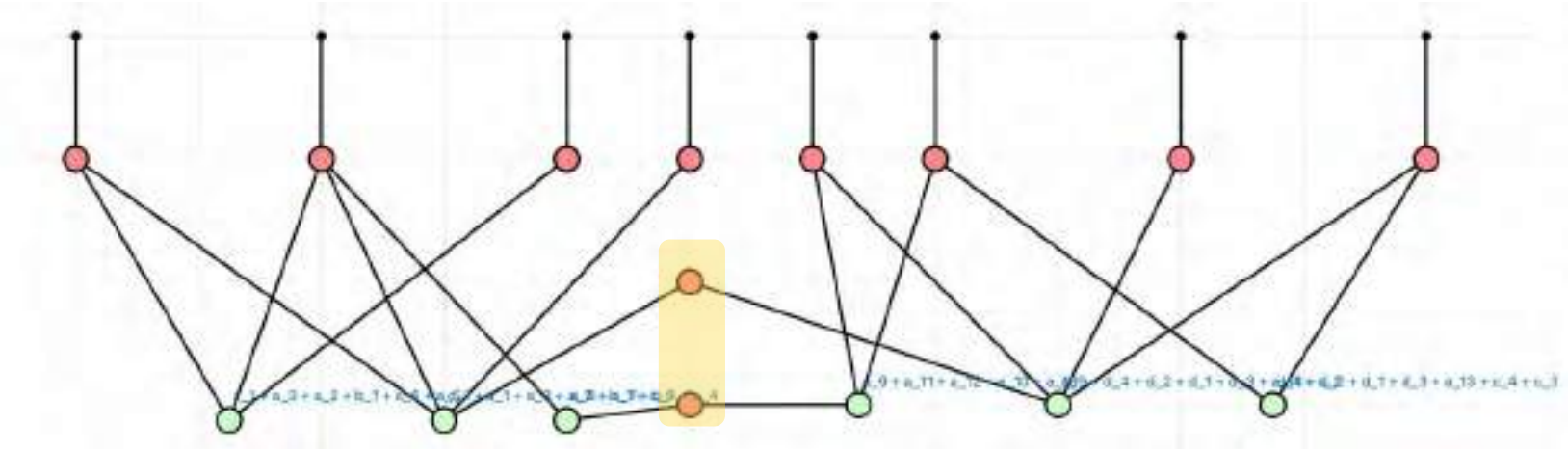
$$B = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \end{pmatrix}$$



Additional Slides/Figures

Contract every routing qubits except column k

$$A' = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \end{pmatrix}$$

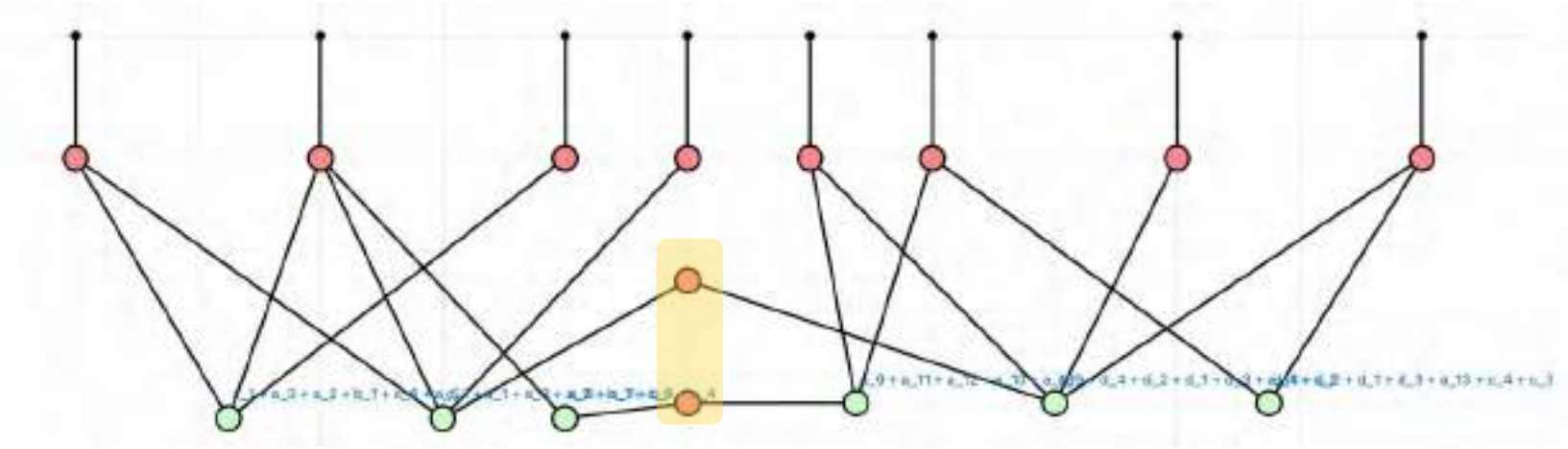


Additional Slides/Figures

Each logical measurement is a linear combination of row with no 1 in the routing columns (the last two)

(to understand as a product of stabilizer with no support remaining on routing qubits)

$$A' = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \end{pmatrix}$$



Additional Slides/Figures

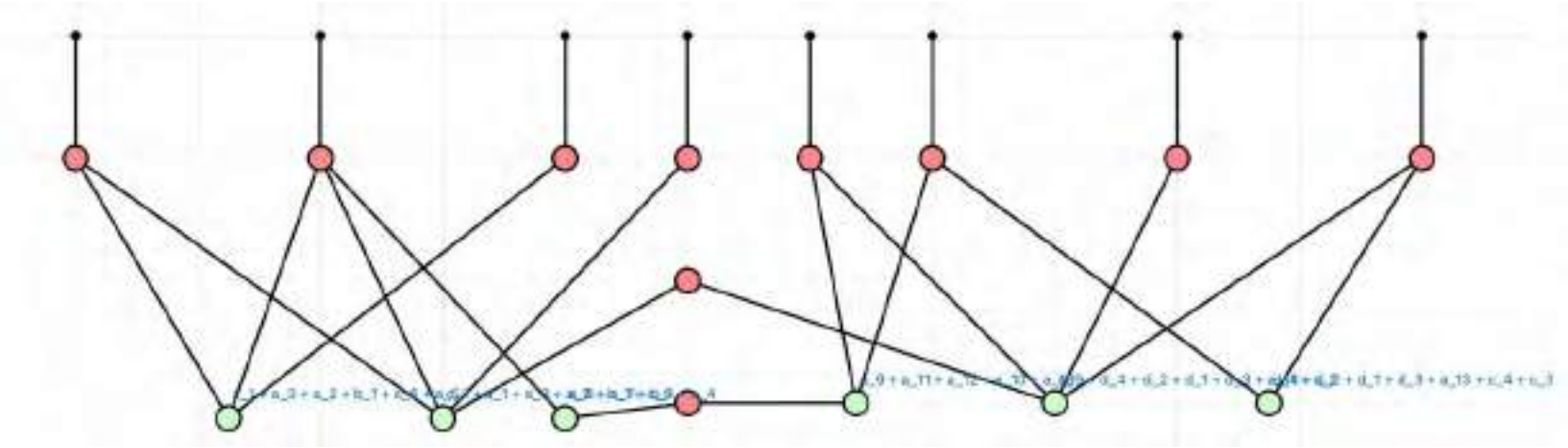
The number of routing on column k limits the number of straddling measurements I can deduce. With k additional columns in the matrix, I can deduce at most k independant equations

$$A' = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \end{pmatrix}$$

Left only

straddling

Right only



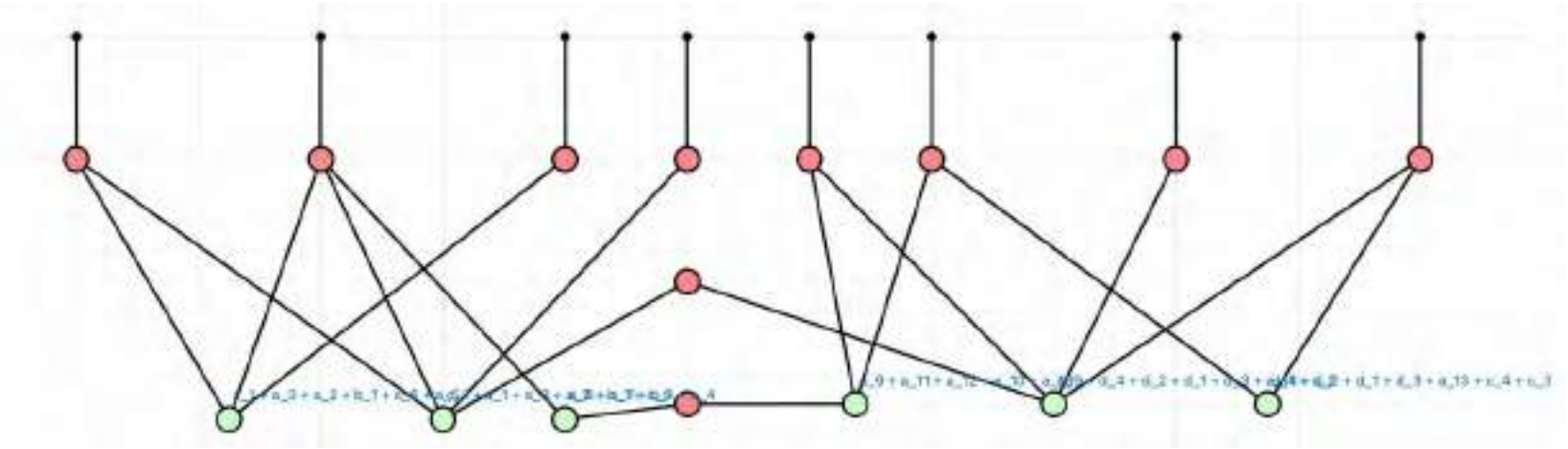
Additional Slides/Figures

$$A = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \end{pmatrix}$$

Minimal number of **independants** straddling measurements at row k determines the number of routing in that column

$$A[C_k] = \begin{pmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \end{pmatrix}$$

straddling



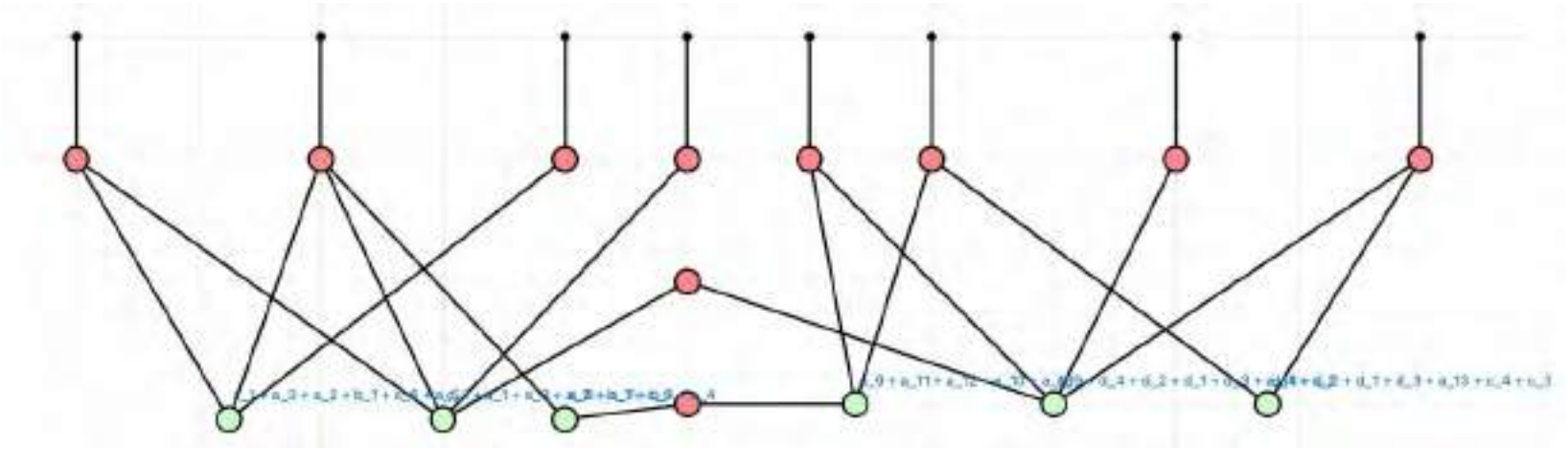
Additional Slides/Figures

$$A = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \end{pmatrix}$$

Minimal number of **independants** straddling measurements at row k determines the number of routing in that column

$$A[C_k] = \begin{pmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \end{pmatrix}$$

straddling



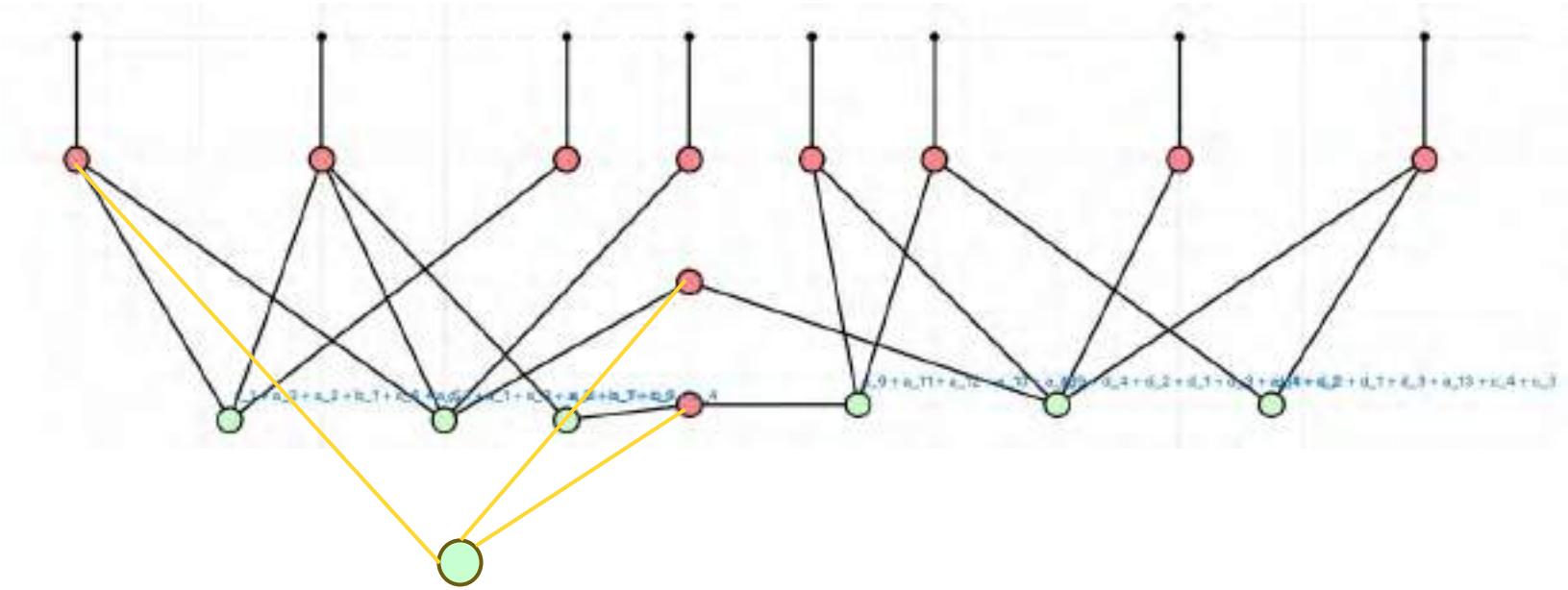
Imagine I need a third straddling measurement but whose right part is linear combination of the two first straddling measurement right part, I can add a green node on the left that connect to the two routing qubits !

Additional Slides/Figures

Minimal number of **independants** straddling measurements at row k determines the number of routing in that column

Imagine I need a third straddling measurement but whose right part is linear combination of the two first straddling measurement right part, I can add a green node on the left that connect to the two routing qubits !

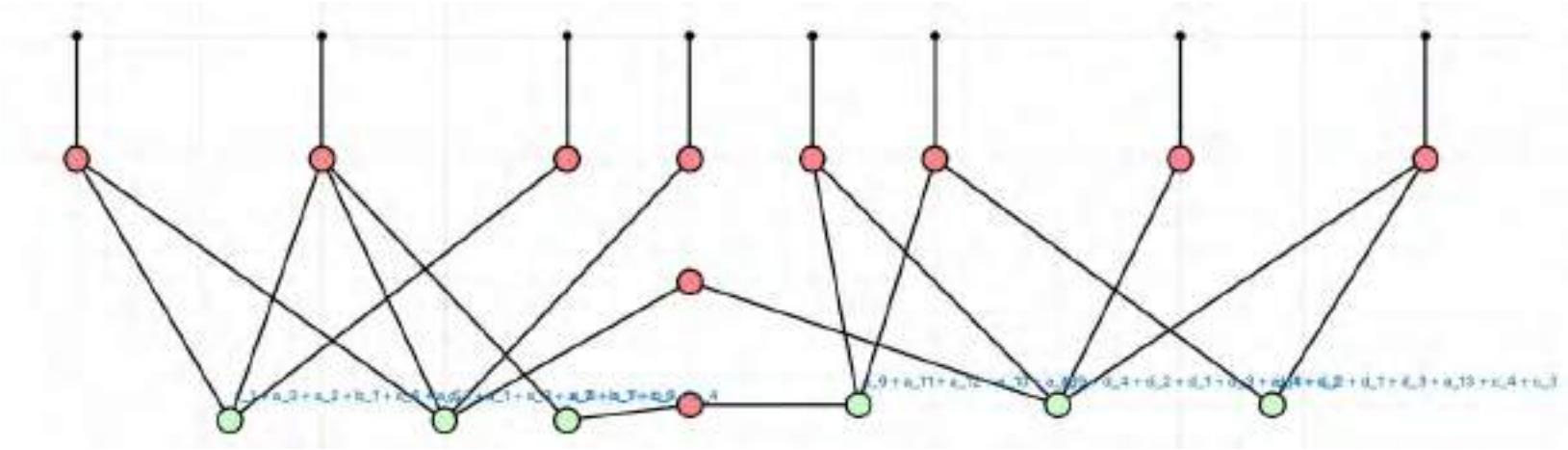
$$\begin{pmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \end{pmatrix}$$



Additional Slides/Figures

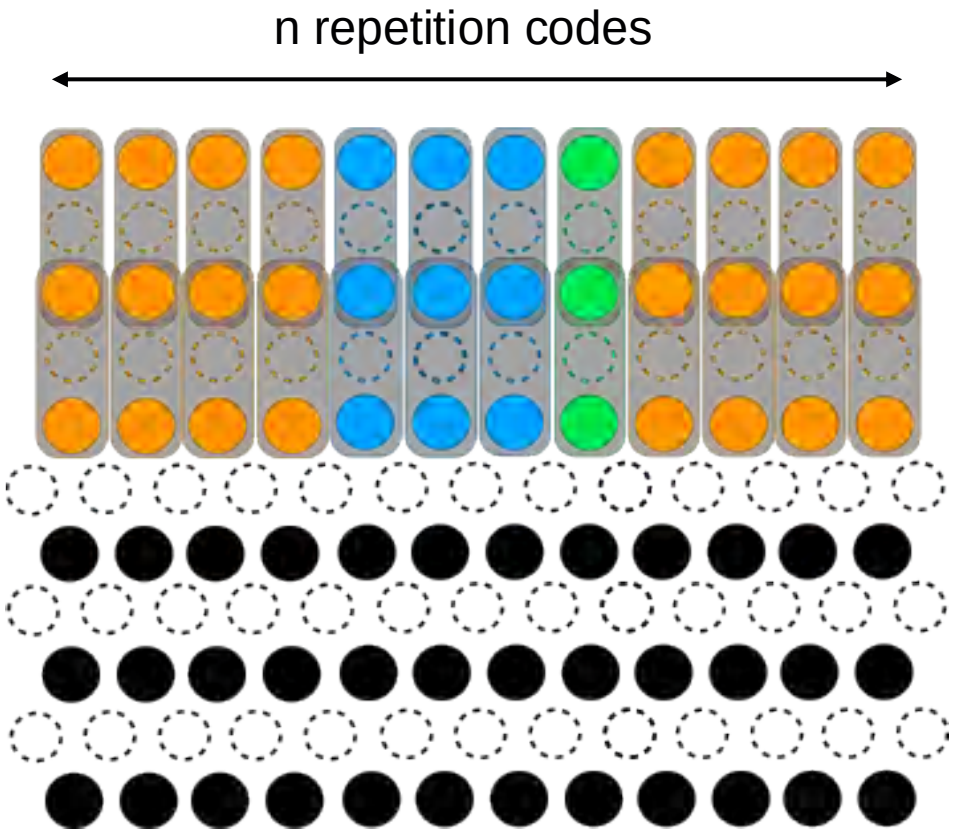
The important quantity is the **rank** of the **right part** of the **straddling measurements** at column k

$$A = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \end{pmatrix} \Rightarrow A[C_k, R_k] = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \end{pmatrix} \Rightarrow$$ Here two !



The important quantity is the **rank** of the **right part** of the **straddling measurements** at column k, we call it $\mathbf{c_k}$

General case for the width of the routing area



Bit-sequences of the 4 X-Pauli strings to measure

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$h \geq \min_{\substack{B \in \mathbb{F}_2^{m \times n} \\ \mathrm{RowSpan}(B) = \mathrm{RowSpan}(A)}} \max_{k=1}^{n-1} c_k(B)$$

$$c_k(B) = \mathrm{rk} B[C_k, R_k]$$

This is the minimal number of routing line you need, doesn't guarantee a square grid embedding !

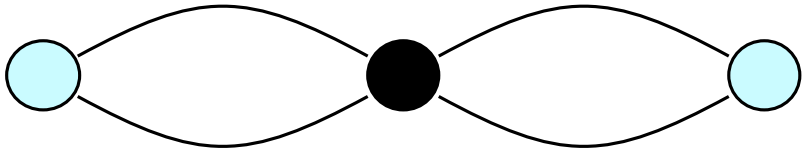
Summary & Outlook

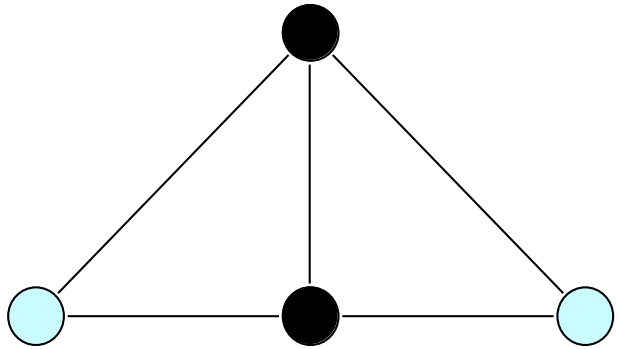


THANK YOU.

Gauge qubits

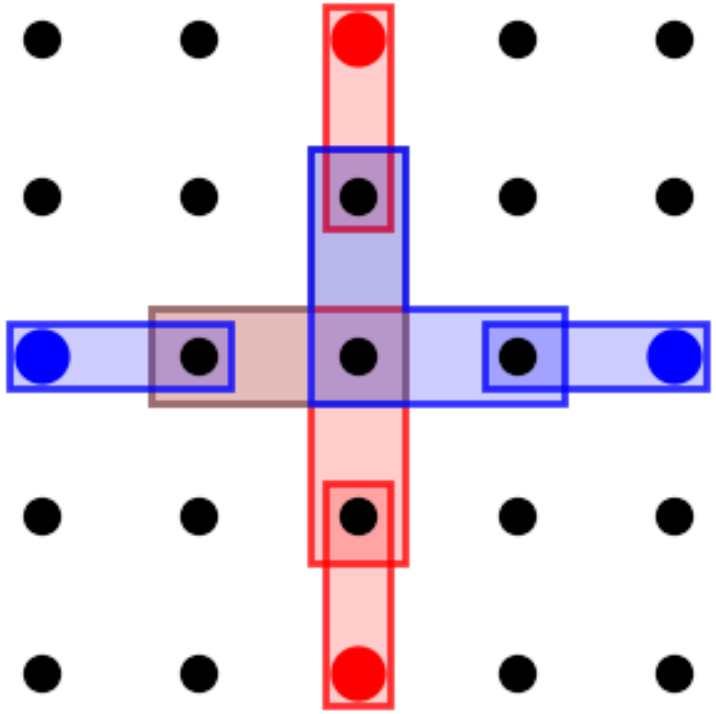
Simple example



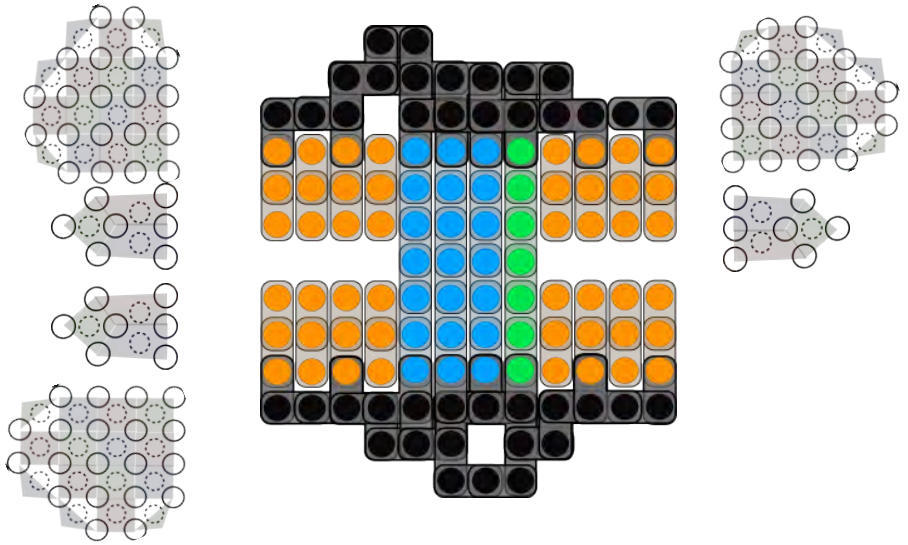


In practice, don’t appear when you minimize the number of routing qubits

Crossing lattice surgery



Resource estimation with simultaneous surgery

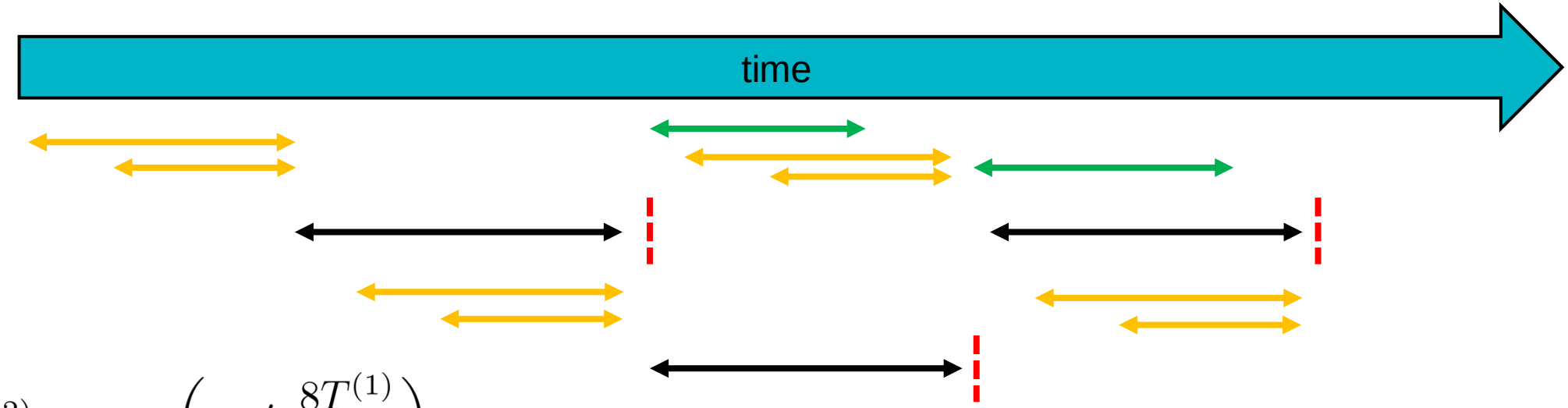


Magic state ready to use

Simultaneous lattice surgery

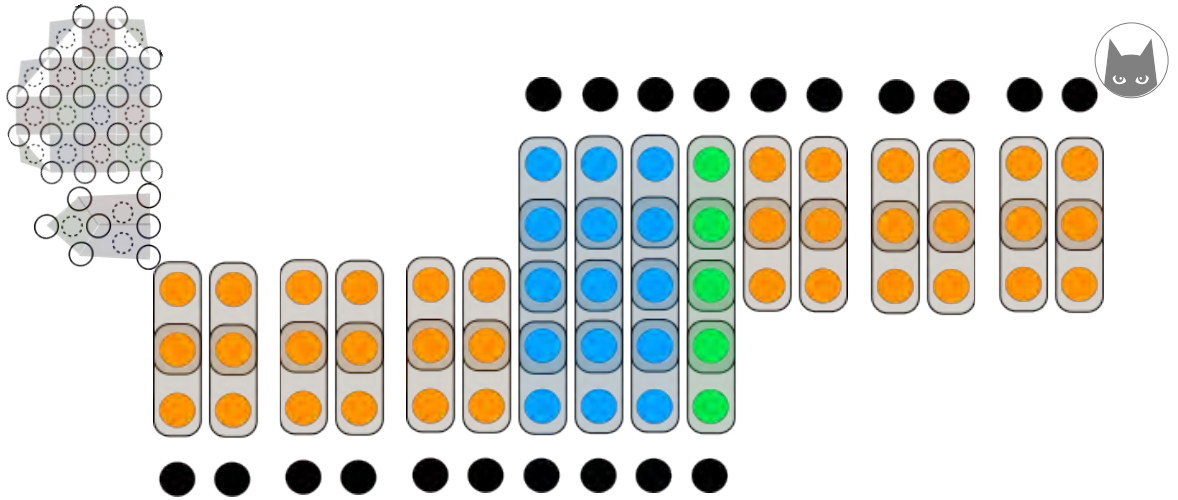
8 Magic states preparation

Reaction time



$$:= T^{(2)} = \max\left(dm', \frac{8T^{(1)}}{n_{L1}}\right)$$

Resource estimation without simultaneous surgery



Magic state ready to use

lattice surgery

2 Magic states preparation

Reaction time

time

$$:= T^{(2)} = \max\left(4 \times dm', 4 \times \frac{2T^{(1)}}{n_{L1}}\right)$$